The `revtex4` document class of the American Physical Society

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This file embodies the implementation of the APS REVTEX 4 document class for electronic submissions to journals.

The distribution point for this work is http://publish.aps.org/revtex4/, which contains fully unpacked, prebuilt runtime files and documentation.

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1 Using REVTEX

The file `README` has retrieval and installation information.

User documentation is presented separately in `revguide.tex`.

The file `template.aps` is a boilerplate file.

2 Overview

REVTEX is a LATEX 2ε document class, aomewhat like a hybrid of the standard LATEX `book` and `article` classes.

Certain packages are (should be) loaded by this class in any case: `amssymb`, `amsmath`, `bm`, `natbib`.

Certain packages are automatically loaded by this class when a corresponding class option has been invoked:

REVTEX option	**package**
`amsfonts`	`amsfonts`
`amssymb`	`amssymb`
`aps`	`overcite`

Certain other packages are to be loaded by the document through explicit use of `\usepackage`. Some mentioned in the user documentation are `graphicx`, `longtable`, and `bm`.

Certain commonly used packages are known to be incompatible with REVTEX, among them `multicol` and `cite`. If such a package is found to be loaded, REVTEX issues an error message and halts the job. Halting might be considered severe punishment for loading an incompatible package, but if we were to proceed, an even weirder error might be encountered further down the road.

This document class implements the substyle: a set of mutually exclusive class options that, in this case, allow the document class to address multiple societies. It also implements a sub-substyle, giving the journal of the given society. The only society currently implemented is `aps`, however, code is in place for another society, the `osa`.

FIXME: should always load the `graphicx` package.

QUERY: since `amsfonts` and `amssymb` extend syntax, why not load them in any case?

QUERY: if you say `\documentclass[prb,osa]{revtex4}`, then the value of `\@society` will be wrong, or? Fixed.

Certain events occur at `\AtEndOfClass` time:

1. Optionally read in packages `amsfonts`, `amssymb`.
2. Read in society substyle `.rtx` file.
3. Read in type size `.rtx` file.
4. Read in the job's style file, the `.rty` file.
5. Define things based on `\secnumarabic@sw`.
6. Install procedures to execute at the very end of the class's `\AtBeginDocument` processing, such as
   (a) closing out the page grid
   (b) putting out the LastPage label.
   (c) issuing a `\bibliographystyle` command, based on the value set by the society substyle.
   (d) setting default values for parameters used in the document. FIXME: differentiate between class's parameters that can wait until they are used in the document, and parameters that are used at `\AtBeginDocument` time.
7. Install procedures to execute the very last at `\AtEndDocument` time, such as the `\clearpage` processing.

Certain events are optionally scheduled for `\AtBeginDocument` time:

1. Setting default values for the Booleans and for other procedures used in formatting.
2. In response to class options options, adjusting parameters and procedures used in formatting.
3. Implementing the `eqsecnum` option, if required.
4. Setting the state engine for data commands.
5. Memorizing procedures for later use.
6. Setting type size and area, for use by later calculations.

Certain events are scheduled for `\class@enddocumenthook` time:

1. Print out the migrated floats or the end notes, if needed.
2. Close out the page grid.
3. Label the last page of the document
4. (`natbib`) prepare to read in the `.aux` file.

3 The installer file

We include here a reference copy of the installer file, `revtex4.ins`. It is not intended that one actually generate that file using `docstrip` via the revtex4 installer file, because that would involve overwriting the file that TEX is typesetting. Instead, the installer file is provided pre-generated.

For the most part, the installer is simple: it generates `revtex4.cls` and a collection of `.rtx` files from `revtex4.dtx`.

However, the process of generating `revtex4.cls` involves the incorporation of portions of two of the `ltx` series of files, `ltxutil.dtx`, and `ltxgrid.dtx`. The former is a collection of macro definitions that serve as the lowest level of extensions to LATEX, the latter implements an alternative output routine and page gridding mechanism that provides for multicolumn page layouts, but without the sorry limitations of the LATEX required package, `multicol`, and is also compatible with the `longtable` package (which `multicol` is not).

```
%<*ins>
% This file requires docstrip version 2.4 or higher available from
% ftp://ctan.tug.org/tex-archive/macros/latex/unpacked/docstrip.tex
\input docstrip
\preamble

This file is part of the APS files in the REVTeX 4 distribution.
For the version number, search on the string %FileInfo

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http://publish.aps.org/revtex4/
mailto:revtex@aps.org

Maintained by Arthur Ogawa (mailto:ogawa@teleport.com)
under contract to the American Physical Society.

See the REVTeX 4 README file for restrictions and more information.

\endpreamble
\keepsilent
\askforoverwritefalse
\generate{%
  \file{revtex4.cls}{%
   \from{revtex4.dtx}{class-pre}%
   \from{ltxutil.dtx}{ltxutil-krn}%
   \from{ltxgrid.dtx}{ltxgrid-krn}%
   \from{textcase.dtx}{package}%
   \from{revtex4.dtx}{class-pst}%
  }%
  \file{aps.rtx}{\from{revtex4.dtx}{aps}}%
  \file{rmp.rtx}{\from{revtex4.dtx}{rmp}}%
  \file{10pt.rtx}{\from{revtex4.dtx}{10pt}}%
  \file{11pt.rtx}{\from{revtex4.dtx}{11pt}}%
```

```
  \file{12pt.rtx}{\from{revtex4.dtx}{12pt}}%
  \file{revsymb.sty}{\from{revtex4.dtx}{revsymb}}%
}%
\endbatchfile
%</ins>
```

4 Writing journal-specifc extensions to REVTEX

With this version of REVTEX, we introduce a somewhat different scheme for adapting REVTEX to the needs of a specific journal.

To create a journal substyle, you create new class options in REVTEX for the society, say `osa`, and any of that society's journals, one of which is, say, `josaa`, using the code for the APS as a guide. In particular, each of your new options should separately define `\@society` and `\@journal`. That for the former will be the same for all options relating to a particular society.

Then, for the society, you create a corresponding `.rtx` file, in our case `osa.rtx`. Within that file, you override procedures and parameter assignments as you see fit. Ideally they will be generally applicable to all of that society's journals (see the file `aps.rtx` for a realization of this scheme). Also within that file, you include a section of code for each journal, that for josaa looks like:

```
% \@ifx{\@journal\journal@josaa}{%
% ⟨code specific to the josaa⟩
% }{}%
%
```

Thus far, the scheme is similar to that used in REVTEX 3.1. However, the new scheme does differ from the old in that the `.rtx` file should define no syntactical extensions to REVTEX.

5 The `revtex4` Document Class

```
%<*class-pre>
\def\class@name{revtex4}%
```

Print a banner in the log:

```
\GenericInfo{}{\space
 Copyright (c) 2001 The American Physical Society.^^J
 mailto:revtex@aps.org^^J
 Licensed under the LPPL:^^Jhttp://www.ctan.org/tex-archive/macros/latex/base/lppl.txt^^J
 Arthur Ogawa <ogawa@teleport.com>^^J
 Based on work by David Carlisle <david@dcarlisle.demon.co.uk>%
 \@gobble
}%
```

5.1 Compatability Processing

If the document has `\documentstyle{revtex4}`, then, instead of attempting to run in compatability mode, just complain and exit.

```
\if@compatibility
  \edef\reserved@a{\errhelp{%
   Change your \string\documentstyle\space statement to
   \string\documentclass\space and rerun.
  }}\reserved@a
  \errmessage{You cannot run \class@name\space in compatability mode}%
  \expandafter\@@end
\fi
```

6 Extensions to the LaTeX Kernel

I am introducing some portions of the `ltxkrnext` package (ftp://ftp.teleport.com/users/ogawa/macros/latex/contrib/supported/ltxkrnext/).

First, de-fang unwanted commands and turn `\ProvidesPackage` into an informative message.

```
\let\DeclareOption@latex\DeclareOption
\let\DeclareOption\@gobbletwo
\let\ProcessOptions@latex\ProcessOptions
\let\ProcessOptions\@empty
\let\ProvidesPackage@latex\ProvidesPackage
\def\ProvidesPackage#1[#2]{%
 \class@info{Incorporating package #1 [#2].}%
}%
%</class-pre>
```

Now load the utility and page grid packages. The `ltxutil` and `ltxgrid` packages are distributed with REVTeX.

```
%\RequirePackage{ltxutil}%
%\RequirePackage{ltxgrid}%
```

The `textcase` package is available from CTAN:/macros/latex/contrib/supported/carlisle/textcase.dtx and is used at the suggestion, and by permission, of David P. Carlisle.

> \ProvidesFiletextcase.dtx [1998/11/12 v0.06 Text only upper/lower case changing (DPC)]

David, in turn, suggests that Donald Arseneau be credited for some of the ideas in the `textcase` package, so let it hereby be so understood.

Note that David's package has a bug, which is fixed as follows: Change:

```
%       \def\NoCaseChange##1{\noexpand\NoCaseChange{\noexpand##1}}
%
```

To:

```
%       \def\NoCaseChange##1{\noexpand\NoCaseChange{\noexpand##1}}%
%
```

```
%\RequirePackage{textcase}%
```

Next, restore those cammands we de-fanged:

```
%<*class-pst>
\let\ProvidesPackage\ProvidesPackage@latex
\let\ProcessOptions\ProcessOptions@latex
\let\DeclareOption\DeclareOption@latex
```

And let the `textcase` commands override those of LaTeX:

```
\expandafter
\let\csname MakeUppercase \expandafter\endcsname
    \csname MakeTextUppercase \endcsname
\expandafter
\let\csname MakeLowercase \expandafter\endcsname
    \csname MakeTextLowercase \endcsname
```

Try to patch `longtable`.

```
\appdef\class@documenthook{%
 \switch@longtable
}%
```

Try to provide a more robust version of `\boldmath`. If Team LaTeX ever gets around to fixing this problem, we will bow out. (This relates to bug #394.)

```
\appdef\class@documenthook{%
 \expandafter\@ifnotrelax\csname boldmath \endcsname{}{%
  \expandafter\let\csname boldmath \endcsname\boldmath
  \expandafter\def\expandafter\boldmath\expandafter{\expandafter\protect\csname boldmath \endcs
 }%
}%
```

7 Options

7.1 Define Booleans Used in Options

The following Booleans are used within the document class to allow the document or the substyle to make selections of formatting. Because an explicit document class option always overrides a substyle, any substyle wishing to assign a value to a flag must first test the flag to confirm that it is still undefined. This further means that the default value of the flag must be assigned late, at the end of `\AtBeginDocument` time, and that the flag may be queried no earlier than that time.

`\class@inithook` The hook `\class@inithook` is installed into `\class@documenthook`, and thereafter serves as the focal point for all code that gives default values to the class's parameters.

A typical code might be:

```
%\appdef\class@inithook{%
% \@ifxundefined\twocolumn@sw{\@booleanfalse\twocolumn@sw}{}%
%}%
%
```

which ensures that the switch \twocolumn@sw is never left undefined.

```
\appdef\class@documenthook{%
 \class@inithook
}%
\def\class@inithook{}%
%
```

\twocolumn@sw The boolean \twocolumn@sw signifies that we are to use a two-column grid.

```
\appdef\class@inithook{%
 \@ifxundefined\twocolumn@sw{\@booleanfalse\twocolumn@sw}{}%
}%
```

\footinbib@sw The boolean \footinbib@sw signifies that footnotes are to be set in the bibliography, as endnotes.

```
\appdef\class@inithook{%
 \@ifxundefined\footinbib@sw{\@booleanfalse\footinbib@sw}{}%
}%
```

\preprintsty@sw The boolean \preprintsty@sw signifies that the document is to be formatted in preprint style.

```
\appdef\class@inithook{%
 \@ifxundefined\preprintsty@sw{\@booleanfalse\preprintsty@sw}{}%
}%
```

\preprint@sw The boolean \preprint@sw signifies that the preprints (cf. \preprint) are to be formatted (usually on the title page). The default is to do so; the option nopreprintnumbers declares not to.

```
\appdef\class@inithook{%
 \@ifxundefined\preprint@sw{\@booleanfalse\preprint@sw}{}%
 \preprintsty@sw{\@booleantrue\preprint@sw}{}%
}%
```

\galley@sw The boolean \galley@sw signifies that the document is to be formatted in galley style.

Asserting both \galley@sw and \preprintsty@sw may produce strange formatting results, but it is not illegal. However, it *is* illegal to assert galley and any twocolumn option.

```
\appdef\class@inithook{%
 \@ifxundefined\galley@sw{\@booleanfalse\galley@sw}{}%
}%
```

\groupauthors@sw The flag \groupauthors@sw signifies that authors are to be grouped. This affects the meaning of \@author@present and \@tempaffil.

FIXME: apparently \@tempaffil and its aliases are never used.

\runinaddress@sw The flag \runinaddress@sw signifies that author addresses are to be run in.

`\@affils@sw` If `\@affils@sw` is false, an address is never recognised as "new" and is therefore always entered into the affiliation list, stopping groups of authors at the same address being amalgamated into the same list: the address will be printed the same number of times it is entered.

`\showPACS@sw` `\showKEYS@sw` If `\showPACS@sw` is true, print the PACS information in the title block, otherwise not. Similarly for `\showKEYS@sw` and the keywords.

```
\appdef\class@inithook{%
 \@ifxundefined\showPACS@sw{\@booleanfalse\showPACS@sw}{}%
 \@ifxundefined\showKEYS@sw{\@booleanfalse\showKEYS@sw}{}%
}%
```

`\floats@sw` The Boolean `\floats@sw` signifies that floats are to be floated; if false, that floats are to be deferred to the end of the document. By default, the former. Note that the state of this Boolean is to be changed by the document class in response to user-selected options.

This boolean and the assignment of its default value is done by the `ltxutil` package.

`\floatp@sw` The Boolean `\floatp@sw` signifies that endfloats are to be set one per page; if false, that endfloats are to be set with multiple floats per page permitted. By default, the latter. Note that the state of this Boolean is to be changed by the document class in response to user-selected options. The default is established here.

`\twoside@sw` The flag `\twoside@sw` signifies that the document is to be formatted for duplex printing. At `\AtBeginDocument` time, we must align the value of the kernel `\newif` switch `\if@twoside` to that of `\twoside@sw`.

```
\appdef\class@inithook{%
 \@ifxundefined\twoside@sw{\@booleanfalse\twoside@sw}{}%
 \twoside@sw{\@twosidetrue}{\@twosidefalse}%
}%
```

`\draft@sw` The flag `\draft@sw` signifies that the document is to be formatted in draft mode. Certain packages may pay attention to the class option `draft` that sets this Boolean.

```
\appdef\class@inithook{%
 \@ifxundefined\draft@sw{\@booleanfalse\draft@sw}{}%
 \draft@sw{\overfullrule 5\p@}{\overfullrule\z@}%
}%
```

`\tightenlines@sw` The boolean `\tightenlines@sw` signifies that the leading is to be made standard amount. If false, it means that the leading is to be set extra open. Has no effect on 10pt size option.

```
\appdef\class@inithook{%
 \@ifxundefined\tightenlines@sw{\@booleanfalse\tightenlines@sw}{}%
}%
```

\lengthcheck@sw The flag \lengthcheck@sw signifies that the length checking is in effect. It is up to the individual journal substyle to alter its formatting accordingly.

```
\appdef\class@inithook{%
 \@ifxundefined\lengthcheck@sw{\@booleanfalse\lengthcheck@sw}{}%
}%
```

\eqsecnum@sw The flag \eqsecnum@sw signifies that equations are to be numbered with the section, e.g., "Eq. (2.13)". This flag is neither set nor queried.

```
\appdef\class@inithook{%
 \@ifxundefined\eqsecnum@sw{\@booleanfalse\eqsecnum@sw}{}%
 \eqsecnum@sw{%
  \@addtoreset{equation}{section}%
  \def\theequation@prefix{\arabic{section}.}%
 }{}%
}%
```

\byrevtex@sw The flag \byrevtex@sw signifies that the document should bear an imprint to the effect that it was formatted by this document class.

```
\appdef\class@inithook{%
 \@ifxundefined\byrevtex@sw{\@booleanfalse\byrevtex@sw}{}%
}%
```

7.2 Declare Options

7.2.1 Checkin: for Editorial Use

A document class option declaring that the document is being processed by the editorial staff.

This option should:

- put date in footer along with folio
- Have the effect of selecting the `preprint` class option.
- Have the effect of selecting the `showpacs` class option.
- specify that when a float is placed h or H, it will be allowed to break over pages. (Note: be sure that if the enclosed `tabular` has an optional argument, you change it to [v], or remove it entirely.)

```
\DeclareOption{checkin}{%
  \@booleantrue\dateinRH@sw
  \@booleantrue\preprintsty@sw
  \def\@pointsize{12}%
  \@booleantrue\showPACS@sw
  \@booleantrue\showKEYS@sw
  \def\fp@proc@h{\allow@breaking@tables}%
  \def\fp@proc@H{\allow@breaking@tables}%
}%
\@booleanfalse\dateinRH@sw
```

```
\def\checkindate{\dateinRH@sw{{\tiny(\today)}}{}}%
\def\allow@breaking@tables{%
 \def\array@default{v}% tabular can break over pages
 \@booleanfalse\floats@sw % table can break over pages
}%
```

7.2.2 Preprint Style

```
\DeclareOption{preprint}{%
  \@booleantrue\preprintsty@sw
  \def\@pointsize{12}%
}%
\DeclareOption{manuscript}{%
 \class@warn{Document class option manuscript is obsolete; use preprint instead}%
 \ExecuteOptions{preprint}%
}%
```

7.2.3 Showing PACS and keywords

```
\DeclareOption{showpacs}{%
  \@booleantrue\showPACS@sw
}%
\DeclareOption{noshowpacs}{%
  \@booleanfalse\showPACS@sw
}%
\DeclareOption{showkeys}{%
  \@booleantrue\showKEYS@sw
}%
\DeclareOption{noshowkeys}{%
  \@booleanfalse\showKEYS@sw
}%
```

7.2.4 Balance the last page when in two-column page grid

If we are in a two-column page grid, we may wish to balance the columns of the last page. This will be done automatically if the `twocolumn` document class option is chosen. This action will be turned off by the `nobalancelastpage`. A complementary class option, `balancelastpage` is also provided.

```
\DeclareOption{balancelastpage}{%
  \@booleantrue\balancelastpage@sw
}%
\DeclareOption{nobalancelastpage}{%
  \@booleanfalse\balancelastpage@sw
}%
\appdef\class@inithook{%
 \@ifxundefined\balancelastpage@sw{%
  \@booleantrue\balancelastpage@sw
 }{}%
}%
```

7.2.5 Showing preprint numbers

```
\DeclareOption{nopreprintnumbers}{\@booleanfalse\preprint@sw}%
\DeclareOption{preprintnumbers}{\@booleantrue\preprint@sw}%
```

7.2.6 Hypertext Option

```
\DeclareOption{hyperref}{%
%\AtEndOfClass{%
% \begingroup
% \edef\@tempa{%
%  \let
%  \noexpand\@clsextension
%  \noexpand\@empty
%  \noexpand\RequirePackage{hyperref}%
%  \def\noexpand
%  \@clsextension{\@clsextension}%
% }%
% \expandafter
% \endgroup
% \@tempa
%}%
```

If you have a hyper-foo enabled browser you may prefer this format which does not print the URL for the home page, but just makes the name a link, but by default print it so it works on paper.

```
% \def\@pointsize{10}%
 \class@warn{Class option "hyperref" is no longer supported.^^JEmploy \string\usepackage{hyperr
}%
```

7.2.7 Type Size

Use `\@pointsize=10` rather than `\@ptsize=0` to allow easy extensions to 9pt or whatever. Note: the three alternatives are mutually exclusive. Furthermore if `\@pointsize` is still undefined at `\AtEndOfClass` time, then the substyle can conclude that the user accepts the default.

```
\DeclareOption{10pt}{\def\@pointsize{10}}
\DeclareOption{11pt}{\def\@pointsize{11}}
\DeclareOption{12pt}{\def\@pointsize{12}}
```

7.2.8 Media Size

```
\DeclareOption{a4paper}
   {\setlength\paperheight {297mm}%
    \setlength\paperwidth  {210mm}}
\DeclareOption{a5paper}
   {\setlength\paperheight {210mm}%
    \setlength\paperwidth  {148mm}}
\DeclareOption{b5paper}
   {\setlength\paperheight {250mm}%
    \setlength\paperwidth  {176mm}}
\DeclareOption{letterpaper}
```

```
   {\setlength\paperheight {11in}%
    \setlength\paperwidth  {8.5in}}
\DeclareOption{legalpaper}
   {\setlength\paperheight {14in}%
    \setlength\paperwidth  {8.5in}}
\DeclareOption{executivepaper}
   {\setlength\paperheight {10.5in}%
    \setlength\paperwidth  {7.25in}}
\DeclareOption{landscape}
   {\setlength\@tempdima   {\paperheight}%
    \setlength\paperheight {\paperwidth}%
    \setlength\paperwidth  {\@tempdima}}
```

7.2.9 Bibnotes

Frontmatter footnotes result from frontmatter commands like `\email`, `\homepage`, `\altaffiliation`, and `\thanks`. The default for `\frontmatter@footnote@produce` is `\frontmatter@footnote@produce@footnote`, which formats the frontmatter footnotes at the foot of the title page. The `bibnotes` class option defers them to the bibliography.

```
\DeclareOption{bibnotes}{\@booleantrue\bibnotes@sw}%
\DeclareOption{nobibnotes}{\@booleanfalse\bibnotes@sw}%
\appdef\class@enddocumenthook{\auto@bib}%
\appdef\class@inithook{%
 \@ifxundefined\bibnotes@sw{\@booleanfalse\bibnotes@sw}{}%
 \bibnotes@sw{%
  \let\frontmatter@footnote@produce\frontmatter@footnote@produce@endnote
 }{%
  \let\frontmatter@footnote@produce\frontmatter@footnote@produce@footnote
 }%
}%
```

7.2.10 Footinbib

```
\DeclareOption{footinbib}{\@booleantrue\footinbib@sw}
\DeclareOption{nofootinbib}{\@booleanfalse\footinbib@sw}
```

7.2.11 altaffilletter

```
\DeclareOption{altaffilletter}{\@booleantrue\altaffilletter@sw}%
\DeclareOption{altaffilsymbol}{\@booleanfalse\altaffilletter@sw}%
\appdef\class@inithook{%
 \@ifxundefined\altaffilletter@sw{\@booleanfalse\altaffilletter@sw}{}%
}%
```

7.2.12 superbib

The document may invoke superscript bibliography style. If not, the journal substyle may make a selection. If neither, the class itself selects a default. This is really a quite deep nesting of defaults.

```
\DeclareOption{superbib}{%
```

```
 \def\place@bibnumber{\place@bibnumber@sup}%
}%
```

7.2.13 citeautoscript

This class option allows you to automatically accomodate a change from non-superscripted, numbered references to superscripted, bunmbered references.

Note: you should always mark up your document with the assumption that references are *not* going to be superscripted. Otherwise this option has no hope of working properly.

```
\DeclareOption{citeautoscript}{%
 \@booleantrue\citeautoscript@sw
}%
\@booleanfalse\citeautoscript@sw
```

7.2.14 Simplex/Duplex Pages

The complementary options `twoside` and `oneside` assert formatting for duplex or simplex printing, respectively. At the same time, we arrange for the selection of the page grid with respect to the marginal column: Because `\if@reversemargin` remains default (false), if duplex printing, this column will always be on the (right), if simplex printing, it will always be on the (outside). QUERY: correct choice?

```
\DeclareOption{twoside}{\@booleantrue\twoside@sw \@mparswitchfalse}
\DeclareOption{oneside}{\@booleanfalse\twoside@sw\@mparswitchtrue}
```

7.2.15 Two-Column Page Grid

`\twocolumn@sw` The flag `\twocolumn@sw` signifies that the document is to be formatted in the two-column page grid.

If no options relating to page grid are invoked by `\AtBeginDcoument` time, we set default values. Up to that point, the class can check if `\twocolumn@sw` is `\undefined` to see if any related options have been invoked.

At `\AtBeginDocument` time, we must align the value of the kernel `\newif` switch `\if@twocolumn` to that of `\twocolumn@sw`.

We install into `\class@enddocumenthook` all tasks that we know must be completed at `\AtEndDocument` time. The first thing it does is end the current paragraph. The last is to return us to one-column page grid. (This task is installed as late as possible, i.e., after items like `\printtables`.)

```
\AtEndOfClass{%
 \def\clear@document{%
  \close@column
 }%
 \appdef\class@inithook{%
  \@ifxundefined\close@column{%
   \let\close@column\close@column@default
  }{}%
 }%
```

```
}%
```

The `twocolumn` class option employs the standard mechanism for changing the column grid: the `ltxgrid` package.

```
\DeclareOption{twocolumn}{%
  \@booleantrue\twocolumn@sw
}%
```

`onecolumn` specifies one-column page grid.

```
\DeclareOption{onecolumn}{%
  \@booleanfalse\twocolumn@sw
}%
```

`galley` emulates setting the galleys of a two-column journal. FIXME: this option should effectively set `\preprintsty@sw` false. NOTE: it makes no sense to assert both galley and twocolumn.

```
\DeclareOption{galley}{%
  \ExecuteOptions{onecolumn}%
  \@booleantrue\galley@sw
  \appdef\set@pica@hook{%
    \advance\textwidth-\columnsep
    \textwidth.5\textwidth
  }%
}%
```

7.2.16 raggedbottom or flushbottom

The class options `raggedbottom` and `flushbottom` determine whether the columns (page) are ragged bottom or flush bottom. Note that we do not select a default here; that is done by the journal substyle.

```
\DeclareOption{raggedbottom}{\@booleantrue\raggedcolumn@sw}
\DeclareOption{flushbottom}{\@booleanfalse\raggedcolumn@sw}
```

7.2.17 tightenlines

This class option specifies that standard leading is to be used to set the type. If lacking, the leading will be loose.

```
\DeclareOption{tightenlines}{\@booleantrue\tightenlines@sw}
```

7.2.18 lengthcheck

This class option specifies that the formatted document should approach as closely as possible the formatting of an actual journal article to facilitate the author's performance of a length check.

```
\DeclareOption{lengthcheck}{%
 \@booleantrue\lengthcheck@sw
 \@booleantrue\twoside@sw
 \@booleantrue\twocolumn@sw
}%
```

7.2.19 Draft and Final

```
\DeclareOption{draft}{%
 \@booleantrue\draft@sw
}%
\DeclareOption{final}{%
 \@booleanfalse\draft@sw
}%
```

7.2.20 eqsecnum

The `eqsecnum` class option signifies that equations are to be numbered within sections.

```
\DeclareOption{eqsecnum}{\@booleantrue\eqsecnum@sw}
```

7.2.21 secnumarabic

The `secnumarabic` class option signifies that sectioning commands are to be numbered arabic.

```
\DeclareOption{secnumarabic}{\@booleantrue\secnumarabic@sw}
```

The code that polls the Boolean `\secnumarabic@sw` appears in Section 22.2.

FIXME: model fleqn after amsfonts. I no longer understand why I said this.

```
\DeclareOption{fleqn}{\input{fleqn.clo}}
```

7.2.22 floats/endfloats

These options control, via the Boolean `\floats@sw`, whether floats are to be migrated to the end of the document.

```
\DeclareOption{floats}{\@booleantrue\floats@sw\@booleanfalse\floatp@sw}
\DeclareOption{endfloats}{\@booleanfalse\floats@sw\@booleanfalse\floatp@sw}
\DeclareOption{endfloats*}{\@booleanfalse\floats@sw\@booleantrue\floatp@sw}
\appdef\class@inithook{%
 \@ifxundefined\floats@sw{\@booleantrue\floats@sw}{}%
 \@ifxundefined\floatp@sw{\@booleantrue\floatp@sw}{}%
}%
```

7.2.23 titlepage/notitlepage

These options control, via `\titlepage@sw`, whether the title block is to be set on a separate page.

`\titlepage@sw` The flag `\titlepage@sw` signifies that a forced page break is to follow the title page: the article title appears on a page by itself.

The Boolean `\titlepage@sw` is in an odd situation: its value is set by one or more substyles, contingent upon the value of another Boolean. Therefore that code must be invoked at `\AtBeginDocument` time. Therefore we must assign the default value at the very end of `\AtBeginDocument` processing: we use `\defaults@hook`. Luckily, this Boolean does not need to be queried at `\AtBeginDocument` time, or we'd be in dire straights.

```
\DeclareOption{titlepage}{\@booleantrue\titlepage@sw}
\DeclareOption{notitlepage}{\@booleanfalse\titlepage@sw}
\appdef\defaults@hook{%
 \@ifxundefined\titlepage@sw{\@booleanfalse\titlepage@sw}{}%
}%
```

7.2.24 Substyle and Sub-substyle

If the society has already been assigned, notify user that `\@society` is being overridden.

```
\def\change@society#1{%
 \def\@tempa{#1}%
 \@ifxundefined\@society{}{%
  \@ifx{\@tempa\@society}{}{%
   \class@warn{Conflicting society \@tempa<>\@society.}%
  }%
 }%
 \let\@society\@tempa
}%
```

Here are the class options relating to the APS

```
\DeclareOption{aps}{\change@society{aps}\def\@journal{}}%
\DeclareOption{pra}{\change@society{aps}\def\@journal{pra}}%
\DeclareOption{prb}{\change@society{aps}\def\@journal{prb}}%
\DeclareOption{prc}{\change@society{aps}\def\@journal{prc}}%
\DeclareOption{prd}{\change@society{aps}\def\@journal{prd}}%
\DeclareOption{pre}{\change@society{aps}\def\@journal{pre}}%
\DeclareOption{prl}{\change@society{aps}\def\@journal{prl}}%
\DeclareOption{prstab}{\change@society{aps}\def\@journal{prstab}}%
\DeclareOption{rmp}{\change@society{aps}\def\@journal{rmp}}%
```

Here are the class options relating to the Optical Society of America

```
\DeclareOption{osa}{\change@society{osa}\def\@journal{}}%
\DeclareOption{osameet}{\change@society{osa}\def\@journal{osameet}}%
\DeclareOption{opex}{\change@society{osa}\def\@journal{opex}}%
\DeclareOption{tops}{\change@society{osa}\def\@journal{tops}}%
\DeclareOption{josa}{\change@society{osa}\def\@journal{josa}}%
```

NOTE: `revguide.tex` does not mention any `\@journal` options besides `prb`.

`amsfonts` The class option `amsfonts` has the same effect as if the document preamble contained a `\usepackage{amsfonts}` statement.

```
\DeclareOption{amsfonts}{%
  \def\class@amsfonts{\RequirePackage{amsfonts}}%
}%
\DeclareOption{noamsfonts}{%
  \let\class@amsfonts\@empty
}%
\AtEndOfClass{%
 \@ifxundefined\class@amsfonts{}{\class@amsfonts}%
}%
```

`amssymb` The class option `amssymb` has the same effect as if the document preamble contained a `\usepackage{amssymb}` statement.

```
\DeclareOption{amssymb}{%
  \def\class@amssymb{\RequirePackage{amssymb}}%
}%
\DeclareOption{noamssymb}{%
  \let\class@amssymb\@empty
}%
\AtEndOfClass{%
 \@ifxundefined\class@amssymb{}{\class@amssymb}%
}%
```

`amsmath` The class option `amsmath` has the same effect as if the document preamble contained a `\usepackage{amsmath}` statement.

We require version 1.2 (datestamped 1997/03/20) or later. The `\ver@amsmath.sty`, will

LaTeX note: Certain LaTeX procedures have an arbitrary and pointless restriction that they may be used only within the preamble. We get around this by preserving the procedures in private `\csname`s.

FIXME note: it is difficult to ensure that an error summary will be printed on the console at the very end, but `ltxgrid` allows accomplishing this via an interrupt, put down at `\AtEndDocument` time.

```
\DeclareOption{amsmath}{%
  \def\class@amsmath{\RequirePackage{amsmath}[\ver@amsmath@prefer]}%
}%
\DeclareOption{noamsmath}{%
  \let\class@amsmath\@empty
}%
\AtEndOfClass{%
 \preserve@LaTeX
 \@ifxundefined\class@amsmath{}{\class@amsmath}%
 \appdef\class@enddocumenthook{\test@amsmath@ver}%
}%
\appdef\preserve@LaTeX{%
 \let\@ifl@aded@LaTeX\@ifl@aded
 \let\@ifpackageloaded@LaTeX\@ifpackageloaded
 \let\@pkgextension@LaTeX\@pkgextension
 \let\@ifpackagelater@LaTeX\@ifpackagelater
 \let\@ifl@ter@LaTeX\@ifl@ter
 \let\@ifl@t@r@LaTeX\@ifl@t@r
 \let\@parse@version@LaTeX\@parse@version
}%
\appdef\restore@LaTeX{%
 \let\@ifl@aded\@ifl@aded@LaTeX
 \let\@ifpackageloaded\@ifpackageloaded@LaTeX
 \let\@pkgextension\@pkgextension@LaTeX
 \let\@ifpackagelater\@ifpackagelater@LaTeX
 \let\@ifl@ter\@ifl@ter@LaTeX
```

```
 \let\@ifl@t@r\@ifl@t@r@LaTeX
 \let\@parse@version\@parse@version@LaTeX
}%
\def\test@amsmath@ver{%
 \begingroup
  \restore@LaTeX
  \@ifpackageloaded{amsmath}{%
   \@ifpackagelater{amsmath}{\ver@amsmath@prefer}{}{%
     \class@warn{%
      You have loaded amsmath, version "\csname ver@amsmath.sty\endcsname",\MessageBreak
      but this class requires version "\ver@amsmath@prefer", or later.\MessageBreak
      Please update your LaTeX installation.
     }%
   }%
  }{%
  }%
 \endgroup
}%
\def\ver@amsmath@prefer{2000/01/15 v2.05 AMS math features}%
```

7.2.25 Presenting Authors and Their Affiliations

The `runinaddress` option is the only one that sets `\runinaddress@sw` to true.

The `unsortedaddress` option is the only one that sets `\@affils@sw` to false.

If we take the `groupedaddress` option as the default, then we can look upon the `superscriptaddress` option as simply turning `\groupauthors@sw` to false.

`groupedaddress`, the default, groups authors above a common address. In order to assert the default and still provide for proper precedence for document-supplied options, we carry out our action at `\AtBeginDocument` time, and only if none of the related class options were invoked.

```
\DeclareOption{groupedaddress}{\clo@groupedaddress}%
\def\clo@groupedaddress{%
 \@booleantrue\groupauthors@sw
 \@booleantrue\@affils@sw
 \@booleanfalse\runinaddress@sw
}%
\appdef\class@inithook{%
 \@ifxundefined\groupauthors@sw{\clo@groupedaddress}{}%
}%
```

`unsortedaddress` is similar to `groupedaddress`, but turns off `\@affils@sw`, with the result that each address that is entered will be printed.

```
\DeclareOption{unsortedaddress}{\clo@unsortedaddress}%
\def\clo@unsortedaddress{%
 \@booleantrue\groupauthors@sw
 \@booleanfalse\@affils@sw
 \@booleanfalse\runinaddress@sw
}%
```

`runinaddress` is similar to `groupedaddress`, with the addition of the Boolean `\runinaddress@sw`, which causes the authors associated with a particular address group to be formatted in a paragraph instead of on separate lines.

```
\DeclareOption{runinaddress}{\clo@runinaddress}%
\def\clo@runinaddress{%
 \@booleantrue\groupauthors@sw
 \@booleantrue\@affils@sw
 \@booleantrue\runinaddress@sw
}%
```

`superscriptaddress` presents author affiliations as superscripts. Authors with like affiliations share the same superscript.

FIXME: if `\groupauthors@sw` is false, would not `\@affils@sw` be a don't care?

```
\DeclareOption{superscriptaddress}{\clo@superscriptaddress}%
\def\clo@superscriptaddress{%
 \@booleanfalse\groupauthors@sw
 \@booleantrue\@affils@sw
 \@booleanfalse\runinaddress@sw
}%
```

7.2.26 Typeset by REVTEX

The class option `byrevtex` signifies that you want the "Typeset by REVTEX" byline to appear on your formatted output. By default, no such byline appears.

```
\DeclareOption{byrevtex}{\@booleantrue\byrevtex@sw}%
```

7.3 Attempt to fix float placement failure

REVTEX uses the `ltxgrid` package, which provides the ability to attempt repairs when LATEX's float placement mechanism is about to fail, but that facility is turned off by default. Users should invoke the `floatfix` document class option to enable this LATEX extension. If not, a helpful message is printed in the log, indicating how to work around the difficulty.

```
\DeclareOption{floatfix}{\@booleantrue\force@deferlist@sw}%
\DeclareOption{nofloatfix}{\@booleanfalse\force@deferlist@sw}%
\appdef\class@inithook{%
 \@ifxundefined\force@deferlist@sw{\@booleanfalse\force@deferlist@sw}{}%
}%
```

`\@fltovf` `\@fltstk` The LATEX kernel error message `\@fltovf` may now be a bit more helpful to the user; likewise for the `\@fltstk` message of `ltxgrid`.

```
\gdef\@fltovf{%
 \@latex@error{%
  Too many unprocessed floats%
  \force@deferlist@sw{}{; try class option [floatfix]}%
 }\@ehb
}%
\def\@fltstk{%
```

```
 \@latex@warning{%
  A float is stuck (cannot be placed)%
  \force@deferlist@sw{}{; try class option [floatfix]}%
 }%
}%
```

`\ltxgrid@info@sw` `\outputdebug@sw`

The two options `ltxgridinfo` and `outputdebug` turn on informative diagnostics within the package `ltxgrid`. Only people who really want to see this output will select these class options. Consult documentation for the `ltxgrid` package to see what output the related switches enable.

```
\DeclareOption{ltxgridinfo}{\@booleantrue\ltxgrid@info@sw}%
\DeclareOption{outputdebug}{\@booleantrue\outputdebug@sw}%
```

7.4 Option to relax page height

The `ltxgrid` package can set text pages to their natural height or force them to the full text height; the latter is the default. If setting the pages with a variable length, the running foot will move up or down with the natural length of the text column. While I recommend against doing so, this option will turn that switch to the latter setting.

```
\DeclareOption{raggedfooter}{\@booleanfalse\textheight@sw}%
\DeclareOption{noraggedfooter}{\@booleantrue\textheight@sw}%
```

7.5 Selecting procedure for processing abstract

There are two different procedures for processing an abstract; the options `newabstract` and `oldabstract` determine which one the user prefers. At present, the default is to use the old, but this may change.

```
\DeclareOption{newabstract}{\@booleanfalse\oldabstract@sw}%
\DeclareOption{oldabstract}{\@booleantrue \oldabstract@sw}%
\appdef\class@inithook{%
 \@ifxundefined\oldabstract@sw{\@booleanfalse\oldabstract@sw}{}%
 \oldabstract@sw{%
  \let@environment{abstract}{abstract-old}%
 }{}%
}%
```

7.6 Default Option

The default handling for a document class option is to treat it as the name of a society and hunt for a corresponding `.rtx` file to read in. This behavior is much like the LaTeX2.09 handling, where one looked for a `.sty` file, except that in this case, we must catch the case where multiple unknown options are declared.

```
\DeclareOption*{%
 \IfFileExists{\CurrentOption\substyle@ext}{%
  \expandafter\change@society\expandafter{\CurrentOption}%
 }{%
```

```
  \OptionNotUsed
 }%
}%
\def\substyle@ext{.rtx}%
\AtEndOfClass{%
 \@ifxundefined\@society{%
  \class@warn{No Society specified, using generic APS style}%
  \def\@society{aps}\def\@journal{}%
 }{}%
 \expandafter\input\expandafter{\@society\substyle@ext}%
}%
```

7.7 Class-Asserted Options

Here we establish the default document class options. Those of the document itself will override these.

```
\ExecuteOptions{10pt}
```

7.8 Execute Options

The document class options are processed in the order they are declared in this document class file.

Precedence is established in the following way:

First, the single `.rtx` file is read in. It may set values to any of the Booleans that represent class options, and it may elect to do so only if that Boolean is undefined, signifying that its corresponding document class options were absent from the document's option list. In this way correct precedence of class options is maintained.

Second, the type size option is acted upon. Type size code should not require altering the values of any of the Booleans carrying class options.

Next, at `\AtEndOfClass` time, certain packages are read in.

Finally, at `\AtBeginDocument` time, the state of the Booleans determines how the formatting will go.

```
\ProcessOptions\relax
```

8 Procedures Dependent Upon Options

Here we introduce `classes.dtx` definitions for the page styles that people will expect to be able to use.

`\ps@headings`
`\ps@myheadings`

```
\def\ps@headings{%
    \let\@oddfoot\@empty\let\@evenfoot\@empty
    \def\@evenhead{\thepage\hfil\slshape\leftmark}%
    \def\@oddhead{{\slshape\rightmark}\hfil\thepage}%
    \let\@mkboth\markboth
```

```
    \def\sectionmark##1{%
      \markboth {\MakeUppercase{%
        \ifnum \c@secnumdepth >\z@
          \thesection\quad
        \fi
        ##1}}{}}%
    \def\subsectionmark##1{%
      \markright {%
        \ifnum \c@secnumdepth >\@ne
          \thesubsection\quad
        \fi
        ##1}}}%
\def\ps@myheadings{%
    \let\@oddfoot\@empty\let\@evenfoot\@empty
    \def\@evenhead{\thepage\hfil\slshape\leftmark}%
    \def\@oddhead{{\slshape\rightmark}\hfil\thepage}%
    \let\@mkboth\@gobbletwo
    \let\sectionmark\@gobble
    \let\subsectionmark\@gobble
    }%
```

\ps@article@twoside
\ps@article@oneside
\ps@preprint

```
\def\ps@article{%
    \@provide\@evenhead{\let\\\heading@cr\thepage\quad\checkindate\hfil{\leftmark}}%
    \@provide\@oddhead{\let\\\heading@cr{\rightmark}\hfil\checkindate\quad\thepage}%
    \@provide\@oddfoot{}%
    \@provide\@evenfoot{}%
    \let\@mkboth\markboth
  \let\sectionmark\@gobble
  \let\subsectionmark\@gobble
}%
\def\ps@article@final{%
    \@provide\@evenhead{\let\\\heading@cr\thepage\quad\checkindate\hfil{\leftmark}}%
    \@provide\@oddhead{\let\\\heading@cr{\rightmark}\hfil\checkindate\quad\thepage}%
    \@provide\@oddfoot{}%
    \@provide\@evenfoot{}%
    \let\@mkboth\markboth
    \def\sectionmark##1{%
      \markboth{%
       \MakeTextUppercase{%
        \@ifnum{\c@secnumdepth >\z@}{\thesection\hskip 1em\relax}{}%
          ##1%
       }%
       }{}%
    }%
    \def\subsectionmark##1{%
      \markright {%
        \@ifnum{\c@secnumdepth >\@ne}{\thesubsection\hskip 1em\relax}{}%
          ##1%
      }%
```

```
    }%
}%
\appdef\class@inithook{%
%\twoside@sw{%
%}{%
%}%
}%
\def\heading@cr{\unskip\space\ignorespaces}%
\def\ps@preprint{%
  \@provide\@oddfoot{\hfil\thepage\quad\checkindate\hfil}%
  \@provide\@evenfoot{\hfil\thepage\quad\checkindate\hfil}%
  \@provide\@oddhead{}%
  \@provide\@evenhead{}%
  \let\@mkboth\@gobbletwo
  \let\sectionmark\@gobble
  \let\subsectionmark\@gobble
}%
\let\@oddhead\@undefined
\let\@evenhead\@undefined
\let\@oddfoot\@undefined
\let\@evenfoot\@undefined
\AtEndOfClass{%
 \appdef\class@documenthook{%
  \@provide\@oddhead{}%
  \@provide\@evenhead{}%
  \@provide\@oddfoot{}%
  \@provide\@evenfoot{}%
 }%
}%
```

`\lastpage@putlabel` Support the default meaning of `\@endpage`. Name of this macro (and the `\label` key) taken from CTAN:/macros/latex/contrib/other/lastpage with code optimised slightly.

```
\def\lastpage@putlabel{%
 \if@filesw
  \begingroup
    \advance\c@page\m@ne
    \immediate\write\@auxout{\string\newlabel{LastPage}{{}{\thepage}}}%
  \endgroup
 \fi
}%
```

Install a late hook into `\class@enddocumenthook` that labels the last page of the document. This is done just before the `.aux` file is closed, and does not require a `\shipout`, because it writes directly to the `.aux` file. Note that we invoke `\clearpage`: this means that no more typesetting should be done past this point. CHECKME: are there any more things installed via `\AtEndOfClass \class@enddocumenthook`?

```
\AtEndOfClass{%
 \appdef\clear@document{%
  \clearpage
  \lastpage@putlabel
 }%
}%
```

9 Required Packages

CTAN:macros/latex/contrib/other/misc/url.sty

```
\RequirePackage{url}%
```

10 Incompatible Packages

Known to be incompatible are `multicol` and `cite`. The former is incompatible with `ltxgrid`, which replaces it. The latter is incompatible with `natbib`, which replaces its functionality.

```
\appdef\class@inithook{%
 \incompatible@package{cite}%
 \incompatible@package{multicol}%
}%
```

11 Society- and Journal-Specific Code

`\@journal` Journal test helper, used as

```
%\@ifx{\@journal\journal@pra}{%
%  ⟨journal-specific setup⟩
%\fi
%
```

Journal code might like to further specify (if as yet undefined) or distinguish on the following Booleans.

Note: the journal substyle code should only alter the value of one of these Booleans if the Boolean is `\undefined`. This convention is what makes the document's options take precedence over the values set by the journal.

FIXME: make this table an exhaustive listing of all the parameters set by the class options.

\@pointsize	(101112), depending on the type size
\footinbib@sw	true if footnotes are to be formatted in the bibliography
\preprintsty@sw	true for preprint and hyperpreprint
\eqsecnum@sw	true means that equations are numbered within sections
\secnumarabic@sw	true means the sections are to be numbered arabic
\groupauthors@sw	true means authors listed separately for each address
\preprint@sw	true means to produce the preprint numbers as part of the title block
\showPACS@sw	true means to produce the PACS as part of the title block
\showKEYS@sw	true means to produce the keywords as part of the title block
\@affils@sw	true means each affiliation is printed, for each author
\runinaddress@sw	true means author addresses are printed run-in
\draft@sw	true implies that PACS will be printed
\tightenlines@sw	true if preprint single spaced
\lengthcheck@sw	true if length checking is in effect
\byrevtex@sw	true means to announce "typeset by REVTEX"
\titlepage@sw	true for title is to be set on a separate page
\twocolumn@sw	true if two-column page grid
\twocolumn@sw	true if we are to automatically balance the columns of the last page
\twoside@sw	true means to format pages for duplex printing
\floats@sw	false means floats are migrated to end of document
\floatp@sw	true means endfloats are set one to a page
\class@amsfonts	if \@empty, means that amsfonts will *not* be loaded
\class@amssymb	if \@empty, means that amssymb will *not* be loaded
\frontmatter@footnote	if \undefined, means that the default (\footnote) will be used
\place@bibnumber	if \undefined, means that the default (inline) will be used

Note: if \twocolumn@sw and \preprintsty@sw are both false, then 'galley' style is in effect. The galley option invokes onecolumn, but does not affect the \preprintsty@sw.

Note: \paperwidth and \paperheight are not integrated into this scheme, and should be selected by the document alone.

12 Front Matter

12.1 The title command

\title
\do@title

The author uses \title to specify the title. As in the AMS classes, \title has an optional argument specifying a short form for use in running heads.

Once the \title is given, you can specify any combination of \thanks, \email, \homepage, and \altaffiliation commands, applying to the title of the document.

```
\renewcommand\title[2][]{%
 \def\@title{#2}%
 \def\@shorttitle{#1}%
 \let\@AF@join\@title@join
}%
```

\@title
\@shorttitle

The arguments to \title are saved in these internal macros, which are set up to produce a warning if \title has not been used before \maketitle.

```
\appdef\class@inithook{%
 \def\@title{\class@warn{No title}}%
 \let\@shorttitle\@empty
 \let\@title@aux\@title@aux@cleared
}%
```

\@title@join
\@title@join@
\@title@aux@cleared

```
\def\@title@join{\expandafter\@title@join@\@title@aux}%
\def\@title@join@#1#2{%
 \def\@title@aux{{\@join{\@separator}{#1}{#2}}}%
}%
\def\@title@aux@cleared{{}}%
```

12.2 The author, affiliation, and related commands

\c@affil

A counter to store the affiliation numbers used for the superscript marks. This command defines \theaffil.

```
\newcounter{affil}
```

12.2.1 Commands for author or collaboration

\author

Each author is given in a *separate* \author command. This is similar to the AMS classes, but the AMS also try to support the article class's \and command for putting multiple authors in one \author command. REVTEX, however, does not support the \and command.

When the \author command is encountered, the previous author, author group, affiliation, and affiliation group are at an end. This procedure moves the previous author to the author group (\move@AU), moves the previous affiliation to the affiliation group (\move@AF), and terminates the previous author/affiliation group (\move@AUAF) as the case may require.

```
\renewcommand\author{% implicit #1
 \@author@def{}% implicit #2
}%
```

\collaboration
\@collaboration@

Specify the collaboration (given after a group of authors).

This procedure does exactly what \author does, and sets a flag signifying that the \collaboration command was given.

Note: earlier versions attempted to ignore the collaboration command if it was considered inappropriate. We must not do this, however, because it throws the state machine out of kilter.

```
\def\collaboration{% implicit #1
 \groupauthors@sw{%
  \class@warn{%
   \string\collaboration\space is only allowed with superscriptaddress option.
   Use superscriptaddress in place of groupedaddress, unsortedaddress, or runinaddress%
```

```
  }%
 }{}%
 \@author@def{\@booleantrue\collaboration@sw}% implicit #2
}%
```

\@author@init
\@author@cleared
\@authorclear@sw
\@author@def
\@author@join@
\@author@join

Here are all the access procedures for the author data structure.

We define a flag value for \@author(private), a procedure for setting \@author to the flag value (\@author@init), a Boolean to test against the flag value (\@authorclear@sw), and a procedure to add new ancillary data to the author.

```
\def\@author@cleared{{}{}}%
\def\@author@init{%
 \let\@author\@author@cleared
}%
\def\@authorclear@sw{\@ifx{\@author\@author@cleared}}%
\appdef\class@inithook{%
 \@author@init
}%
\def\@author@def#1#2{%
 \move@AU\move@AF\move@AUAF
 \let\@AF@join\@author@join
 #1%
 \def\@author{{#2}{}}%
}%
\def\@author@join@#1#2#3{%
  \def\@author{{#1}{\@join{\@separator}{#2}{#3}}}%
}%
\def\@author@join{\expandafter\@author@join@\@author}%
\def\@author@gobble#1#2{}%
```

\move@AU

Move the most recent author to the list of current authors. If we recently did a \@author@init, this is a no-op. If we recently encountered a \collaboration, we preface the author with the \CO@opr intercollated operator, otherwise we apply \AU@opr.

```
\def\move@AU{%
 \@authorclear@sw{}{%
  \collaboration@sw{%
    \expandafter\appdef
    \expandafter\AU@grp
    \expandafter{%
    \expandafter \CO@opr
                  \@author
                 }%
  }{%
    \expandafter\appdef
    \expandafter\AU@grp
    \expandafter{%
    \expandafter \AU@opr
                  \@author
                 }%
```

```
  }%
 }%
 \@author@init
 \@booleanfalse\collaboration@sw
}%
```

`\@author@finish` We define a procedure to terminate author/affiliation parsing just before the title block is formatted. We detect the case where an author group is not followed by an affiliation group.

We first finish off any author- or affiliation groups in progress. Next, we detect the case where there is an author group in progress, but no affiliation group. If so, we emit a diagnositic message and act as if a `\noaffiliation` command had been given. In any case, we finish off any author/affiliation pair that may remain. Formatting of the title block may now proceed.

```
\def\@author@finish{%
 \move@AU\move@AF
 \@ifx{\AU@grp\@empty}{}{%
  \@ifx{\AF@grp\@empty}{%
   \begingroup
    \let\href\@secondoftwo
    \let\AU@opr\@firstoftwo
    \let\CO@opr\@firstoftwo
    \class@warn{Assuming \string\noaffiliation\space for \AU@grp}%
   \endgroup
   \begingroup\@affiliation{\blankaffiliation}%\endgroup in \@affiliation
   \move@AF
%
%  \appdef     \@AAC@list{\AF@opr}%
%  \appdef@eval\@AAC@list{\AF@grp}%
%  \appdef@e   \@AAC@list{\AU@grp}%
%  \let\AU@grp\@empty
%  \let\AF@grp\@empty
  }{}%
 }%
 \move@AUAF
}%
```

`\@join` The procedure `\@join` inserts a separator between two tokens, or, if the first token is nil, elides both that token and the separator.

```
\def\@join#1#2#3{%
  \@if@empty{#2}{#3}{#2#1#3}%
}%
\def\@separator{;\space}%
```

`\surname` `\firstname` No-op to allow better post-processing of the file.

```
\let\surname\@firstofone
\let\firstname\@firstofone
```

\AU@grp The control sequence name \AU@grp accretes consecutive \author entries (with \AU@opr as the intercollated operator) and \collaboration entries (with \CO@opr as the intercollated operator). Ultimately, its contents are appended to the author/affiliation list \@AAC@list. It must be initialized to \@empty.

Note on \AU@opr and \CO@opr: it is essential to treat these two operators to the greatest extent possible on an equal footing. Therefore we invariably assign values to the pair of them within the same procedure.

```
\let\AU@grp\@empty
```

\and The original LaTeX idea of using one \author command to capture all authors and their address just doesn't work with multiple authors possibly sharing addresses, so in this class disable \and.

```
\renewcommand\and{\class@err{\protect\and\space is not supported}}
```

12.2.2 Commands for affiliation

\affiliation After each group of authors with the same address, give that address in \affiliation. If later in the list you have an author with the address of an earlier author, repeat the \affiliation command: the system will detect the similarity and (if using the superscriptaddress option) only print the affiliation once, reusing the superscript marker.

When the \affiliation command is encountered, the current author and author group (if any) are at an end. Also, the current affiliation is at an end.

FIXME: changes to catcode required elsewhere now?

```
\def\cat@comma@active{\catcode`\,\active}%
{\cat@comma@active\gdef,{\active@comma}}%
\def\active@comma{,\penalty-300\relax}%
\newcommand\affiliation{%
 \move@AU\move@AF
 \begingroup
  \cat@comma@active
  \@affiliation
}%
\def\@affiliation#1{%
 \endgroup
 \let\@AF@join\@affil@join
 \@affil@def{#1}%
}%
```

\noaffiliation User-level command signifying that the current group of authors has *no* affiliation.

This is implemented by acting as if the \affiliation command has been given, and using a flag value of \blankaffiliation, which \@affil@group can recognize as a no-op.

```
\newcommand\noaffiliation{%
  \move@AU\move@AF
% \edef\AF@grp{\AF@grp}%
  \begingroup\@affiliation{\blankaffiliation}%
```

```
% \move@AF
}%
\def\blankaffiliation{}%
```

\@affil@cleared
\@affil@init
\@affilclear@sw
\@affil@def
\@affil@join@
\@affil@join

Here are all the access procedures for the affiliation data structure. Note the similarity to those of the \@author data structure.

We define a flag value for \@affil (private), a procedure for setting \@affil to the flag value (\@affil@init), a Boolean to test against the flag value (\@affilclear@sw), and a procedure to add new ancillary data to the affiliation.

```
\def\@affil@cleared{{}{}}%
\def\@affil@init{%
 \let\@affil\@affil@cleared
}%
\def\@affilclear@sw{\@ifx{\@affil\@affil@cleared}}%
\appdef\class@inithook{%
 \@affil@init
}%
\def\@affil@def#1{\def\@affil{{#1}{}}}%
\def\@affil@join@#1#2#3{%
  \def\@affil{{#1}{\@join{\@separator}{#2}{#3}}}%
}%
\def\@affil@join{\expandafter\@affil@join@\@affil}%
```

\move@AF

Move the most recent affiliation to the affiliation group. If we recently did a \@affil@init, this is a no-op.

```
\def\move@AF{%
 \@affilclear@sw{}{%
```

Set the \temp@sw to false, then execute the affiliation list (\@AFF@list) with the intercollated operator set to \@affil@match and with \@tempa holding the first component of \@affil, the current affiliation, which is to be matched. If the current affiliation has been seen before, \temp@sw will be set to true, \@tempc will be set to the existing affiliation number, and \@tempd will be set to the affiliation's ancillary data. The Boolean \@affils@sw being false prevents this test from ever returning a positive result.

This procedure uses \@tempa and sets \@tempc, \@tempd, and \temp@sw.

```
  \@booleanfalse\temp@sw
  \let\@tempd\@empty
  \@affils@sw{%
    \expandafter\@firstoftwodef\expandafter\@tempa\@affil
    \def\AFF@opr{\@affil@match\@tempa}%
    \@AFF@list
  }{}\temp@sw
```

True clause: This affiliation has been seen before. If ancillary data for the affiliation have been given but are not identical to those seen before, give a warning.

```
  {%
   \expandafter\@secondoftwodef\expandafter\@tempb\@affil
   \@ifx{\@tempb\@empty}{}{%
```

```
    \@ifx{\@tempb\@tempd}{}{%
     \class@warn{%
      Ancillary information for \@tempa\space must not be different!
      Please put it on the first instance%
     }%
    }%
   }%
   \appdef@eval\AF@grp\@tempc
  }%
```

False clause: This is a new address: increment the counter to get a unique ID for the new affiliation and append the new number, address, and footnote to the list `\@AFF@list`.

```
  {%
    \@if@empty{\expandafter\@firstoftwo\@affil}{%
\edef\@tempc{\z@}%
    }{%
\stepcounter{affil}%
\edef\@tempc{\theaffil}%
    }%
\appdef     \@AFF@list{\AFF@opr}%
\appdef@eval\@AFF@list\@tempc
\appdef@e   \@AFF@list{\@affil}%
\appdef@eval\AF@grp\@tempc
  }%
```

Append the ID of the affiliation to the affiliation group of the current author group.

```
  \@affil@init
 }%
}%
\def\@firstoftwodef#1#2#3{%
 \def#1{#2}%
}%
\def\@secondoftwodef#1#2#3{%
 \def#1{#3}%
}%
```

`\move@AUAF` Append `\AF@grp` and `\AU@grp` to `\@AAC@list` and merge any new `\AF@grp` to `\@AFG@list`.

```
\def\move@AUAF{%
```

Entire procedure is predicated on something non-trivial to move.

If both author and affiliation are nil, bail out.

No, try that again:

If we have seen any affiliations, then it is time to move the current affiliation group and author group to `\@AAC@list`. If not, we are picking up authors into an author group, and we should bail out.

```
%\@ifx{\AU@grp\@empty}{\@ifx{\AF@grp\@empty}{\true@sw}{\false@sw}}{\false@sw}%
 \@ifx{\AF@grp\@empty}%
```

```
{}{%
```

First, append current affiliation group (expansion of `\AF@grp`) to the author/affiliation list (`\@AAC@list`), using `\AF@opr` as the intercollated operator; also append the current authors (`\AU@grp`).

```
    \appdef     \@AAC@list{\AF@opr}%
    \appdef@eval\@AAC@list{\AF@grp}%
    \appdef@e   \@AAC@list{\AU@grp}%
```

Next, append the current affiliation group to `\@AFG@list`, (if it is not already there) using `\AFS@opr` as the intercollated operator.

Note that `\@AFG@list` is a list of *unique* affiliation groups.

```
    \@ifx{\@empty\AU@grp}{}{%
     \@ifx{\@empty\AF@grp}{}{%
      \@booleanfalse\temp@sw
      \def\AFS@opr{\x@match\AF@grp}%
      \@AFG@list
      \temp@sw{}{%
        \appdef     \@AFG@list{\AFS@opr}%
        \appdef@eval\@AFG@list{\AF@grp}%
      }%
     }%
    }%
```

Finally, null out the macros holding the author group and affiliation group, respectively.

```
    \let\AU@grp\@empty
    \let\AF@grp\@empty
}%
}%
```

`\@AAC@list` The `\@AAC@list` macro accretes authors (using `\AU@opr` as the intercollated operator), collaborations (using `\CO@opr` as the intercollated operator), and affiliations (using `\AF@opr` as the intercollated operator).

```
\appdef\class@inithook{%
 \let\@AAC@list\@empty
}%
```

`\@AFG@list` The macro `\@AFG@list` accretes `\AF@grp`, with `\AFS@opr` as the intercollated operator.

This token list is employed when and only when `groupedaddress` is in effect. It must be empty at the beginning of the document.

```
\appdef\class@inithook{%
 \let\@AFG@list\@empty
}%
```

`\@AFF@list` The macro `\@AFF@list` accretes the list of affiliations with `\AFF@opr` as the intercollated operator. It must be empty at the beginning of the document.

Each `\affiliation` command in the document contributes to this list, as long as the argument of that instance of the command is "new".

Each entry in this list consists of the `\AFF@opr` intercollated operator followed by three brace-delimited tokens, representing, in order:

1. the affiliation's unique ID, a number increasing monotonically for each new entry,
2. the optional argument of the `\affiliation` command, the footnote to the affiliation, and
3. the text of the affiliation.

At various junctures in the code, the control sequence name `\AFF@opr` assumes the meaning of `\@affil@footnote` (FIXME: never used), `\@affil@group`, `\affil@script`, or is defined to expand to `\@affil@match\@tempa`. and the list `\@AFF@list` is expanded.

```
\appdef\class@inithook{%
 \let\@AFF@list\@empty
}%
```

`\AF@grp` The macro `\AF@grp` accretes affiliation IDs into an affiliation group. It must be empty at the beginning of the document.

```
\let\AF@grp\@empty
```

`\@affil@match` The definition for the `\AFF@opr`, the intercollated operator for `\@AFF@list`, when adding a new affiliation: seeks a match with any earlier affiliation. When the affiliation list is executed, this procedure tests each entry against the 'new' affiliation (which has been stored in `\@tempa`).

Uses `\temp@sw`, `\@tempc`, and `\@tempd` to communicate back: if it returns `\temp@sw` true, a match was found, then `\@tempc` will be the footnote number of the matching entry and `\@tempd` will be the auxiliary information of the matching entry.

```
\def\@affil@match#1#2#3#4{%
 \temp@sw{}{%
  \def\@tempifx{#3}%
  \@ifx{\@tempifx#1}{%
    \@booleantrue\temp@sw
    \def\@tempc{#2}%
    \def\@tempd{#4}%
  }{%
  }%
 }%
}%
```

`\@tempaffil@script` Give a superscript marker to the address, and junk the text argument (`#2`).

```
%\def\@tempaffil@script#1#2{%
% ,\,#1%
%}%
```

\@tempaffil@group Execute if we are grouping authors above 'main addresses' (groupedaddress). Put temp address in a footnote, with explanatory text. Note: this procedure uses \@tempb and \@tempc to communicate to \AFF@opr.

```
%\def\@tempaffil@group#1#2{%
% \def\@tempc{#1}%
% \def\@tempb{#2}%
% \begingroup
%  \let\AFF@opr \@affil@footnote
%  \@AFF@list
% \endgroup
%}%
```

\@affil@footnote The alias of \AFF@opr when \@AFF@list is executed by \@tempaffil@group. If we find a match of the affiliation number to \@tempc, then footnotes are formatted with the relevant content and \@tempb.

```
%\def\@affil@footnote#1#2#3{%
%  \def\@tempifx{#1}%
%  \@ifx{\@tempifx\@tempc}{%
%   \comma@space
%   \expandafter\frontmatter@footnote\expandafter{\@tempb#2}%
%   \frontmatter@footnote{#3}%
%  }{%
%  }%
%}%
```

\@AF@join The \csname \@AF@join is a procedure used within \email, \homepage, \thanks, and \altaffiliation. When such a command appears after an \author, \collaboration, or \affiliationcommand, the \@AF@join procedure appends the argument given to the appropriate macro. Its default is to give an error message, since these commands are legal only within the particular context mentioned.

```
\appdef\class@inithook{%
 \let\@AF@join\@AF@join@error
}%
\def\@AF@join@error#1{%
 \class@warn{%
  \string\email, \string\homepage, \string\thanks, or \string\altaffiliation\space
  appears in wrong context.
 }%
}%
```

12.2.3 Commands for auxiliary information

\email Just tacks the email address on to the current author or affiliation.

```
\def\sanitize@url{%
 \@makeother\%%
 \@makeother\~%
 \@makeother\_%
}%
```

```
\newcommand*\email[1][Electronic address: ]{\begingroup\sanitize@url\@email{#1}}%
\def\@email#1#2{%
 \endgroup
 \@AF@join{#1\href{mailto:#2}{#2}}%
}%
```

`\homepage` Just tacks the URL on to the current author or affiliation. Note: group opened in `\homepage` is closed in `\@homepage`.

```
\newcommand*\homepage[1][URL:~]{\begingroup\sanitize@url\@homepage{#1}}%
\def\@homepage#1#2{%
 \endgroup
 \@AF@join{#1\url{#2}}%
}%
```

`\url`
`\href`

```
\appdef\class@documenthook{%
 \providecommand\url{\texttt}%
 \providecommand\href[2]{\url{#2}}%
}%
```

`\thanks` The operative version of `\thanks` appends an item to `\@author`, or `\@affil`'s auxiliary data.

```
\def\thanks{% implicit #1
  \@AF@join
}%
```

`\altaffiliation` Implemented more or less like `\thanks` but shares the affiliation counter. Optional argument may be used to give explanatory text eg 'currently staying at' This will be placed before the address, if used in a footnote.

```
\newcommand*\altaffiliation[2][]{%
  \@AF@join{#1#2}%
}%
```

12.2.4 Procedures for author, collaboration, and affiliation

Macros that list off, say, authors, and which require punctuation, like "A, B, and C", will use `\set@listcomma@list`, `\set@listcomma@count`, `\@listand`, and `\@listcomma`. These macros use the `\@tempcnta` register to keep track of how many items remain to be listed off.

At present, `\@author@present` will decrement that counter.

`\set@listcomma@list`
`\set@listcomma@count` Pop the author count for this collaboration.

Note: by side effect, it assigns `\@listcomma`: for a list of length two, suppress comma addition.

```
\def\set@listcomma@list#1{%
  \expandafter\@reset@ac\expandafter#1#1{0}\@reset@ac{%
   \let\@listcomma\relax
  }{%
```

```
    \let\@listcomma\@listcomma@comma
  }%
}%
\def\set@listcomma@count#1{%
\@ifnum{#1=\tw@}{%
\let\@listcomma\relax
}{%
\let\@listcomma\@listcomma@comma
}%
}%
```

Does the actual pop operation, then generates a Boolean which selects one of the two assignments for `\@listcomma`.

Note: this procedure sets `\@tempcnta` to reflect the number of items to list off.

```
\def\@reset@ac#1#2#3\@reset@ac{%
  \def#1{#3}%
  \@tempcnta#2\relax
  \@ifnum{#2=\tw@}%
}%
```

`\listand` Might need extending with penalties etc.

Note: this procedure expects `\@tempcnta` to reflect the current number of items remaining to list off.

```
\def\@listand{\@ifnum{\@tempcnta=\tw@}{\andname\space}{}}%
```

`\@listcomma` This definition, with `\@ne` puts a comma before and.

David Carlisle says: In UK English (at least) would have `\tw@` here, which would then implement the convention of indicating conjunction with comma or "and", but not both.

Note: this procedure expects `\@tempcnta` to reflect the current number of items remaining to list off.

```
\def\@listcomma@comma{\@ifnum{\@tempcnta>\@ne}{,}{}}%
\def\@listcomma@comma@UK{\@ifnum{\@tempcnta>\tw@}{,}{}}%
```

`\@collaboration@err` This command will be executed if are in a `groupedaddress` situation and the document contained a `\collaboration` command.

There already has been issued a warning message explaining that `\collaboration` is wrong in this case, so here we simply gobble.

```
\def\@collaboration@err#1#2{%
%\class@warn{An illegal \string\collaboration\ command has been entered!}%
}%
```

`\doauthor` Main control over how authors are typeset.

`#1` is loaded by `\author`

`#2` is loaded by `\email`, `\homepage`, `\altaffiliation`, or `\thanks`

`#3` is the superscript affiliation, if at all used.

First, the author name is formatted, followed by a comma, then come any marks relating to affiliation (if present), then come the homepage URL and email address, if any, with appropriate punctuation.

```
\def\doauthor#1#2#3{%
  \ignorespaces#1\unskip\@listcomma
  \def\@tempifx{#3}%
  \@ifx{\@tempifx\@empty}{%
   \def\@tempifx{#2}%
   \@ifx{\@tempifx\@empty}{}{\frontmatter@footnote{#2}}%
  }{%
   #3%
   \def\@tempifx{#2}%
   \@ifx{\@tempifx\@empty}{}{\comma@space\frontmatter@footnote{#2}}%
  }%
  \space \@listand
}%
```

`\AF@opr` DPC: The number(s) associated to the affiliation address(es).

`\@thanks` Society-specific options may change this. `\comma@space` is a separator between adjacent footnotes.

```
\def\@thanks{\comma@space\footnote}
```

`\x@match` Procedure `\x@match` is used as an alias `\AFS@opr`, probing the list of affiliation groups and setting a Boolean if a match is found to #1 (usually the current affiliation group).

Procedure `\y@match` is used as an alias to `\@TBN@opr`, probing the list of unique title block footnotes for a match to the footnote text: when the first match is found it sets a Boolean, defining `\@tempb` to the number of that footnote.

```
\def\x@match#1#2{%
 \temp@sw{}{%
  \def\@tempifx{#2}%
  \@ifx{\@tempifx#1}{%
    \@booleantrue\temp@sw
  }{%
  }%
 }%
}%
\def\y@match#1#2#3{%
 \temp@sw{}{%
  \def\@tempifx{#3}%
  \@ifx{\@tempifx#1}{%
    \@booleantrue\temp@sw
    \def\@tempb{#2}%
  }{%
  }%
 }%
}%
```

`\frontmatter@footnote` Used when typesetting a title block footnote. (Item 290: merge duplicates.)

Maintains a list of frontmatter footnotes along with associated device. Uniquifies each footnote by traversing the list and looking for a match to the footnote

text. If found, define `\@tempb` to the corresponding footnote device, and typeset it with `\@footnotemark`. If never found, manually increment footnote counter, determine the corresponding footnote device, and add footnote text and device to the list of frontmatter footnotes.

The list `\@FMN@list` is processed later in the title block production.

Note that this method of making footnotes runs afoul of `hyperref`'s patches of the LaTeX kernel. Therefore, we avoid trouble by refraining from using `\@footnotemark` and by creating our own hypertext links.

```
\def\frontmatter@footnote#1{%
 \begingroup
  \@booleanfalse\temp@sw
  \def\@tempa{#1}%
  \let\@tempb\@empty
  \def\@TBN@opr{\y@match\@tempa}%
  \@FMN@list
  \temp@sw{%
   \expandafter\frontmatter@footnotemark
   \expandafter{\@tempb}%
  }{%
   \stepcounter\@mpfn
   \protected@xdef\@tempb{\the\csname c@\@mpfn\endcsname}%
   \expandafter\frontmatter@footnotemark
   \expandafter{\@tempb}%
   \expandafter\g@addto@macro
   \expandafter\@FMN@list
   \expandafter{%
   \expandafter \@TBN@opr
   \expandafter {\@tempb}{#1}%
                }%
  }%
 \endgroup
}%
\appdef\class@inithook{%
 \global\let\@FMN@list\@empty
}%
\def\frontmatter@footnotemark#1{%
 \leavevmode
 \ifhmode\edef\@x@sf{\the\spacefactor}\nobreak\fi
  \begingroup
   \hyper@linkstart {link}{frontmatter@#1}%
    \csname c@\@mpfn\endcsname#1\relax
    \def\@thefnmark{\thempfn}%
    \@makefnmark
   \hyper@linkend
  \endgroup
 \ifhmode\spacefactor\@x@sf\fi
 \relax
}%
```

12.3 The keywords command

\keywords
\@keywords

Usual game, save text in a macro for processing by \maketitle.

```
\def\keywords#1{%
  \aftermaketitle@chk{\keywords}%
  \gdef\@keywords{#1}%
}%
\def\@keywords{}%
```

12.4 The \date command and related commands

\date
\received
\revised
\accepted
\published
\@date
\@received
\@revised
\@accepted
\@published

Use the \date command to specify the document date, the \received command to specify the date received, \revised for date revised, \accepted for date accepted, and \published for date published. Normally only \date to be used by author, the remainder used only by editors.

DPC: As for \keywords, but this time don't flag a warning if there is no revision date specified.

In each case the user-level command defines the value of a macro which buffers the data entered by the user. For instance, \received causes \@received to acquire a value. The optional argument allows the user to override the text that will be typeset along with the date; the default value of that text is itself a localized macro.

```
\renewcommand*\date[2][\Dated@name]{\def\@date{#1#2}}%
\def\@date{}%
\newcommand*\received[2][\Received@name]{\def\@received{#1#2}}%
\def\@received{}%
\newcommand*\revised[2][\Revised@name]{\def\@revised{#1#2}}%
\def\@revised{}%
\newcommand*\accepted[2][\Accepted@name]{\def\@accepted{#1#2}}%
\def\@accepted{}%
\newcommand*\published[2][\Published@name]{\def\@published{#1#2}}%
\def\@published{}%
```

12.5 The pacs command

PACS, the Physics and Astronomy Classification Scheme.

\pacs
\@pacs

As for \keywords

```
\def\pacs#1{%
  \aftermaketitle@chk{\pacs}%
  \gdef\@pacs{#1}%
}%
\def\@pacs{}%{\class@warn{No PACS numbers}}%
```

12.6 The \preprint command

```
\def\preprint#1{\gappdef\@preprint{\preprint{#1}}}%
\let\@preprint\@empty
```

12.7 draft

DPC: Name clash with draft option. LaTeX doesn't care, but it confuses me.
This command is a 3.1 relic, now removed.

```
%\newcommand\draft{\@booleantrue\draft@sw}%
```

12.8 The abstract environment

\absbox

```
\newbox\absbox
```

abstract Abstract, as in AMS classes, must be specified *before* \maketitle. It just saves everything up in \absbox.

```
% \changes{v4.e}{1998/01/31}
%    {hookify}
% Note that the specifics of how the abstract is to be
% formatted are set by \cmd\frontmatter@abstractwidth,
% \cmd\frontmatter@abstractheading, and \cmd\frontmatter@abstractfont.
%
% Here we wish to set the abstract into type but save it away in a box, much like
% the \cmd\minipage\ command does.
%
% Note that the \cmd\endabstract\ portion of the environment
% begins with code that mostly duplicates \cmd\endminipage, but without the \cmd\@iiiparbox.
% At the end, we simply transfer the contents of the \cmd\minipage\
% into a box of our own.
%
% However, we also have to end the gratuitous
% paragraph begun by \cmd\minipage.
% We ensure that no trace of that paragraph is left,
% by absorbing the \cmd\parindent\ box.
% Doing so destroys the paragraph entirely, except
% for the \cmd\parskip\ glue, and that morsel is pruned, because we are at the top
% of a page.
%
% This mechanism is vulnerable, however, because anything placed into the horizontal list
% after the \cmd\parindent\ box will confound it.
% And that is exactly what happens under Mik\TeX:
% There seems to be an extension to the \TeX\ standard operating here which inserts a \cmd\spec
% (The mechnism is \emph{not} based on \cmd\everypar.)
%
% To accomodate this state of affairs,
% we have a new mechanism that offers an even more complete rewrite of \env{minipage}.
%
% The user can elect to use this new mechanism via the \classoption{newabstract};
% likewise the old mechanism via the \classoption{oldabstract}.
%
%    \begin{macrocode}
\newenvironment{abstract}{%
```

```
  \aftermaketitle@chk{\begin{abstract}}%
  \global\setbox\absbox\vbox\bgroup
   \color@begingroup
   \columnwidth\textwidth
   \hsize\columnwidth
   \@parboxrestore
   \def\@mpfn{mpfootnote}\def\thempfn{\thempfootnote}\c@mpfootnote\z@
   \let\@footnotetext\frontmatter@footnotetext
   \minipagefootnote@init
   \let\@listdepth\@mplistdepth \@mplistdepth\z@
   \@minipagerestore
   \@setminipage
    \frontmatter@abstractheading
    \frontmatter@abstractfont
    \let\footnote\mini@note
}{%
    \par
    \unskip
    \minipagefootnote@here
    \@minipagefalse   %% added 24 May 89
    \color@endgroup
  \egroup
}%
```

`\frontmatter@footnotetext` We reimplement `\@mpfootnotetext` under a new name (so as not to be overridden by the `hyperref` package) and extend it to accomodate hyperrefs.

Note that this procedure is very like `ltxutil`'s `\mp@footnotetext`, except that it uses `p@mpfootnote` instead of `p@mpfootnote`, and `\frontmatter@makefntext` instead of `\@makefntext`.

```
\long\def\frontmatter@footnotetext#1{%
  \minipagefootnote@pick
    \reset@font\footnotesize
    \interlinepenalty \interfootnotelinepenalty
    \splittopskip \footnotesep
    \splitmaxdepth \dp \strutbox
    \set@footnotewidth
    \@parboxrestore
    \protected@edef\@currentlabel{\csname p@mpfootnote\endcsname\@thefnmark}%
    \color@begingroup
      \frontmatter@makefntext{%
        \rule\z@\footnotesep\ignorespaces#1\@finalstrut\strutbox
      }%
    \color@endgroup
  \minipagefootnote@drop
}%
```

`\frontmatter@abstractheading` The default abstract head; journals will override this procedure.

```
\def\frontmatter@abstractheading{%
 \begingroup
```

```
  \centering\large
  \abstractname
  \par
 \endgroup
}%
```

`\frontmatter@abstractfont` The default type specification for the body of the abstract. Journals will override this setting.

```
\def\frontmatter@abstractfont{}%
```

`\frontmatter@abstractwidth` The default setting is the full text width; journals can override this setting.

```
\def\frontmatter@abstractwidth{\textwidth}
```

`\frontmatter@abstract@produce` This procedure determines how the abstract is incorporated into the title block. We split this out in anticipation of the need to lift the limitation that the abstract not break over pages.

If we are in preprint style, we provide an easy pagebreak point immediately above the abstract. This means that the abstract will either fit entirely on the same page as the title block, or it will start a page of its own (which may itself break onto a subsequent page if necessary).

```
\def\frontmatter@abstract@produce{%
  \par
% \tracingall
  \preprintsty@sw{%
   \do@output@MVL{%
    \vskip\frontmatter@preabstractspace
    \vskip200\p@\@plus1fil
    \penalty-200\relax
    \vskip-200\p@\@plus-1fil
   }%
  }{%
   \addvspace{\frontmatter@preabstractspace}%
  }%
% \leavevmode
   \begingroup
    \dimen@\baselineskip
    \setbox\z@\vtop{\unvcopy\absbox}%
    \advance\dimen@-\ht\z@\advance\dimen@-\prevdepth
    \@ifdim{\dimen@>\z@}{\vskip\dimen@}{}%
   \endgroup
   \unvbox\absbox
  \@ifx{\@empty\mini@notes}{}{\mini@notes\par}%
% \par
  \addvspace\frontmatter@postabstractspace
% \showlists
% \tracingplain
}%
```

`\frontmatter@preabstractspace` `\frontmatter@postabstractspace` Space above and space below abstract in title block

```
\def\frontmatter@preabstractspace{.5\baselineskip}
\def\frontmatter@postabstractspace{.5\baselineskip}
```

12.9 Formatting the title block

`titlepage` This is LaTeX 2ε's `article` class version, with modifications.

```
\newenvironment{titlepage}{%
      \twocolumn@sw{\onecolumngrid}{\newpage}%
      \thispagestyle{titlepage}%
      \setcounter{page}\@ne
}{%
     \twocolumn@sw{\twocolumngrid}{\newpage}%
     \twoside@sw{}{%
        \setcounter{page}\@ne
     }%
}%
```

`\maketitle` Put it all together to format the title block.

```
\def\maketitle{%
  \@author@finish
  \title@column\titleblock@produce
  \suppressfloats[t]%
```

Now save some memory.

```
  \let\and\relax
  \let\affiliation\@gobble
% \let\address\affiliation
  \let\author\@gobble
  \let\@AAC@list\@empty
  \let\@AFF@list\@empty
  \let\@AFG@list\@empty
% \let\keywords\@gobble
% \let\@keywords\@empty
  \let\@AF@join\@AF@join@error
  \let\email\@gobble
  \let\@address\@empty
  \let\maketitle\relax
  \let\thanks\@gobble
  \titlepage@sw{%
   \vfil
   \clearpage
  }{}%
}%
```

`\titleblock@produce` This procedure produces the title block.

It effectively executes inside a group and always returns us to vertical mode.

Note: we assume that it is OK to set the footnote counter to zero at this point. Is this a safe assumption?

```
\def\titleblock@produce{%
 \begingroup
```

Special footnote version [of footnotes] for `\thanks`, copied from `article` class `\maketitle`.

```
  \let\footnote\footnote@latex
  \let\@makefnmark\@makefnmark@latex
  \let\@footnotemark\@footnotemark@latex
  \let\thefootnote\frontmatter@thefootnote
% \let\@makefntext\frontmatter@makefntext
  \global\c@footnote\z@
  \let\@makefnmark\frontmatter@makefnmark
  \frontmatter@setup
  \thispagestyle{titlepage}\label{FirstPage}%
```

Produce the title:

```
  \frontmatter@title@produce
```

Produce the author list:

```
  \groupauthors@sw{%
   \frontmatter@author@produce@group
  }{%
   \frontmatter@author@produce@script
  }%
```

Produce the dates:

```
  \frontmatter@RRAPformat{%
   \expandafter\produce@RRAP\expandafter{\@date}%
   \expandafter\produce@RRAP\expandafter{\@received}%
   \expandafter\produce@RRAP\expandafter{\@revised}%
   \expandafter\produce@RRAP\expandafter{\@accepted}%
   \expandafter\produce@RRAP\expandafter{\@published}%
  }%
```

Produce the abstract, PACS, and keywords, and end any paragraph.

```
  \frontmatter@abstract@produce
  \@pacs@produce\@pacs
  \@keywords@produce\@keywords
  \par
  \frontmatter@finalspace
 \endgroup
}%
```

`\frontmatter@title@produce` The default formatting procedure for the article title. This procedure should take care of the vertical space below the title.

Journals may override this procedure, but it is more likely that they will use the hooks `\frontmatter@title@above`, `\frontmatter@title@format`, and `\frontmatter@title@below`.

At this point, we patch into `\let@mark` so that the common markup idiom `\\` is taken care of.

```
\def\frontmatter@title@produce{%
 \begingroup
  \frontmatter@title@above
```

```
  \frontmatter@title@format
  \@title
  \@ifx{\@title@aux\@title@aux@cleared}{}{%
   \expandafter\frontmatter@footnote\expandafter{\@title@aux}%
  }%
  \par
% \@ifx{\@shorttitle\@empty}{%
%  \markboth{\@title}{\@title}%
% }{%
%  \markboth{\@shorttitle}{\@shorttitle}%
%% \class@warn{Unused short title ignored}%
% }%
  \frontmatter@title@below
 \endgroup
}%
\appdef\let@mark{\let\\\relax}%
```

`\frontmatter@title@above` `\frontmatter@title@format` `\frontmatter@title@below`

The default values for formatting specifications for the article title.

The procedure `\frontmatter@title@above` should take care of the vertical space above the title; `\frontmatter@title@below` below. The procedure `\frontmatter@title@format` should invoke any font switches, etc. that may apply to the title.

Journals will override this procedure.

```
\def\frontmatter@title@above{}%
\def\frontmatter@title@format{}%
\def\frontmatter@title@below{\addvspace{\baselineskip}}%
```

12.9.1 Authors and affiliations in superscriptaddress style

`ontmatter@author@produce@script`

`\frontmatter@author@produce@script` is an alias for `\frontmatter@author@produce`, the procedure that formats the author/affiliation list.

In this case, the authors affiliations are being superscripted (class option `superscriptaddress`).

This procedure must ensure that any paragraph that it starts gets ended finally.

```
\def\frontmatter@author@produce@script{%
  \begingroup
    \let\@author@present\@author@present@script
```

When the author/affiliation list `\@AAC@list` is executed, `\@tempcnt` counts each author, and `\@tempa` stores a list of author indices for the current collaboration.

Note: this procedure uses `\AU@temp` to hold a list of author counts for each collaboration. Note: also uses `\@tempcnta` to communicate between procedures. This usage ot `\@tempcnta` is OK, because the expansion of `\@AAC@list` will trigger only the execution of our own procedures.

```
    \let\AU@temp\@empty
    \@tempcnta\z@
```

We wish to know how many authors are in each collaboration, for the purposes of `\listcomma` and `\listand`.

Here we assign values for the intercollated operators appearing within `\@AAC@list`, then execute that macro, registering the authors in each collaboration. Afterwards, clean up by emulating an empty collaboration.

The result, stored in `\AU@temp`, is a list of brace-delmited tokens, each a number representing the number of authors in that collaboration.

```
    \let\AF@opr \@gobble
    \def\AU@opr{\@author@count\@tempcnta}%
    \def\CO@opr{\@collaboration@count\AU@temp\@tempcnta}%
    \@AAC@list
    \expandafter\CO@opr\@author@cleared
```

Next, present the authors and their affiliations.

We assign values for the intercollated operators appearing within `\@AAC@list`. Next, assign `\@listcomma` based on the number of authors in the first collaboration. Then expand `\@AAC@list`.

Upon completion, there will be two space tokens following the last author because of `\listand` processing; remove them, then end the last paragraph.

FIXME: this juncture would be a good time to effectively eliminate the unused affiliations in `\@AFF@list`.

```
    \begingroup
\frontmatter@authorformat
\let\AF@opr \@affilID@def
\let\AU@opr \@author@present
\def\CO@opr{\@collaboration@present\AU@temp}%
\set@listcomma@list\AU@temp
\@AAC@list
\unskip\unskip
\par
\endgroup
```

We now list out the affiliations in the order they appeared.

```
    \begingroup
\frontmatter@above@affiliation@script
% \let\@AFU@list\@empty
% \let\AFS@opr \affils@present@script
% \@AFG@list
% \let\AF@opr  \affil@present@script
% \@AFU@list
\let\AFF@opr \@affil@script
% \say\@AFF@list
     \@AFF@list
\frontmatter@footnote@produce
\par
\endgroup
  \endgroup
}%
```

`\@author@count` This version of `\AU@opr` counts the number of authors it processes.

```
\def\@author@count#1{\advance#1\@ne\@author@gobble}%
```

`\@collaboration@present` Format a collaboration. Note that we immediately end the paragraph thus begun, because we only support `superscriptaddress`-style processing.

Note also that the execution of the production procedures takes place inside of a group; for reasons of protective programming.

Bug note: originally, the number of authors in each collaboration was stored in `\@tempa`, but this exposed us to procedures in our production that modified the meaning of `\@tempa`.

The last action done in this procedure is to assign `\@listcomma` based on the number of authors in the next collaboration.

```
\def\@collaboration@present#1#2#3{%
 \par
 \begingroup
  \frontmatter@collaboration@above
  \@author@present{(\ignorespaces#2\unskip)}{#3}%
  \par
 \endgroup
 \set@listcomma@list#1%
}%
\def\frontmatter@collaboration@above{}%
```

`\@collaboration@count` Simply register each author in this collaboration. Note: #1 is a `\csname` to hold the value, #2 is the `\count` register holding the count of interest.

```
\def\@collaboration@count#1#2{%
 \appdef@eval#1{\the#2}#2\z@
 \@author@gobble
}%
```

`\@affilID@def` The `\@affilID@def` procedure, an alias of `\AF@opr` used during `\frontmatter@author@produce@script` processing, memorizes the affiliation in `\@affilID@temp`.

```
\def\@affilID@def{\def\@affilID@temp}%
```

`\@affilID@temp` The macro `\@affilID@temp` is used to communicate between `\doauthor` and instances of `\AF@opr` within the author list.

```
\let\@affilID@temp\@empty
```

`\affils@present@script` An alias of `\AFS@opr`, this procedure is applied to `\@AFG@list` and builds `\@AFU@list`, a list of unique affiliations found within affiliation groups.

```
%\def\affils@present@script#1{%
% \get@affil#1{}%
%}%
%\def\get@affil#1{%
% \def\@tempa{#1}%
% \@ifx{\@empty\@tempa}{}{%
%  \@booleanfalse\temp@sw
%  \def\AF@opr{\x@match\@tempa}%
%  \@AFU@list
%  \temp@sw{}{%
%   \appdef\@AFU@list{\AF@opr{#1}}%
```

```
%  }%
%  \get@affil
% }%
%}%
```

`\affil@present@script` An alias of `\AF@opr`, this procedure is applied to `\@AFU@list` when formatting the affiliations under `superscriptaddress`. It in turn applies `\affil@script` to `\@AFF@list`, thereby formatting the affiliation associated with this affiliation index.

Note: it traverses the `\@AFF@list` for each entry in the `\@AFU@list`, thereby making this portion of code execute in N^2 time.

```
%\def\affil@present@script#1{%
% \def\@tempa{#1}%
% \begingroup
%  \frontmatter@affiliationfont
%  \let\AFF@opr \affil@script
%  \@AFF@list
% \endgroup
%}%
```

`\affil@script` Alias of `\AFF@opr`, this procedure is applied to `\@AFF@list` when formatting the affiliations under `superscriptaddress`. It rejects all entries not matching the affiliation index held in `\@tempa`.

```
\def\affil@script#1#2#3{%
 \def\@tempifx{#1}\@ifx{\@tempifx\@tempa}{%
  \@if@empty{#2}{}{%
   \par
   \begingroup
    \textsuperscript{\normalfont#1}%
    #2%
    \@if@empty{#3}{}{\frontmatter@footnote{#3}}%
    \par
   \endgroup
  }%
 }{}%
}%
```

`\@affil@script` Alias of `\AFF@opr`: The affiliations are being formatted in the case where affiliations are being superscripted: make a list out of the affiliations with the numbers printed. Society-specific code can change the formatting by overriding the definition of `\frontmatter@affiliationfont`.

FIXME: dead code.

```
\def\@affil@script#1#2#3{%
 \@ifnum{#1=\z@}{}{%
  \par
  \begingroup
   \frontmatter@affiliationfont
%  \textsuperscript{\normalfont#1}%
```

```
    \textsuperscript{#1}%
    #2%
    \@if@empty{#3}{}{\frontmatter@footnote{#3}}%
    \par
   \endgroup
  }%
}%
```

`\@author@present@script` This version of `\AU@opr` applies to the `superscriptaddress` class option. Need to add commas between groups of address numbers, which are passed in the macro `\@affilID@temp` in the form `{3}{4}{7}` if this set of authors is related to addresses, 3, 4 and 7.

```
\def\@author@present@script#1#2{%
 \begingroup
% \parskip\@author@parskip\relax
  \gdef\comma@space{\textsuperscript{,\,}}%
% \@ifx{\@affilID@temp\@empty}{%
%  \doauthor{#1}{#2}{}%
% }{%
%  \expandafter\set@count@\@affilID@temp{}{}\@@nil
%  \@ifnum{\count@=\z@}{%
%   \doauthor{#1}{#2}{}%
%  }{%
%   \doauthor{#1}{#2}{%
%    \textsuperscript{\expandafter\@affilcomma\@affilID@temp\relax\relax}%
%   }%
%  }%
% }%
 \doauthor{#1}{#2}{%
  \let\@tempa\@empty
  \expandafter\@affil@present@script\@affilID@temp\relax
 }%
 \endgroup
 \advance\@tempcnta\m@ne
}%
%\def\set@count@#1#2\@@nil{%
% \count@#1\relax
%}%
\def\@affilcomma#1#2{%
 \@ifx{\z@#1}{%
  \@ifx{\relax#2}{}{%
   \@affilcomma{#2}%
  }%
 }{%
  #1%
  \@ifx{\relax#2}{}{%
   \@ifx{\z@#2}{%
    \@affilcomma
   }{%
    ,\,\@affilcomma{#2}%
```

```
    }%
   }%
  }%
}%
\def\@affil@present@script#1{%
 \@ifx{\relax#1}{%
  \@ifx{\@tempa\@empty}{}{%
   \textsuperscript{\expandafter\@affilcomma\@tempa\relax\relax}%
  }%
 }{%
  \@ifx{\z@#1}{}{\appdef\@tempa{{#1}}}%
  \@affil@present@script
 }%
}%
\@provide\@author@parskip{\z@skip}%
```

12.9.2 Authors and affiliations in groupedaddress style

`\frontmatter@author@produce@group` `\frontmatter@author@produce@group` is an alias for `\frontmatter@author@produce`, the procedure that formats the author/affiliation list when `groupedaddress` is in effect.

In this case, the authors are being grouped above their shared addresses. How it works: the `\@AFG@list`, a list of all unique affiliation groups, is expanded. For each such group, all authors belonging to that group are formatted above the affiliation group. Collaborations are silently passed over.

This procedure must ensure that any paragraph that it starts gets ended finally.

```
\def\frontmatter@author@produce@group{%
  \begingroup
    \let\@author@present\@author@present@group
    \frontmatter@authorformat
    \let\AFS@opr \affils@present@group
    \let\@listcomma\relax
    \@AFG@list
    \frontmatter@footnote@produce
    \par
  \endgroup
  \frontmatter@authorbelow
}%
\@provide\frontmatter@authorbelow{}%
```

`\affils@present@group` Alias of `\AFS@opr`, this procedure is used on every element of `\@AFG@list`, the list of unique affiliation groups, when formatting the author/affiliation block: It presents all authors sharing this affiliation group.

This procedure is called via the procedure `\frontmatter@author@produce@group`, an alias for `\frontmatter@author@produce`,

Its use pertains to the class options `groupedaddress`, `unsortedaddress`, and `runinaddress` (that is, `\groupauthors@sw` is true).

For each invocation, the entire `\@AAC@list` is run through (twice), making this process execute in N^2 time. (Is this really true?)

```
\def\affils@present@group#1{%
```

Count (using `\@tempcnta`) the authors in the author list whose affiliation is `#1`.

Note that `\AF@temp` is used as a storage register for the argument of the procedure, for purposes of comparison using `\ifx`.

Note: here we use `\@tempcnta` to communicate accumulate the number of authors in the current affiliation set.

QUERY: does the value of `\AF@temp` and `\AF@opr` need to persist?

QUERY: what value does `\AU@opr` have at this point? ANSWER: the `\@affilID@count` and `\@affilID@match` procedures assign meanings to `\AU@opr` and `\CO@opr` dynamically. We initialize the two operators to `\undefined` so as to catch the pathological cases.

```
\begingroup
  \def\AF@temp{#1}%               % 0B13
  \@tempcnta\z@
  \let\AU@opr \undefined          % 0B13
  \let\CO@opr \undefined          % 0B13
  \def\AF@opr{\@affilID@count\AF@temp\@tempcnta}% 0B13
  \@AAC@list
```

If no authors using this affiliation group, skip it.

```
  \@ifnum{\@tempcnta=\z@}{}{%
```

DPC: If we have a list of length two, need to locally zap a comma.

```
   \begingroup
    \frontmatter@above@affilgroup
    \set@listcomma@count\@tempcnta
```

Run through the author list, presenting (with `\@author@present`) those authors whose affiliation matched the given one.

```
    \let\AU@opr \undefined          % 0B13
    \let\CO@opr \undefined          % 0B13
    \def\AF@opr{\@affilID@match\AF@temp}%
    \@AAC@list
   \endgroup
```

Now that the authors have all been presented, present the affiliations, grouped.

Note: `\@tempcnta` is used to count the number of affiliations for this set of authors.

```
   \begingroup
    \par
    \frontmatter@above@affiliation
    \frontmatter@affiliationfont
    \let\\\frontmatter@addressnewline
    \@tempcnta\z@
    \@tfor\AF@temp:=#1\do{%
     \expandafter\@ifx\expandafter{\z@\AF@temp}{}{%
      \advance\@tempcnta\@ne
```

```
      }%
     }%
     \@ifnum{\@tempcnta=\tw@}{%
      \let\@listcomma\relax
     }{}%
     \def\after@address{\@listcomma\ \@listand}%
     \runinaddress@sw{%
     }{%
      \tightenlines@sw{%
      }{%
       \parskip\z@ %space between affilations in grouped mode
      }%
      \appdef\after@address\par
     }%
     \let\AFF@opr \@affil@group
     \do@affil@fromgroup\@AFF@list#1\relax
    \endgroup
```

End of branch handling authors.

```
   }%
   \par
 \endgroup
%\smallskip
}%
```

`\@affilID@count` `\@affilID@match`

The control sequence name `\AF@opr` is placed in the author/affiliation list (`\@AAC@list`) for each (new) affiliation, with that affiliation as its argument.

The control sequence name `\AF@opr` is an alias of a procedure to count the number of authors at a given affiliation (`\@affilID@count`) or to present the given authors (`\@affilID@match`). These two procedures are executed when formatting under the `groupedaddress` class option.

```
\def\@affilID@count#1#2#3{%
  \def\@tempifx{#3}%
  \@ifx{\@tempifx#1}{%
    \def\AU@opr{\@author@count#2}%
  }{%
    \let\AU@opr \@author@gobble
  }%
  \let\CO@opr \@collaboration@err
}%
\def\@affilID@match#1#2{%
 \def\@tempifx{#2}%
 \@ifx{\@tempifx#1}{%
   \let\AU@opr \@author@present
 }{%
   \let\AU@opr \@author@gobble
 }%
  \let\CO@opr \@collaboration@err
}%
```

`\do@affil@fromgroup` Executes iteratively: selects the next address ID to print, and then re-execute the list of addresses to print the text of that address.

Note: an argument of `\relax` breaks out of this iteration, while an argument of `\z@` (`\noaffiliation`) is a no-op.

Note: we have createed an N^2 computing process. FIXME: can use hashing instead, you know.

```
\def\do@affil@fromgroup#1#2{%
  \@ifx{\relax#2}{}{%
\count@#2\relax
\@ifnum{\count@=\z@}{}{#1}%
    \do@affil@fromgroup#1%
  }%
}%
```

`\@affil@group` The alias of `\AFF@opr` when the list `\@AFF@list` is executed by `\do@affil@fromgroup`(`groupedaddress`): Print the address if its number matches `\count@` (value set by `\do@affil@fromgroup`).

Note that we detect a `\noaffiliation` here by the second parameter's being the token `\blankaffiliation`.

```
\def\@affil@group#1#2#3{%
  \@ifnum{\count@=#1\relax}{%
    \@ifx{#2\blankaffiliation}{}{%
#2%
\@if@empty{#3}{}{%
\frontmatter@footnote{#3}%
}%
\after@address
    }%
    \advance\@tempcnta\m@ne
  }{}%
}%
```

`\@author@present@group` This version of `\AU@opr` applies to the `groupedaddress` class option. Traverse the list calling the `\doauthor` hook on each author.

```
\def\@author@present@group#1#2{%
  \gdef\comma@space{\gdef\comma@space{\textsuperscript{,\,}}}%
  \doauthor{#1}{#2}{}%
  \advance\@tempcnta\m@ne
}%
```

`\@pacs@produce` PACS, keywords and dates.

```
\def\@pacs@produce#1{%
 \showPACS@sw{%
  \begingroup
   \frontmatter@PACS@format
   \@pacs@name#1\par
  \endgroup
 }{%
  \@if@empty{#1}{}{%
```

```
    \class@warn{\PACS@warn}%
   }%
  }%
}%
\def\PACS@warn{If you want your PACS to appear in your output, use document class option showpa
```

`\@keywords@produce`

```
\def\@keywords@produce#1{%
 \showKEYS@sw{%
  \begingroup
   \frontmatter@keys@format
   \@keys@name#1\par
  \endgroup
 }{%
  \@if@empty{#1}{}{%
   \class@warn{If you want your keywords to appear in your output, use document class option sh
  }%
 }%
}%
```

`atter@footnote@produce@footnote` `matter@footnote@produce@endnote` Produce any footnotes to appear at the bottom of the title page.

If frontmatter footnotes have been redirected to the bibliography, this will be a no-op. ref.: `\present@bibnote`.

```
\def\frontmatter@footnote@produce@footnote{%
 \let\@TBN@opr\present@FM@footnote
 \@FMN@list
 \global\let\@FMN@list\@empty
}%
\def\present@FM@footnote#1#2{%
 \begingroup
  \csname c@\@mpfn\endcsname#1\relax
  \def\@thefnmark{\thempfn}%
  \frontmatter@footnotetext{#2}%
 \endgroup
}%
\def\frontmatter@footnote@produce@endnote{%
}%
```

`\frontmatter@thefootnote` The in the title block, the footnote symbol (used for alternate affiliation and sundry) is a `\@fnsymbol` or a lowercase letter, as selected by a boolean.

```
\def\frontmatter@thefootnote{%
 \altaffilletter@sw{\@alph}{\@fnsymbol}\c@footnote
}%
```

`\frontmatter@makefnmark` The default procedure for setting the footnote mark within the frontmatter. Journals will override this procedure.

```
\def\frontmatter@makefnmark{%
 \@textsuperscript{%
  \normalfont\@thefnmark
```

```
 }%
}%
```

`\frontmatter@makefntext` This procedure is used when setting footnotes within the title block.

```
\long\def\frontmatter@makefntext#1{%
 \parindent 1em
 \noindent
 \Hy@raisedlink{\hyper@anchorstart{frontmatter@\the\c@footnote}\hyper@anchorend}%
 \@makefnmark
 #1%
}%
```

`\frontmatter@setup` The default type specification for the title page. Journals will override this definition.

```
\def\frontmatter@setup{}%
```

`\frontmatter@RRAPformat` `\punct@RRAP` `\produce@RRAP` The procedure `\frontmatter@RRAPformat` is the formatting procedure for the "Received, Revised, etc." block on the title page. The macro `\punct@RRAP` produces the punctuation between the entries. Journals will override these two definitions.

The procedure `\produce@RRAP` is fairly general, and should be used by most journal substyles.

The procedure `\frontmatter@RRAP@format` will be overridden by the journal.

```
\def\frontmatter@RRAPformat#1{%
 \removelastskip
 \begingroup
  \frontmatter@RRAP@format
  #1\par
 \endgroup
}%
\def\punct@RRAP{; }%
\def\produce@RRAP#1{%
  \@if@empty{#1}{}{%
   \@ifvmode{\leavevmode}{\unskip\punct@RRAP\ignorespaces}%
   #1%
  }%
}%
```

`\frontmatter@authorformat` The default type specification for the author list. Journals will override this definition.

```
\def\frontmatter@authorformat{}%
```

`\frontmatter@above@affilgroup` The default amount of space above an affiliation group (in a `\affils@present@group` production). Journals will override this definition.

```
\def\frontmatter@above@affilgroup{}%
```

`\frontmatter@above@affiliation` `matter@above@affiliation@script` The `\frontmatter@above@affiliation` is the amount of space above affiliations in the `groupedaddress` style, and `\frontmatter@above@affiliation@script` is that for `superscriptaddress`. Journals will override this definition.

```
\def\frontmatter@above@affiliation{}%
\def\frontmatter@above@affiliation@script{}%
```

`\frontmatter@affiliationfont` The default type specification for the affiliation. Journals will override this definition.

```
\def\frontmatter@affiliationfont{}%
```

`\frontmatter@RRAP@format` The default type specification for the dates. Journals will override this definition.

```
\def\frontmatter@RRAP@format{}%
```

`\frontmatter@PACS@format` Specify the formatting of the title page PACS statement. Journals will override.

```
\def\frontmatter@PACS@format{}%
```

`\frontmatter@keys@format` Specify the formatting of the title page keywords statement. Journals will override.

```
\def\frontmatter@keys@format{}%
```

`\frontmatter@finalspace`

```
\def\frontmatter@finalspace{\addvspace{18\p@}}
```

`\frontmatter@addressnewline` The definition of \\ for address handling. Default puts all the 'lines' on a run-in line, separated by comma and space. DPC: was space between lines of addresss.

```
\def\frontmatter@addressnewline{%
  \@ifhmode{\skip@\lastskip\unskip\unpenalty\break\hskip\skip@}{}%
  % was: \vskip-.5ex
}%
```

`\frontmatter@preabstractspace` `\frontmatter@postabstractspace` Space above and space below abstract in title block

```
\def\frontmatter@preabstractspace{5.5\p@}
\def\frontmatter@postabstractspace{6.5\p@}
```

`\aftermaketitle@chk` Error-checking code.

```
\def\aftermaketitle@chk#1{%
  \@ifx{\maketitle\relax}{%
    \class@err{\protect#1 must be used before \protect\maketitle}%
  }{}%
}%
```

`\ps@titlepage` Default page style for title page. Journals will override this procedure.

```
\def\ps@titlepage{\ps@empty}%
```

FIXME: there is a limitation to the default meaning of `\@startpage`, the label "FirstPage" is only defined if the document has a `\maketitle` command.

```
\def\volumeyear#1{\gdef\@volumeyear{#1}}%
\def\@volumeyear{}%
\def\volumenumber#1{\gdef\@volumenumber{#1}}%
```

```
\def\@volumenumber{}%
\def\issuenumber#1{\gdef\@issuenumber{#1}}%
\def\@issuenumber{}%
\def\eid#1{\gdef\@eid{#1}}%
\def\@eid{}%
%
\def\startpage#1{\gdef\@startpage{#1}\c@page#1\relax}%
\def\@startpage{\pageref{FirstPage}}%
\def\endpage#1{\gdef\@endpage{#1}}%
\def\@endpage{\pageref{LastPage}}%
```

12.10 Printing out the "list-of" elements

FIXME: The `\appendix@toc` procedure should change the meaning of `\l@section` so that the `\section`s can be appropriately formatted, reflecting their status as appendices.

```
\def\print@toc#1{%
 \begingroup
% \c@secnumdepth-\maxdimen
  \expandafter\section
  \expandafter*%
  \expandafter{%
              \csname#1name\endcsname
              }%
  \let\appendix\appendix@toc
  \@starttoc{#1}%
 \endgroup
}%
\def\appendix@toc{}%
```

13 Body

13.1 counters

The following definitions are probably identical to those in `classes.dtx`

```
\def\labelenumi{\theenumi.}
\def\theenumi{\arabic{enumi}}

\def\labelenumii{(\theenumii)}
\def\theenumii{\alph{enumii}}
\def\p@enumii{\theenumi}

\def\labelenumiii{\theenumiii.}
\def\theenumiii{\roman{enumiii}}
\def\p@enumiii{\theenumi(\theenumii)}

\def\labelenumiv{\theenumiv.}
\def\theenumiv{\Alph{enumiv}}
\def\p@enumiv{\p@enumiii\theenumiii}
```

```
\def\labelitemi{\textbullet}
\def\labelitemii{\normalfont\bfseries\textendash}
\def\labelitemiii{\textasteriskcentered}
\def\labelitemiv{\textperiodcentered}

\pagenumbering{arabic}
```

13.2 float parameters

from the old aps.sty. (DPC: same as article I think) AO: here, LaTeX's standard classes fail very poorly (the price of backward compatability): the values for `\floatpagefraction` and `\dblfloatpagefraction` need to be raised to avoid creating extremely short float pages.

```
\setcounter{topnumber}{2}
\def\topfraction{.9}
\setcounter{bottomnumber}{1}
\def\bottomfraction{.9}
\setcounter{totalnumber}{3}
\def\textfraction{.1}
\def\floatpagefraction{.9}
\setcounter{dbltopnumber}{2}
\def\dbltopfraction{.9}
\def\dblfloatpagefraction{.9}
```

13.3 List Environments

```
\newenvironment{verse}{%
  \let\\=\@centercr
  \list{}{%
    \itemsep\z@ \itemindent -1.5em\listparindent \itemindent
    \rightmargin\leftmargin\advance\leftmargin 1.5em}\item[]%
}{%
 \endlist
}%

\newenvironment{quotation}{%
  \list{}{%
    \listparindent 1.5em
    \itemindent\listparindent
    \rightmargin\leftmargin \parsep \z@ \@plus\p@}\item[]%
}{%
 \endlist
}%

\newenvironment{quote}{%
  \list{}{%
  \rightmargin\leftmargin}\item[]%
}{%
 \endlist
}%
```

```
\def\descriptionlabel#1{%
  \hspace\labelsep \normalfont\bfseries #1%
}%
\newenvironment{description}{%
 \list{}{%
    \labelwidth\z@ \itemindent-\leftmargin
    \let\makelabel\descriptionlabel
 }%
}{%
 \endlist
}%
```

13.4 Sectioning Commands

13.4.1 Counters and Their Productions

```
\newcounter{part}
\newcounter{section}
\newcounter{subsection}[section]
\newcounter{subsubsection}[subsection]
\newcounter{paragraph}[subsubsection]
\newcounter{subparagraph}[paragraph]
```

```
\def\thepart          {\Roman{part}} %
\def\thesection       {\Roman{section}}
\def\p@section        {}
\def\thesubsection    {\Alph{subsection}}
\def\p@subsection     {\thesection\,}
\def\thesubsubsection {\arabic{subsubsection}}
\def\p@subsubsection  {\thesection\,\thesubsection\,}
\def\theparagraph     {\alph{paragraph}}
\def\p@paragraph      {\thesection\,\thesubsection\,\thesubsubsection\,}
\def\thesubparagraph  {\arabic{subparagraph}}
\def\p@subparagraph   {\thesection\,\thesubsection\,\thesubsubsection\,\theparagraph\,}
```

13.4.2 The Acknowledgments Environment

This user-level markup produces a head introducing the acknowledgments, and acts as a wrapper for the text. In this implementation, it is an unnumbered section, but appears within the toc.

For compatiability's sake, we implement it under the alternative spelling `acknowledgements`.

```
\newenvironment{acknowledgments}{%
 \acknowledgments@sw{%
  \section*{\acknowledgmentsname}%
 }{%
  \par
 \phantomsection
  \addcontentsline{toc}{section}{\protect\numberline{}\acknowledgmentsname}%
 }%
}{%
```

```
 \par
}%
\@booleantrue\acknowledgments@sw
\newenvironment{acknowledgements}{%
 \replace@environment{acknowledgements}{acknowledgments}%
}{%
 \endacknowledgments
}%
```

13.4.3 Part Opener

section setup copied verbatim from revtex3 aps/osa. Does not explicitly depend on pointsize options.

```
\def\part{\par
   \addvspace{4ex}%
   \@afterindentfalse
   \secdef\@part\@spart}%

\def\@part[#1]#2{%
 \@ifnum{\c@secnumdepth >\m@ne}{%
        \refstepcounter{part}%
        \addcontentsline{toc}{part}{\thepart\hspace{1em}#1}%
 }{%
      \addcontentsline{toc}{part}{#1}%
 }%
 \begingroup
    \parindent \z@ \raggedright
    \interlinepenalty\@M
    \@ifnum{\c@secnumdepth >\m@ne}{%
      \Large \bf \partname~\thepart%
      \par\nobreak
    }{}%
    \huge \bf
    #2%
    \markboth{}{}\par
 \endgroup
   \nobreak
   \vskip 3ex
   \@afterheading
}%

\def\@spart#1{{\parindent \z@ \raggedright
    \interlinepenalty\@M
    \huge \bf
    #1\par}
    \nobreak
    \vskip 3ex
    \@afterheading}
```

13.4.4 Stacked Heads

Here are the class default definitions for sectioning commands. A society or a journal substyle will likely override these definitions.

In doing so, you can customize the formatting for a particular level by defining, e.g., `\@hangfrom@section` or `\@subsectioncntformat`.

```
\def\section{%
  \@startsection
    {section}%
    {1}%
    {\z@}%
    {0.8cm \@plus1ex \@minus .2ex}%
    {0.5cm}%
    {\normalfont\small\bfseries}%
}%
\def\subsection{%
  \@startsection
    {subsection}%
    {2}%
    {\z@}%
    {.8cm \@plus1ex \@minus .2ex}%
    {.5cm}%
    {\normalfont\small\bfseries}%
}%
\def\subsubsection{%
  \@startsection
    {subsubsection}%
    {3}%
    {\z@}%
    {.8cm \@plus1ex \@minus .2ex}%
    {.5cm}%
    {\normalfont\small\itshape}%
}%
```

13.4.5 Runin Heads

```
\def\paragraph{%
  \@startsection
    {paragraph}%
    {4}%
    {\parindent}%
    {\z@}%
    {-1em}%
    {\normalfont\normalsize\itshape}%
}%
\def\subparagraph{%
  \@startsection
    {subparagraph}%
```

```
    {5}%
    {\parindent}%
    {3.25ex \@plus1ex \@minus .2ex}%
    {-1em}%
    {\normalfont\normalsize\bfseries}%
}%
```

13.5 Math

\theequation We change the production of the equation counter so that we can accomodate the `eqsecnum` option.

```
\def\theequation{%
 \theequation@prefix\arabic{equation}%
}%
\def\theequation@prefix{}%
```

13.6 Type Size-Dependent Settings

Note that many parameters, as well as the meaning of `\normalsize`, are deferred until `\AtEndOfClass` time. Therefore, code elsewhere in this class *should not* assume these things' values have been set!

```
\AtEndOfClass{%
 \expandafter\input\expandafter{\@pointsize pt\substyle@ext}%
}%
```

13.7 All Point Sizes

```
\setcounter{secnumdepth}{4}

\lineskip 1pt
\normallineskip 1pt
\def\baselinestretch{1}%

\@lowpenalty   51
\@medpenalty  151
\@highpenalty 301

\@beginparpenalty -\@lowpenalty
\@endparpenalty   -\@lowpenalty
\@itempenalty     -\@lowpenalty

\arraycolsep 3pt
\tabcolsep 2pt
\arrayrulewidth .4pt
\doublerulesep 2pt

\skip\@mpfootins = 0pt

\fboxsep  = 3.0pt
\fboxrule = 0.4pt
```

13.8 Figures

`figure` We define the `figure` environment. Later, we will horse around with its meaning in order to accomodate `\floats@sw`.

```
\newenvironment{figure}
               {\@float{figure}}
               {\end@float}
\newenvironment{figure*}
               {\@dblfloat{figure}}
               {\end@dblfloat}
```

`\@makecaption` If caption is one line long, to be centered; if lines turn, then set justified.

```
\newlength\abovecaptionskip
\newlength\belowcaptionskip
\setlength\abovecaptionskip{10\p@}
\setlength\belowcaptionskip{2\p@}

\long\def\@makecaption#1#2{%
  \par
% \nobreak
  \vskip\abovecaptionskip
  \begingroup
   \small\rmfamily
   \sbox\@tempboxa{%
    \let\\\heading@cr
    #1: #2%
   }%
   \@ifdim{\wd\@tempboxa >\hsize}{%
    \begingroup
     \samepage
     \flushing
     \let\footnote\@footnotemark@gobble
     #1: #2\par
    \endgroup
   }{%
     \global \@minipagefalse
     \hb@xt@\hsize{\hfil\unhbox\@tempboxa\hfil}%
   }%
  \endgroup
  \vskip\belowcaptionskip
}%
\def\@footnotemark@gobble{%
 \@footnotemark
 \@ifnextchar[{\@gobble@opt@i}{\@gobble}%
}%
\def\@gobble@opt@i[#1]#2{}%
\def\@mpmakefntext#1{%
 \flushing
 \parindent=1em
 \noindent
```

```
\hb@xt@1em{\hss\@makefnmark}%
#1%
}%
```

`\thefigure` The figure counter and float placement defaults.

```
\newcounter{figure}
\renewcommand \thefigure {\@arabic\c@figure}
```

Note that we give the '!' modifier by default. This is an effort to avoid the syndrome wherein a deferred float finds itself unqualified for placement, thereby getting carried until `\clearpage`.

```
\def\fps@figure{tbp}
\def\ftype@figure{1}
\def\ext@figure{lof}
\def\fnum@figure{\figurename~\thefigure}
```

13.8.1 Deferring `figure` Floats

We install a hook at `\AtBeginDocument` time which determines if figures will float or will be deferred until `\printfigures` time.

```
\appdef\class@documenthook{%
 \do@if@floats{figure}{.fgx}%
}%
\appdef\class@enddocumenthook{%
 \printfigures\relax
}%
```

`\printfigures` The user-level command `\printfigures` determines where the figures are to appear in a document in which `\floats@sw` is false. If the user invokes the `endfloats` class option and fails to insert a `\printfigures` command, the figures will be printed at the end of the document. If the command is given, but floats are not being deferred, it amounts to a no-op.

```
\newcommand\printfigures{%
 \@ifstar{\true@sw}{\floatp@sw{\true@sw}{\false@sw}}%
 {%
  \print@float{figure}{\oneapage}%
 }{%
  \print@float{figure}{}%
 }%
}%
```

`\@xfloat@prep` We patch into the procedure `\@xfloat@prep`. This patch applies to all floats (not `figure` alone) and makes the type center.

```
\appdef\@xfloat@prep{%
 \appdef\@parboxrestore{\centering}%
 \let\@makefnmark\@makefnmark@latex
 \let\@footnotemark\@footnotemark@latex
}%
```

13.9 Tables

DPC: More or less taken from `revtex2` `aps.sty`, but using dcolumn for decimal alignment.

`table` We define the `table` environment. Later, we will horse around with its meaning in order to accomodate `\floats@sw`.

```
\newenvironment{table}
               {\@float{table}}
               {\end@float}
\newenvironment{table*}
               {\@dblfloat{table}}
               {\end@dblfloat}
```

`\thetable` Table counter and default float placement declarations.

```
\newcounter{table}
\renewcommand\thetable{\@Roman\c@table}
```

Note that we give the '!' modifier by default. This is an effort to avoid the syndrome wherein a deferred float finds itself unqualified for placement, thereby getting carried until `\clearpage`.

```
\def\fps@table{tbp}
\def\ftype@table{2}
\def\ext@table{lot}
\def\fnum@table{\tablename~\thetable}
```

`\table@hook` Assign a meaning to the hook installed into float processing.

`\squeezetable` By default floats are `\small`. The `\squeezetable` declaration makes them smaller (`\scriptsize`). In general you can locally redefine `\table@hook` to be whatever you like. (DPC: `\Huge\color{magenta}`...?)

```
\def\table@hook{\small}%
\def\squeezetable{\def\table@hook{\scriptsize}}%
\appdef\@floatboxreset{\table@hook}%
```

13.9.1 Deferring `table` Floats

We install a hook at `\AtBeginDocument` time which determines if tables will float or will be deferred until `\printtables` time.

We also deal with the possibility of `longtable` environments.

```
\appdef\class@documenthook{%
 \floats@sw{}{%
  \let@environment{longtable@float}{longtable}%
  \let@environment{longtable}{longtable@write}%
  \let@environment{longtable*@float}{longtable*}%
  \let@environment{longtable*}{longtable*@write}%
  \let@environment{turnpage@float}{turnpage}%
  \let@environment{turnpage}{turnpage@write}%
 }%
 \do@if@floats{table}{.tbx}%
```

```
}%
\appdef\class@enddocumenthook{%
 \printtables\relax
}%
\newenvironment{longtable@write}{%
 \write@@float{longtable}{table}%
}{%
 \endwrite@float
}%
\newenvironment{longtable*@write}{%
 \write@@float{longtable*}{table}%
}{%
 \endwrite@float
}%
\newenvironment{turnpage@write}{%
 \immediate\write\tablewrite{\string\begin{turnpage}}%
}{%
 \immediate\write\tablewrite{\string\end{turnpage}}%
}%
```

`\printtables` The user-level command `\printtables` determines where the tables are to appear in a document in which `\floats@sw` is false. If the user invokes the `nofloats` and fails to insert a `\printtables` command, the tables will be printed at the end of the document. If the command is given, but floats are not being deferred, it amounts to a no-op.

```
\newcommand\printtables{%
 \begingroup
  \let@environment{longtable}{longtable@float}%
  \let@environment{longtable*}{longtable*@float}%
  \let@environment{turnpage}{turnpage@anchored}%
  \prepdef\longtable{\trigger@float@par}%
  \expandafter\prepdef\csname longtable*\endcsname{\trigger@float@par}%
  \expandafter\prepdef\csname table@floats\endcsname{%
   \onecolumngrid@push
  }%
  \expandafter\appdef\csname endtable@floats\endcsname{%
   \onecolumngrid@pop
  }%
  \@ifstar{\true@sw}{\floatp@sw{\true@sw}{\false@sw}}%
  {%
   \print@float{table}{\oneapage}%
  }{%
   \print@float{table}{}%
  }%
 \endgroup
}%
\newenvironment{turnpage@anchored}{%
 \onecolumngrid@push
 \setbox\z@\vbox to\textwidth\bgroup
```

```
  \columnwidth\textheight
}{%
  \vfil
 \egroup
 \rotatebox{90}{\box\z@}%
 \onecolumngrid@pop
}%
```

14 Tabular

Every tabular has a double (Scotch) rule above and below. The column specifier "d" is implemented using the `dcolumn` package, if available. FIXME: always load dcolumn!

\tabular@hook
\endtabular@hook
ruledtabular

```
\def\endtabular@hook{}%
%\RequirePackage{dcolumn}%
\appdef\class@documenthook{%
 \@ifpackageloaded{dcolumn}{%
  \newcolumntype{d}{D{.}{.}{-1}}%
 }{}%
}%
\def\toprule{\hline\hline}%
\def\colrule{\hline}%
\def\botrule{\hline\hline}%
\newenvironment{ruledtabular}{%
 \def\array@default{v}%
 \appdef\tabular@hook{\def\@halignto{to\hsize}}%
 \let\tableft@skip@default\tableft@skip
 \let\tableft@skip\tableft@skip@float
 \let\tabmid@skip@default\tabmid@skip
 \let\tabmid@skip\tabmid@skip@float
 \let\tabright@skip@default\tabright@skip
 \let\tabright@skip\tabright@skip@float
 \let\array@row@pre@default\array@row@pre
 \let\array@row@pre\array@row@pre@float
 \let\array@row@pst@default\array@row@pst
 \let\array@row@pst\array@row@pst@float
 \appdef\array@row@rst{%
  \let\array@row@pre\array@row@pre@default
  \let\array@row@pst\array@row@pst@default
  \let\tableft@skip\tableft@skip@default
  \let\tabmid@skip\tabmid@skip@default
  \let\tabright@skip\tabright@skip@default
  \appdef\tabular@hook{\let\@halignto\@empty}%
 }%
}{%
}%
```

15 Footnote Text

\@makefntext

```
\def\@makefntext#1{%
  \def\baselinestretch{1}%
  \reset@font\footnotesize
  \parindent 1em%
  \noindent
  \hb@xt@1.8em{%
   \Hy@raisedlink{\hyper@anchorstart{footnote@\the\c@footnote}\hyper@anchorend}%
   \hss\@makefnmark
  }%
  #1\par
}%
```

15.1 Citations, Bibliography, Endnotes

15.1.1 Bibliography

Load Patrick Daly's `natbib` package, ftp://ctan.tug.org/macros/latex/contrib/supported/natbib

Note that `natbib` assumes that it loads over a document class, such as the `article` class, that has already defined `thebibliography` and `\@listi`.

Note also that `natbib` also installs a command `\NAT@set@cites` into `\AtBeginDocument` which presumes that the proper `\bibpunct` command has been issued.

We have a problem with `natbib` in that it does not accomodate late binding of its package options. This means that those options must be known at the time the package loads. Therefore, we select `sort&compress` now, and thereby foreclose on alternative uses of `natbib`.

Certain societies may prefer an alternative; they are out of luck. If you find yourself in this position, please let the author of `natbib` know that his package should be modified to accomodate late binding of package options.

To remedy the situation, note that the macro `\NAT@sort` controls whether citations are left alone (`\NAT@sort`=0), sorted (`\NAT@sort`=1), or sorted and compressed (`\NAT@sort`=2). Since we give `natbib` the `sort&compress` option, if you prefer `sort`, you need only define `\NAT@sort` to be 1. However, if you prefer the effect of having neither `sort` nor `sort&compress`, you must define `\NAT@sort` to be 0 and you must also define `\NAT@sort@cites`:

```
%\renewcommand\NAT@sort@cites[1]{\edef\NAT@cite@list{#1}}%
%
```

For other `natbib` customizations, you may proceed as if you were going to use the `natbib.cfg` file: anything that you can modify by this means is fair game. Once REVTEX is finished loading, you can assert any definitions for `natbib` that you wish.

```
\newenvironment{thebibliography}{}{}%
```

```
\let\@listi\@empty
\RequirePackage[sort&compress]{natbib}[1999/05/28 7.0]%
\let\NAT@thebibliography\thebibliography
\let\NAT@endthebibliography\endthebibliography
\let\bibliographystyle@latex\bibliographystyle
```

`\NAT@citesuper` We redefine `natbib`'s procedure `\NAT@citesuper`, which is executed when setting a superscript citation. The `\hspace` is removed: in any case, it should really be `\hspace*`, to prevent an unwanted pagebreak.

```
\renewcommand\NAT@citesuper[3]{%
 \ifNAT@swa
  \leavevmode
  \unskip
% \hspace{1\p@}%
  \textsuperscript{#1}%
  \if*#3*\else\ (#3)\fi
 \else
  #1%
 \fi
 \endgroup
}%
```

`\@makefnmark@cite` We define a procedure that will set a footnote mark the same way that a citation is set. If footnotes are put in the bibliography with `\footinbib@sw`, then the corresponding mark should look the same as the result of a `\cite`. This is how we do it.

```
\def\@makefnmark@cite{\begingroup\NAT@swatrue\@cite{{\@thefnmark}}{}{}}%
```

`\bibliographystyle` We arrange for the selection of bibliography style to occur either due to the document's explicit `\bibliographystyle` statement or via the journal substyle.

Note that REVTEX is now incompatible with any package that patches `\bibliographystyle`, except for those (like `natbib`) that load before here.

```
\let\bibliographystyle@latex\bibliographystyle
\def\bibliographystyle{\def\@bibstyle}%
```

`\bibsection` `\NAT@bibsetnum` `\NAT@bibsetup` `\bibpreamble` `\newblock` `\bibnumfmt` We define the sectioning command to use when starting the bibliography and gently coax `natbib` into using the formatting procedures that *we* want it to use.

This way of setting up `thebibliography` automatically sets the label width based on the largest number used within the bibliography. This scheme will not work properly using the author/year style of bib entry, though.

We define `\bibnumfmt` to be `\place@bibnumber`, which is a macro managed by REVTEX. If the document defines `\bibnumfmt`, then that definition will be used instead, which is what the `natbib` package gives as its programming interface.

And if the substyle does not give a meaning to `\place@bibnumber`, we give it that of the original `\bibnumfmt`.

```
\def\bibsection{%
 \@ifx@empty\refname{%
  \par
```

```
 }{%
  \let\@hangfroms@section\@hang@froms
  \section*{\refname}%
  \@nobreaktrue
 }%
}%
\let\NATx@bibsetnum\NAT@bibsetnum
\def\NAT@bibsetnum#1{%
 \setlength{\topsep}{\z@}%
 \NATx@bibsetnum{\ref{LastBibItem}}%
}%
\let\NATx@bibsetup\NAT@bibsetup
\def\NAT@bibsetup{%
 \setlength{\labelwidth}{\z@}%
 \setlength{\labelsep}{\z@}%
 \setlength{\itemindent}{\z@}%
 \setlength{\listparindent}{\z@}%
 \setlength{\topsep}{\z@}%
 \setlength{\parsep}{\z@}%
 \NATx@bibsetup
}%
\let\bibpreamble\@empty
\def\newblock{\ }%
\let\NATx@bibnumfmt\bibnumfmt
\def\bibnumfmt{\place@bibnumber}%
\AtEndOfClass{%
 \appdef\class@documenthook{%
  \@ifxundefined\place@bibnumber{%
   \let\place@bibnumber\NATx@bibnumfmt
  }{}%
 }%
}%
```

`\NAT@onlinecite` We anticipate using numerical citations. If superscript citations are used, we nonetheless need a way to use numerical citations as required by the author. These should be accessible via the `\citet` command.

Therefore we remember how to do a numerical citation even when the superscript citation has been selected.

Can you say "inelegant hack"? I knew you could!

```
\let\NAT@citeyear\citeyear
\def\NAT@onlinecite#1{%
 \begingroup
  \let\@cite\NAT@citenum
  \let\NAT@mbox\mbox
  \let\citeyear\NAT@citeyear
  \def\NAT@space{ }%
  \cite{#1}%
 \endgroup
}%
```

+ We extend natbib's syntax with two commands to set a citation on the baseline
\textcite (as opposed to superscripted) and as text (rather than parenthetical), respectively. A journal substyle that makes citations be superscripted or parenthetical as the case may be, should ensure that the author has continued access to these two styles.

```
\appdef\class@inithook{%
 \@ifxundefined\@onlinecite{\def\@onlinecite{\citealp}}{}%
 \@ifxundefined\@textcite{\def\@textcite{\citet}}{}%
}%
\DeclareRobustCommand\onlinecite{\@onlinecite}%
\DeclareRobustCommand\textcite{\@textcite}%
```

thebibliography We put a tail patch into \thebibliogrphy and a headpatch into \endthebibliography.
\present@bibnote We also apply a fix to the \list procedure so that, at least within the bibliography, items are not formatted via \parshape, which caused problems during column balancing. (Other list-like environments may also require fixing, but we will save that for another day.)

Here we provide a default treatment for frontmatter notes deferred to the bibliography; a journal substyle might want to override the definition of \present@bibnote.

```
\renewenvironment{thebibliography}[1]{%
 \NAT@thebibliography{#1}%
 \@clubpenalty\clubpenalty
 \let\@TBN@opr\present@bibnote
 \@FMN@list
}{%
 \@endnotesinbib
 \edef\@currentlabel{\arabic{NAT@ctr}}%
 \label{LastBibItem}%
 \NAT@endthebibliography
 \global\let\auto@bib\@empty
}%
\def\present@bibnote#1#2{%
 \item[%
  \textsuperscript{%
   \normalfont
   \Hy@raisedlink{\hyper@anchorstart{frontmatter@#1}\hyper@anchorend}%
   \begingroup
    \c@footnote#1\relax
    \frontmatter@thefootnote
   \endgroup
  }%
 ]#2\par
}%
```

\place@bibnumber@sup For producing the \bibitem device, we define two procedures to select from.
\place@bibnumber@inl Note that we could have used natbib's \ifNAT@super switch, but it does not allow for altering the meaning of \bibnumfmt.

```
\def\place@bibnumber@sup#1{\textsuperscript{#1}}%
\def\place@bibnumber@inl#1{[#1]}%
```

\@cite
\nocite
write@bibliographystyle

We wish to delay committing the \bibliographystyle until as late as possible. The journal substyle will define a default bibliography style, and the document's explicit \bibliographystyle command, if any, will override that default. For the sake of BibTEX, we must execute the \bibliographystyle command (putting a \bibstyle command into the auxiliary file) somewhere before the first \cite command (or any other like command), which puts a \citation command into the auxiliary file.

Therefore, we headpatch the two procedures in natbib that can result in the appearance in the auxiliary file of a \citation command. These two procedures are \@cite and \nocite.

Not!

The \bibstyle command may appear quite late in the revtex4.dtx.aux file. We now delay the automatic writing of the \bibstyle command to the end of the job.

The procedure \write@bibliographystyle tests whether a \bibliographystyle command has already been given. If not, it effectively executes the needed \bibliographystyle command, then neutralizes itself (we only need to do this once per job).

```
%\AtBeginDocument{%
% \let\natbib@@cite\@cite
% \def\@cite{\write@bibliographystyle\natbib@@cite}%
% \let\natbib@nocite\nocite
% \def\nocite{\write@bibliographystyle\natbib@nocite}%
%}%
\def\write@bibliographystyle{%
 \@ifxundefined\@bibstyle{}{%
  \expandafter\bibliographystyle@latex\expandafter{\@bibstyle}%
 }%
 \global\let\write@bibliographystyle\relax
}%
\AtEndDocument{\write@bibliographystyle}%
```

\super@cite
\super@cite@

The \super@cite procedure takes the place of \@citex and implements the feature of the citeautoscript class option. This means that the document should be marked up as if citations were *not* superscripted, and then if you select a journal substyle that has superscripted citations, REVTEX will do its best to alter the formatting of the \cites to accomodate superscript style.

Note that the commands \onlinecite and \textcite are not affected by this procedure.

Here's a subtle one: when is the argument of \super@cite@@ not the same as the token \@let@token? Answer: when the latter is \@sptoken! This case has to be handled separately.

```
\def\super@cite[#1][#2]#3{%
 \leavevmode
```

```
 \skip@\lastskip
 \unskip
 \begingroup
  \def\@tempa{[#1][#2]{#3}}%
  \super@cite@
}%
\def\super@cite@{%
 \futurelet\@let@token\super@cite@@
}%
\long\def\super@cite@@#1{%
 \@ifx{\@let@token\@sptoken}{%
  \aftergroup\@sptoken\false@sw
 }{%
  \@ifx{\@let@token.}{\true@sw}{%
   \@ifx{\@let@token,}{\true@sw}{%
    \@ifx{\@let@token'}{\true@sw}{%
     \@ifx{\@let@token)}{\true@sw}{%
      \false@sw
     }%
    }%
   }%
  }%
 }%
 {%
  #1%
  \super@cite@
 }{%
  \expandafter\@citex@NAT\@tempa\endgroup
  #1%
 }%
}%
\AtBeginDocument{%
 \citeautoscript@sw{%
  \@ifx{\place@bibnumber\place@bibnumber@sup}{%
   \let\@citex@NAT\@citex\let\@citex\super@cite
  }{}%
 }{}%
}%
```

Resolve an incompatability between `natbib` and `listings`. The latter package tests `\chapter`(which has now been `\let` to `\relax` as a side effect `natbib`'s use of LaTeX's `\@ifundefined`).

We couch our fix in such terms that will not be disruptive if `\chapter` is actually defined at this point.

```
\@ifx{\chapter\relax}{\let\chapter\@undefined}{}%
```

15.1.2 `\endnotes` and `\rtx@bibnotes`

FIXME: check hyperref compatibility!

\mini@note
\save@note
```
\def\mini@note{\save@note\mini@notes}%Implicit #1
\def\save@note#1#2{%
  \stepcounter{footnote}%
  \protected@xdef\@thefnmark{\the\c@footnote}%
  \@footnotemark
  \expandafter\g@addto@macro
  \expandafter#1%
  \expandafter{%
  \expandafter \@@footnotetext
  \expandafter {\@thefnmark}{#2}%
               }%
}%
\long\def\@@footnotetext#1{\def\@thefnmark{#1}\@footnotetext}%
\let\mini@notes\@empty
```

\endnote A version of footnote that appears at the end of the document. Actually it just appears where \printendnotes appears.

```
\def\endnote{%
 \begingroup
  \aftergroup\@footnotemark
  \aftergroup\@endnotetext
  \@ifnextchar[{%
   \@xendnote
  }{%
   \stepcounter{footnote}%
   \protected@xdef\@tempa{\thefootnote}%
   \expandafter\@xendnote\expandafter[\@tempa]%
 }%
}%
```

\@xendnote

```
\def\@xendnote[#1]{%
  \begingroup
   \c@footnote#1\relax
   \unrestored@protected@xdef\@thefnmark{endnote\thefootnote}%
  \endgroup
 \endgroup
}%
\def\@endnotemark{%
 \expandafter\cite\expandafter{\@thefnmark}%
}%
```

\endnote@ext

```
\def\endnote@ext{.end}%
```

\@endnotetext The commands \label, \index, and \glossary, which are robustified for \markright and \addcontentsline, are likewise robustified here.

Note the similarity to \protected@write.

```
\long\def\@endnotetext#1{%
  \@ifxundefined\@endnoteout{%
    \newwrite\@endnoteout
    \gdef\endnote@stream{\jobname\endnote@ext}%
    \immediate\openout\@endnoteout\endnote@stream\relax
  }{}%
  \begingroup
    \let\label\relax \let\index\relax \let\glossary\relax
    \let\cite \relax \let\ref  \relax \let\pageref \relax
    \let\(    \relax \let\)    \relax \let\\    \relax
%   \let\protect\noexpand
    \let \protect \@unexpandable@protect
    \newlinechar`\^^M%
%   \newlinechar`\ %
    \immediate\write\@endnoteout{\string\@doendnote{\@thefnmark}{#1}}%
  \endgroup
}%
```

\endnotetext

```
\def\endnotetext{%
  \@ifnextchar[{%
   \@xendnotenext
  }{%
   \protected@xdef\@thefnmark{\thefootnote}%
   \@endnotetext
  }%
}%
```

\@xendnotenext

```
\def\@xendnotenext[#1]{%
  \begingroup
     \c@footnote#1\relax
     \unrestored@protected@xdef\@thefnmark{\thefootnote}%
  \endgroup
  \@endnotetext
}%
```

\printendnotes

```
\def\printendnotes{%
 \@ifx{\@endnoteout\@undefined}{}{%
  \begingroup
   \c@secnumdepth-\maxdimen
   \section{%
    \notesname
   }%
   \immediate\closeout\@endnoteout
   \global\let\@endnoteout\@undefined
   \begin{enumerate}%
   \makeatletter
```

```
    \input{\endnote@stream}%
    \end{enumerate}%
   \endgroup
  }%
}%
\appdef\class@enddocumenthook{%
  \printendnotes
}%
```

`\firstnote@num` `\set@firstnote` In REVTEX, we have permanently altered the way that the footnote counter gets reset at the beginning of the document.

If footnotes appear in the bibliography, we initialise the footnote counter to number of cites (found last time) via `\set@firstnote`.

```
\mathchardef\firstnote@num\z@
\appdef\class@documenthook{%
  \global\c@footnote\firstnote@num
  \appdef\maketitle{\global\c@footnote\firstnote@num}%
}%
\def\set@firstnote#1{%
  \@ifnum{\firstnote@num=#1\relax}{}{%
   \class@warn@end{Endnote numbers changed: rerun LaTeX}%
  }%
  \immediate\write\@mainaux{%
    \global\chardef\string\firstnote@num#1\relax
  }%
}%
```

`\@endnotesinbib` `\@endnotesinbibliography` We define a function `\@endnotesinbib`, and a variant `\@endnotesinbibliography`. The former is invoked at the start of the end processing for `\end{thebibliography}`; the latter is a synonym.

The procedure typesets the footnotes that are to appear in the bibliography; the default is to simply arrange for the footnote counter to be reset at the start of the document.

The switchover to setting footnotes in the bibliography is done via `\AtBeginDocument` code that changes the meaning of `\footnote` and that substitutes the synonym for `\@endnotesinbib`.

Note that this code make the assumption that the counter used in `thebibliography` is `\c@NAT@ctr`.

```
\def\@endnotesinbib{%
  \set@firstnote\z@
}%
\def\@endnotesinbibliography{%
  \expandafter\set@firstnote\expandafter{\the\c@NAT@ctr}%
  \@ifx{\@endnoteout\@undefined}{}{%
   \immediate\closeout\@endnoteout
   \global\let\@endnoteout\@undefined
   \makeatletter
   \input{\endnote@stream}%
```

```
 }%
}%
\appdef\class@documenthook{%
 \let\@makefnmark@latex\@makefnmark
 \let\@footnotemark@latex\@footnotemark
 \footinbib@sw{%
  \let\footnote\endnote
  \let\@footnotemark\@endnotemark
  \let\@makefnmark\@makefnmark@cite
  \let\@endnotesinbib\@endnotesinbibliography
  \appdef\class@enddocumenthook{\auto@bib}%
  \let\printendnotes\relax
 }{}%
}%
```

`\auto@bib` Under some circumstances, we must typeset the bibliography automatically. For instance, the document specifies footnotes in the bibliography, but contains no bibliography.

Note that this facility is not able to work more than once per document. If multiple bibliographys are required (e.g., per article), it will be the responsibility of the journal style to restore `\auto@bib` to its original meaning so it can be re-invoked.

```
\def\auto@bib{%
 \@ifx@empty\@FMN@list{%
  \@ifxundefined\@endnoteout{%
   \false@sw
  }{%
   \true@sw
  }%
 }{%
  \true@sw
 }{%
  \begin{thebibliography}{}\end{thebibliography}%
 }%
}%
```

`\@doendnote`

```
\def\@doendnote#1#2{\bibitem{#1}#2}
```

16 Initial setup

Note that this code is executed at `\AtBeginDocument` time in order to handle a particular timing problem. Please do not move this code any earlier!

```
\appdef\class@documenthook{%
```

We assert our own page style only if nobody else has done so. Users wishing to customize their documents will be able to invoke a `\pagestyle` command anywhere in the preamble.

```
\@ifxundefined\@evenhead{%
\@ifxundefined\@oddhead{%
\@ifxundefined\@oddfoot{%
\@ifxundefined\@evenfoot{%
\preprintsty@sw{%
\ps@preprint
}{%
\ps@article
}%
}{}%
}{}%
}{}%
}{}%
 \pagenumbering{arabic}%
 \normalsize
 \tabbingsep \labelsep
 \leftmargin\leftmargini
 \labelwidth\leftmargin\advance\labelwidth-\labelsep
 \let\@listi\@listI
 \@listi
%\twoside@sw{}{%
% \raggedbottom
%}%
```

We make sure that the "environment" component mark is set.

```
 \def\curr@envir{document}%
 \mark@envir{\curr@envir}%
```

When setting the column grid, we have to alter the procedure for formatting lists. Because `\twocolumngrid` requires rebalancing columns at some points, lists must *not* involve the use of `\moveleft`, `\moveright`, or `\parshape`. The alternative procedure only involves the manipulation of `\leftskip` and `\rightskip`.

FIXME: Always sloppy??

```
 \twocolumn@sw{%
  \open@column@mlt\tw@
  \set@colht
  \@floatplacement
  \@dblfloatplacement
  \sloppy
  \@booleanfalse\raggedcolumn@sw
  \let\set@listindent\set@listindent@
 }{%
  \open@column@one\@ne
  \set@colht
  \@floatplacement
  \@dblfloatplacement
  \sloppy
% \raggedbottom
  \let\set@listindent\set@listindent@parshape
 }%
```

```
}%
```

17 \appendix

```
%\newif\ifappendixon
```

Note that, within appendices, equations are numbered within sections (appendices).

```
\def\appendix{%
 \par
%\appendixontrue
 \setcounter{section}\z@
 \setcounter{subsection}\z@
 \setcounter{subsubsection}\z@
 \def\thesubsection{\arabic{subsection}}%
 \def\thesubsubsection{\alph{subsubsection}}%
 \@addtoreset{equation}{section}%
 \def\theequation@prefix{\thesection}%
 \addtocontents{toc}{\protect\appendix}%
 \@ifstar{%
  \def\thesection{\unskip}%
  \def\theequation@prefix{A.}%
 }{%
  \def\thesection{\Alph{section}}%
 }%
}%
```

18 Changing the page grid

18.1 Avoiding Grid Changes

In preprint styles, "wide text" is a no-op, and the title page processing involves no grid change.

`\title@column@default`
`\close@column@default`

```
\def\title@column@default#1{%
 \minipagefootnote@init
 #1%
 \minipagefootnote@foot
}%
\def\close@column@default{%
 \newpage
}%
```

18.2 Galley Style: Margin Changes

A variant of preprint processing. Emulate journal appearance somewhat.

`widetext@galley` DPC: We're in galley style so do a lob sided display environment.

QUERY: How can we be sure that we are in galley style? ANSWER: as noted elsewhere, require both `\twocolumn@sw` and `\preprintsty@sw` to be false.

```
\def\galley@outdent{\rightmargin-\columnwidth\advance\rightmargin-\columnsep}%
\let\widetext@outdent\@empty
\newenvironment{widetext@galley}{%
  \list{}{%
    \topsep        \z@skip
    \listparindent \parindent
    \itemindent    \parindent
    \leftmargin    \z@
    \parsep        \z@\@plus\p@
    \widetext@outdent
    \relax
  }%
  \item\relax
}{
  \endlist
}%
```

18.3 Grid Changing Via `ltxgrid`

In case `twocolumngrid` has been invoked, switch column grid using the column grid-changing commands. Supply stub definitions of those commands here.

`\title@column@grid`
`\close@column@grid`

The title block always starts at the top of a new page.

Note that, for the procedure `\close@column@grid`, we balance columns by switching to the one-column page grid, then we return to a two-column page grid in anticipation of the upcoming `\clearpage`.

```
\def\title@column@grid#1{%
 \minipagefootnote@init
  \onecolumngrid
  \begingroup
   \let\@footnotetext\frontmatter@footnotetext
   \let\set@footnotewidth\set@footnotewidth@ii
   #1%
  \endgroup
  \twocolumngrid
 \minipagefootnote@foot
}%
\def\close@column@grid{%
 \balancelastpage@sw{%
  \onecolumngrid
% \twocolumngrid
 }{}%
}%
```

`widetext@grid`

We slip into the one-column page grid within the scope of this environment.

Note that we set adornments above and below the `widettext`. These are set as leaders, so they will disappear at a page break.

```
\newenvironment{widetext@grid}{%
  \par\ignorespaces
  \setbox\widetext@top\vbox{%
   \vskip15\p@
   \hb@xt@\hsize{%
    \leaders\hrule\hfil
    \vrule\@height6\p@
   }%
   \vskip6\p@
  }%
  \setbox\widetext@bot\hb@xt@\hsize{%
    \vrule\@depth6\p@
    \leaders\hrule\hfil
  }%
  \onecolumngrid
  \dimen@\ht\widetext@top\advance\dimen@\dp\widetext@top
  \cleaders\box\widetext@top\vskip\dimen@
  \let\set@footnotewidth\set@footnotewidth@ii
}{%
  \par
  \setbox\widetext@bot\vbox{%
   \hb@xt@\hsize{\hfil\box\widetext@bot}%
   \vskip14\p@
  }%
  \dimen@\ht\widetext@bot\advance\dimen@\dp\widetext@bot
  \cleaders\box\widetext@bot\vskip\dimen@
  \twocolumngrid\global\@ignoretrue
  \@endpetrue
}%
\newbox\widetext@top
\newbox\widetext@bot
```

Decide, finally, how the page grid is to be manipulated.

```
\appdef\class@documenthook{%
 \twocolumn@sw{%
  \let@environment{widetext}{widetext@grid}%
  \let\title@column\title@column@grid
  \let\close@column\close@column@grid
 }{%
  \let@environment{widetext}{widetext@galley}%
  \preprintsty@sw{%
```

Change the page grid not at all.

```
  }{%
```

If we are galley style, change the page margin only.

```
   \galley@sw{%
    \let\widetext@outdent\galley@outdent
   }{%
   }%
```

```
  }%
 }%
}%
```

`\title@column` Provide a default meaning for `\title@column` in case it was never defined.

```
\appdef\class@inithook{%
 \@ifxundefined\title@column{%
  \let\title@column\title@column@default
 }{}%
}%
```

19 Old font commands

```
\DeclareOldFontCommand{\rm}{\normalfont\rmfamily}{\mathrm}
\DeclareOldFontCommand{\sf}{\normalfont\sffamily}{\mathsf}
\DeclareOldFontCommand{\tt}{\normalfont\ttfamily}{\mathtt}
\DeclareOldFontCommand{\bf}{\normalfont\bfseries}{\mathbf}
\DeclareOldFontCommand{\it}{\normalfont\itshape}{\mathit}
\DeclareOldFontCommand{\sl}{\normalfont\slshape}{\@nomath\sl}
\DeclareOldFontCommand{\sc}{\normalfont\scshape}{\@nomath\sc}
\DeclareRobustCommand*\cal{\@fontswitch\relax\mathcal}
\DeclareRobustCommand*\mit{\@fontswitch\relax\mathnormal}
```

20 English-Language Texts

As this class is just for English language journals, we could hardwire these texts, but to make it easier to use this as a basis for the code for similar journal styles, separate out all the fixed text strings into babel-style macros of the form `\...name`

Note: for babel compatability, use version 1999/05/05 v3.6x or later.
Some of these might need changing in the society-specific code.

`\today` Procedure `\today` is used in the article class, but not in this document class.

```
\def\today{\ifcase\month\or
  January\or February\or March\or April\or May\or June\or
  July\or August\or September\or October\or November\or December\fi
  \space\number\day, \number\year}
```

`\notesname` Text entity `\notesname` is used in `\printendnotes`.

```
\def\notesname{Notes}
```

`\partname` Text entity `\partname` is used in `\@part`.

```
\def\partname{Part}
```

`\tocname` Text entity `\tocname` is used in `\tableofcontents`, as defined in the standard LaTeX book class.

```
\def\tocname{Contents}
```

`\lofname` Text entity `\lofname` is used in `\listoffigures`, as defined in the standard LaTeX book class.

```
\def\lofname{List of Figures}
```

`\lotname` Text entity `\lotname` is used in `\listoftables`, as defined in the standard LaTeX book class.

```
\def\lotname{List of Tables}
```

`\refname` Text entity `\refname` is used in `thebibliography`.

```
\def\refname{References}
```

`\indexname` Text entity `\indexname` is used in `theindex`, as defined in the standard LaTeX book class.

```
\def\indexname{Index}
```

`\figurename` Text entity `\figurename` is used in `figure`, `\figuresname` in `\printfigures`.

```
\def\figurename{FIG.}
\def\figuresname{Figures}%
```

`\tablename` Text entity `\tablename` is used in `table`, `\tablesname` in `\printtables`.

```
\def\tablename{TABLE}
\def\tablesname{Tables}%
```

`\abstractname` Text entity `\abstractname` is used in `abstract`.

```
\def\abstractname{Abstract}
```

`\appendixesname` `\appendixname` Text entity `\appendixesname` is used in TOC.

```
\def\appendixesname{Appendixes}%
\def\appendixname{Appendix}%
```

`\acknowledgmentsname` Text entity `\acknowledgmentsname` is used in `acknowledgments`.

```
\def\acknowledgmentsname{Acknowledgments}
```

`\journalname` This should be set by the society journal options, eg 'pra'.

```
\def\journalname{??}
```

`\copyrightname` Default layout does not assign copyright, but a journal that wants to might use this.

```
\def\copyrightname{??}
```

`\andname` The text string "and" for use in author lists.

```
\def\andname{and}
```

`\@pacs@name` `\@keys@name` The text string prepended to PACS numbers, resp. to keywords.

```
\def\@pacs@name{PACS numbers: }%
\def\@keys@name{Keywords: }%
```

\ppname The text string “pp” for use in page ranges.

```
\def\ppname{pp}
```

\numbername The text string “number” for use in article reference.

```
\def\numbername{number}
```

\volumename The text string “volume” for use in article reference.

```
\def\volumename{volume}
```

\Dated@name These texts are used in the \date, et al. commands.
\Received@name
\Revised@name
\Accepted@name
\Published@name

```
\def\Dated@name{Dated: }%
\def\Received@name{Received }%
\def\Revised@name{Revised }%
\def\Accepted@name{Accepted }%
\def\Published@name{Published }%
```

21 Legacy Commands

We deffine some commands left over from version 3.1, or give default meanings. Some definitions can be overridden in the document preamble or in included packages.

Note on the namespace: command names like \REV@*name* are used here, because it is not clear that any of this code is generally useful.

```
\def\address{\replace@command\address\affiliation}%
\def\altaddress{\replace@command\altaddress\altaffiliation}%
\newenvironment{references}{%
 \class@warn@end{The references environment is not supported; use thebibliography instead.}
 \gdef\references{\thebibliography{}}\references
}{%
 \endthebibliography
}%
\def\draft{%
 \class@warn@end{Command \string\draft\space is obsolete;^^JInvoke option draft instead.}%
 \@booleantrue\draft@sw
}%
\def\tighten{%
 \class@warn@end{Command \string\tighten\space is obsolete;^^JInvoke option tightenlines instea
 \@booleantrue\tightenlines@sw
}%
\def\tableline{%
 \noalign{%
  \class@warn@end{Command \string\tableline\space is obsolete;^^JUse \string\colrule\space inst
  \global\let\tableline\colrule
 }%
 \tableline
}%
\def\case{\replace@command\case\frac}%
```

```
\def\slantfrac{\replace@command\slantfrac\frac}%
\def\tablenote{\replace@command\tablenote\footnote}%
\def\tablenotemark{\replace@command\tablenotemark\footnotemark}%
\def\tablenotetext{\replace@command\tablenotetext\footnotetext}%
% Lose the following definition:
\DeclareRobustCommand\REV@text[1]{%
 \relax
 \ifmmode
  \mathchoice
   {\hbox{{\everymath{\displaystyle     }#1}}}%
   {\hbox{{\everymath{\textstyle        }#1}}}%
   {\hbox{{\everymath{\scriptstyle      }\let\f@size\sf@size\selectfont#1}}}%
   {\hbox{{\everymath{\scriptscriptstyle}\let\f@size\ssf@size\selectfont#1}}}%
  \glb@settings
 \else
  \mbox{#1}%
 \fi
}%
% Lose the following definition:
\DeclareRobustCommand\REV@bbox[1]{%
 \relax
 \ifmmode
  \mathchoice
   {\hbox{{\everymath{\displaystyle     }\boldmath$#1$}}}%
   {\hbox{{\everymath{\textstyle        }\boldmath$#1$}}}%
   {\hbox{{\everymath{\scriptstyle      }\boldmath$#1$}}}%
   {\hbox{{\everymath{\scriptscriptstyle}\boldmath$#1$}}}%
  \glb@settings
 \else
  \mbox{#1}%
 \fi
}%
\DeclareRobustCommand\REV@bm[1]{%
 \class@warn@end{To use \string\bm, please load the bm package!}%
 \global\let\bm\relax
}%
\def\FL{\obsolete@command\FL}%
\def\FR{\obsolete@command\FR}%
\def\narrowtext{\obsolete@command\narrowtext}%
\def\mediumtext{\obsolete@command\mediumtext}%
\newenvironment{quasitable}{%
 \let@environment{tabular}{longtable}%
}{%
}%
```

\text If not otherwise defined, give default meanings to certain commands.

\bm
\bibinfo
\eprint
\url

```
\appdef\class@inithook{%
 \@ifxundefined\text{\let\text\REV@text}{}%
 \@ifxundefined\bm{\let\bm\REV@bm}{}%
 \providecommand\bibinfo[2]{#2}%
```

```
 \providecommand\eprint[2][]{#2}%
 \providecommand\url[1]{#1}%
}%
```

\bbox

```
\def\bbox#1{%
 \class@warn@end{\string\bbox\space is obsolete,^^Jload the bm package and use \string\bm\space
 \global\let\bbox\relax
}%
```

\mathletters

```
\newenvironment{mathletters}{%
 \class@warn@end{Environment {mathletters} is obsolete;^^Jload the amsmath package and use {sub
 \global\let\mathletters\@empty
}{%
}%
```

\eqnum

```
\def\eqnum#1{%
 \class@warn@end{\string\eqnum\space is obsolete, load the amsmath package and use \string\tag!
 \global\let\eqnum\@gobble
}%
```

We read in the symbol definitions.

```
\RequirePackage{revsymb}%
\appdef\class@inithook{\revsymb@inithook}%
```

22 Endgame for the Document Class

This section contains definitions related to the end-of-class processing: we want to control exactly what happens next.

We provide for a "job macro package" that can override definitions made by the class or any other packages it loads. We act on the value of `\secnumarabic@sw`; this code is a model for dealing with the job macro package. We also install a the very last procedures into `\AtEndOfClass`, ones that need to tail on to any processing performed either by the class, by any of its included packages, or by the job macro package and any packages it may have loaded.

22.1 Job Macro Package

You can create a "job macro package" for your document that will be read in automatically every time your document is processed. Thus, if your job is a file called `myarticle.tex`, then the file `myarticle.rty` will be read in just the same as if you had placed a `\usepackage{myarticle.rty}` statement immediately following your `\documentclass` statement.

Within your `.rty` file, you can define and use control sequence names that use the `@` character and you can override any of the definitions or assignments made

by the REVTEX document class or the selected journal substyle. That is, you have the power to really mess things up badly.

If you choose to have a job macro package, you are well advised to read the LATEX guide to document classes, revtex4.dtxclsguide.tex or read up on the subject in a book line the LATEX Companion.

The file revtex4.dtxtemplate.rty contains a template for creating your own job macro package.

```
\AtEndOfClass{%
 \InputIfFileExists{\jobname.rty}{}{}%
}%
```

22.2 \secnumarabic@sw

\secnumarabic@sw The flag \secnumarabic@sw signifies that sectioning commands are to be numbered arabic. The effect of this command should override any settings made by the journal substyle or by the job macro package. However, it should be lower in precedence to definitions appearing in the document preamble. Therefore, we install its code into \AtEndOfClass, but do it *after* the corresponding code for the journal substyle and the job macro package.

Note: \thesection must *not* be redefined, say, at \AtBeginDocument time by some delayed code.

```
\AtEndOfClass{%
 \@ifxundefined\secnumarabic@sw{\@booleanfalse\secnumarabic@sw}{}%
 \secnumarabic@sw{%
  \def\p@section        {}%
  \def\thesection       {\arabic{section}}%
  \def\p@subsection     {}%
  \def\thesubsection    {\thesection.\arabic{subsection}}%
  \def\p@subsubsection  {}%
  \def\thesubsubsection {\thesubsection.\arabic{subsubsection}}%
 }{}%
}%
```

22.3 Hook for default values of Booleans

This portion of the code for this class file *must* appear at the very end.

\defaults@hook The procedure \defaults@hook should hereby be the very last to be installed into \AtBeginDocument. We provide it with a default in case it has not been used up til now.

CHECKME: is any other hook installed later than \defaults@hook?

```
\AtEndOfClass{%
 \appdef\class@documenthook{%
  \defaults@hook
 }%
}%
\@ifxundefined\defaults@hook{\let\defaults@hook\@empty}{}%
```

End of the class file.

```
%</class-pst>
```

23 Symbols: the revsymb module

We immediately define a utility command: this module's warning.

```
%<*revsymb>
\def\REVSYMB@warn#1{\PackageWarningNoLine{revsymb}{#1}}%
```

\lambdabar

```
\DeclareRobustCommand\lambdabar{%
  \bgroup
    \def\@tempa{%
      \hbox{%
        \raise.73\ht\z@
        \hb@xt@\z@{%
          \kern.25\wd\z@
          \vrule \@width.5\wd\z@\@height.1\p@\@depth.1\p@
          \hss
        }%
        \box\z@
      }%
    }%
    \mathchoice
      {\setbox\z@\hbox{$\displaystyle     \lambda$}\@tempa}%
      {\setbox\z@\hbox{$\textstyle        \lambda$}\@tempa}%
      {\setbox\z@\hbox{$\scriptstyle      \lambda$}\@tempa}%
      {\setbox\z@\hbox{$\scriptscriptstyle\lambda$}\@tempa}%
  \egroup
}%
```

\openone DPC: Really should use a font that includes this glyph. Unfortunately not in AMS ones, but is in bbold, cmbb. (I think, must check), FIXME: check for bbold.

```
\DeclareRobustCommand\openone{\leavevmode\hbox{\small1\normalsize\kern-.33em1}}%
```

\corresponds \overdots \overcirc Jörg Knappen suggests the replacements: replace \corresponds with \triangleq, source amssymb; replace \overcirc with \mathring, source latex2e; replace \overdots with \dddot, source amsmath.

Any use of any of these commands will result in a warning message at the end of the log file. If the corresponding package is not loaded, a definition will quietly be provided.

```
\DeclareRobustCommand\corresponds{\replace@command\corresponds\triangleq}%
\DeclareRobustCommand\overcirc{\replace@command\overcirc\mathring}%
\DeclareRobustCommand\overdots{\replace@command\overdots\dddot}%
\DeclareRobustCommand\REV@triangleq{%
 {\lower.2ex\hbox{=}}{\kern-.75em^\triangle}%
}%
```

```
\DeclareRobustCommand\REV@dddot[1]{%
 \@ontopof{#1}{\cdots}{1.0}\mathord{\box2}%
}%
```

\succsim \precsim \lesssim \gtrsim \alt \agt

These version 3.1 commands are always supplied, but the definitions in amssymb are preferred.

```
\DeclareRobustCommand\altsuccsim{\succ\kern-.9em_\sim\kern.3em}%
\DeclareRobustCommand\altprecsim{\prec\kern-1em_\sim\kern.3em}%
\let\REV@succsim\altsuccsim
\let\REV@precsim\altprecsim
\DeclareRobustCommand\REV@lesssim{\mathrel{\mathpalette\vereq{<}}}%
\DeclareRobustCommand\REV@gtrsim{\mathrel{\mathpalette\vereq{>}}}%
\DeclareRobustCommand\alt{\lesssim}
\DeclareRobustCommand\agt{\gtrsim}
\def\vereq#1#2{%
 \lower3\p@\vbox{%
  \baselineskip1.5\p@
  \lineskip1.5\p@
  \ialign{$\m@th#1\hfill##\hfil$\crcr#2\crcr\sim\crcr}%
 }%
}%
```

\tensor \overstar \loarrow \roarrow

```
\DeclareRobustCommand\tensor[1]{\@ontopof{#1}{\leftrightarrow}{1.15}\mathord{\box2}}
\DeclareRobustCommand\overstar[1]{\@ontopof{#1}{\ast}{1.15}\mathord{\box2}}
\DeclareRobustCommand\loarrow[1]{\@ontopof{#1}{\leftarrow}{1.15}\mathord{\box2}}
\DeclareRobustCommand\roarrow[1]{\@ontopof{#1}{\rightarrow}{1.15}\mathord{\box2}}
```

\@ontopof

```
\def\@ontopof#1#2#3{%
 {%
  \mathchoice
    {\@@ontopof{#1}{#2}{#3}\displaystyle     \scriptstyle      }%
    {\@@ontopof{#1}{#2}{#3}\textstyle        \scriptstyle      }%
    {\@@ontopof{#1}{#2}{#3}\scriptstyle      \scriptscriptstyle}%
    {\@@ontopof{#1}{#2}{#3}\scriptscriptstyle\scriptscriptstyle}%
 }%
}%
```

\@@ontopof Same as REVTEX3, more or less.

```
\def\@@ontopof#1#2#3#4#5{%
  \setbox\z@\hbox{$#4#1$}%
  \setbox\f@ur\hbox{$#5#2$}%
  \setbox\tw@\null\ht\tw@\ht\z@ \dp\tw@\dp\z@
  \@ifdim{\wd\z@>\wd\f@ur}{%
    \setbox\f@ur\hb@xt@\wd\z@{\hss\box\f@ur\hss}%
    \mathord{\rlap{\raise#3\ht\z@\box\f@ur}\box\z@}%
  }{%
    \setbox\f@ur\hb@xt@.9\wd\f@ur{\hss\box\f@ur\hss}%
```

```
    \setbox\z@\hb@xt@\wd\f@ur{\hss$#4\relax#1$\hss}%
    \mathord{\rlap{\copy\z@}\raise#3\ht\z@\box\f@ur}%
  }%
}%
```

\frak Deal with legacy \frak: if amsfonts not loaded, defined in such a way as to ask for that package. Also, says to use \mathfrak instead.

```
\DeclareRobustCommand\frak{%
 \REVSYMB@warn{%
  Command \string\frak\space unsupported:^^J%
  please use \string\mathfrak\space instead.%
 }%
 \global\let\frak\mathfrak
 \frak
}%
\DeclareRobustCommand\REV@mathfrak{%
 \REVSYMB@warn{%
  Command \string\mathfrak\space undefined:^^J%
  please specify the amsfonts or amssymb option!%
 }%
 \global\let\mathfrak\@firstofone
 \mathfrak
}%
```

\Bbb Deal with legacy \Bbb: if amsfonts not loaded, defined in such a way as to ask for that package. Also, says to use \mathbb instead.

```
\DeclareRobustCommand\Bbb{%
 \REVSYMB@warn{%
  Command \string\Bbb\space unsupported:^^J%
  please use \string\mathbb\space instead.%
 }%
 \global\let\Bbb\mathbb
 \Bbb
}%
\DeclareRobustCommand\REV@mathfrak{%
 \REVSYMB@warn{%
  Command \string\mathbb\space undefined:^^J%
  please specify the amsfonts or amssymb option!%
 }%
 \global\let\mathbb\@firstofone
 \mathbb
}%
```

\Bigglb Deal with legacy bold delimiters. Each of the following takes an implicit argument consisting of the delimiter to be made big and bold. FIXME: \DeclareBoldMathCommand is not the right tool!

```
\def\Bigglb{\REV@boldopen \Bigg}%
\def\Biglb {\REV@boldopen \Big }%
\def\bigglb{\REV@boldopen \bigg}%
```

```
\def\biglb {\REV@boldopen \big }%
\def\Biggrb{\REV@boldclose\Bigg}%
\def\Bigrb {\REV@boldclose\Big }%
\def\biggrb{\REV@boldclose\bigg}%
\def\bigrb {\REV@boldclose\big }%
\def\REV@pmb#1{%
 \hbox{%
  \setbox\z@=\hbox{#1}%
  \kern-.02em\copy\z@\kern-\wd\z@
  \kern .04em\copy\z@\kern-\wd\z@
  \kern-.02em
  \raise.04em\copy\z@
 }%
}%
\def\REV@boldopen #1#2{\mathopen {\REV@pmb{$#1#2$}}}%
\def\REV@boldclose#1#2{\mathclose{\REV@pmb{$#1#2$}}}%
```

`\AtBeginDocument` Package dependencies are taken care of `\AtBeginDocument` time.

```
\def\revsymb@inithook{%
 \@ifxundefined\dddot{\let\dddot\REV@dddot}{}%
 \@ifxundefined\triangleq{\let\triangleq\REV@triangleq}{}%
 \@ifxundefined\succsim{\let\succsim\altsuccsim}{}%
 \@ifxundefined\precsim{\let\precsim\altprecsim}{}%
 \@ifxundefined\lesssim{\let\lesssim\REV@lesssim}{}%
 \@ifxundefined\gtrsim {\let\gtrsim \REV@gtrsim }{}%
 \@ifxundefined\mathfrak{\let\mathfrak\REV@mathfrak}{}%
 \@ifxundefined\mathbb{\let\mathbb\REV@mathbb}{}%
}%

%</revsymb>
```

24 The `10pt` class option: the `10pt` module

The file `10pt.rtx` is read in by the `revtex4` document class if `\@pointsize` has the value 10.

```
%<*10pt>

\def\normalsize{%
   \@setfontsize\normalsize\@xpt{11.5}%
   \abovedisplayskip 10\p@ \@plus2\p@ \@minus5\p@
   \belowdisplayskip \abovedisplayskip
   \abovedisplayshortskip  \abovedisplayskip
   \belowdisplayshortskip \abovedisplayskip
   \let\@listi\@listI
}%

\def\small{%
  \@setfontsize\small\@ixpt{10.5}%
  \abovedisplayskip 8.5\p@ \@plus3\p@ \@minus4\p@
  \belowdisplayskip \abovedisplayskip
```

```
  \abovedisplayshortskip \z@ \@plus2\p@
  \belowdisplayshortskip 4\p@ \@plus2\p@ \@minus2\p@
  \def\@listi{%
    \leftmargin\leftmargini
    \topsep 4\p@ \@plus2\p@ \@minus2\p@
    \parsep 2\p@ \@plus\p@ \@minus\p@
    \itemsep \parsep
  }%
}%
\def\footnotesize{%
  \@setfontsize\footnotesize\@viiipt{9.5pt}%
  \abovedisplayskip 6\p@ \@plus2\p@ \@minus4\p@
  \belowdisplayskip \abovedisplayskip
  \abovedisplayshortskip \z@ \@plus\p@
  \belowdisplayshortskip 3\p@ \@plus\p@ \@minus2\p@
  \def\@listi{%
    \leftmargin\leftmargini
    \topsep 3\p@ \@plus\p@ \@minus\p@
    \parsep 2\p@ \@plus\p@ \@minus\p@
    \itemsep \parsep
  }%
}%
\def\scriptsize{%
 \@setfontsize\scriptsize\@viipt\@viiipt
}%
\def\tiny{%
 \@setfontsize\tiny\@vpt\@vipt
}%
\def\large{%
 \@setfontsize\large\@xiipt{14pt}%
}%
\def\Large{%
 \@setfontsize\Large\@xivpt{18pt}%
}%
\def\LARGE{%
 \@setfontsize\LARGE\@xviipt{22pt}%
}%
\def\huge{%
 \@setfontsize\huge\@xxpt{25pt}%
}%
\def\Huge{%
 \@setfontsize\Huge\@xxvpt{30pt}%
}%
%\appdef\class@documenthook{%
 \@ifxundefined\twoside@sw{\@booleanfalse\twoside@sw}{}%
 \twoside@sw{%
%   \oddsidemargin  -.1in
%   \evensidemargin -.4in
    \oddsidemargin  -20pt
```

```
    \evensidemargin -20pt
    \marginparwidth 107pt
 }{%
    \oddsidemargin  -.25in
    \evensidemargin -.25in
    \marginparwidth 30pt
 }%
%}%
\marginparsep 6pt
\topmargin -61pt
\headheight 25pt
\headsep 16pt
\topskip 10pt
\footskip 30pt
```

Was: 57\baselineskip+ \topskip

```
%\appdef\set@pica@hook{%
 \textheight = 56pc
%}%
\textwidth42.5pc
\columnsep 1.5pc
\columnseprule 0pt
\footnotesep 1pt
\skip\footins 39pt plus 4pt minus 2pt
\def\footnoterule{\kern-19pt\hrule width.5in\kern18.6pt\relax}%
\floatsep 12pt plus 2pt minus 2pt
\textfloatsep 20pt plus 2pt minus 4pt
\intextsep 12pt plus 2pt minus 2pt
\dblfloatsep 12pt plus 2pt minus 2pt
\dbltextfloatsep 20pt plus 2pt minus 4pt
\@fptop 0pt plus 1fil
\@fpsep 8pt plus 2fil
\@fpbot 0pt plus 1fil
\@dblfptop 0pt plus 1fil
\@dblfpsep 8pt plus 2fil
\@dblfpbot 0pt plus 1fil
\marginparpush 5pt
\parskip 0pt plus 1pt
\parindent 10pt
\emergencystretch8\p@
\partopsep 2pt plus 1pt minus 1pt
\leftmargini 25pt
\leftmarginii 22pt
```

```
\leftmarginiii 18.7pt
\leftmarginiv 17pt
\leftmarginv 10pt
\leftmarginvi 10pt

\def\@listI{%
  \leftmargin\leftmargini
  \parsep 4\p@ plus2\p@ minus\p@
  \topsep 8\p@ plus2\p@ minus4\p@
  \itemsep 4\p@ plus2\p@ minus\p@
}%

\labelsep 4pt

\def\@listii{%
  \leftmargin\leftmarginii
  \labelwidth\leftmarginii
  \advance\labelwidth-\labelsep
  \topsep 4\p@ plus2\p@ minus\p@
  \parsep 2\p@ plus\p@ minus\p@
  \itemsep \parsep
}%

\def\@listiii{%
  \leftmargin\leftmarginiii
  \labelwidth\leftmarginiii
  \advance\labelwidth-\labelsep
  \topsep 2\p@ plus\p@ minus\p@
  \parsep \z@
  \partopsep \p@ plus\z@ minus\p@
  \itemsep \topsep
}%

\def\@listiv{%
  \leftmargin\leftmarginiv
  \labelwidth\leftmarginiv
  \advance\labelwidth-\labelsep
}%

\def\@listv{%
  \leftmargin\leftmarginv
  \labelwidth\leftmarginv
  \advance\labelwidth-\labelsep
}%

\def\@listvi{%
  \leftmargin\leftmarginvi
  \labelwidth\leftmarginvi
  \advance\labelwidth-\labelsep
}%

%</10pt>
```

25 The 11pt class option: the 11pt module

The file 11pt.rtx is read in by the revtex4 document class if \@pointsize has the value 11.

```
%<*11pt>
\def\normalsize{%
    \@setfontsize\normalsize\@xipt{13.6}%
    \abovedisplayskip 11\p@ \@plus3\p@ \@minus6\p@
    \belowdisplayskip \abovedisplayskip
    \abovedisplayshortskip  \abovedisplayskip
    \belowdisplayshortskip \abovedisplayskip
    \let\@listi\@listI
}%
\def\small{%
   \@setfontsize\small\@xpt\@xiipt
   \abovedisplayskip 10\p@ \@plus2\p@ \@minus5\p@
   \abovedisplayshortskip \z@ \@plus3\p@
   \belowdisplayshortskip 6\p@ \@plus3\p@ \@minus3\p@
   \def\@listi{\leftmargin\leftmargini
               \topsep 6\p@ \@plus2\p@ \@minus2\p@
               \parsep 3\p@ \@plus2\p@ \@minus\p@
               \itemsep \parsep
   }%
   \belowdisplayskip \abovedisplayskip
}%
\def\footnotesize{%
   \@setfontsize\footnotesize\@ixpt{11}%
   \abovedisplayskip 8\p@ \@plus2\p@ \@minus4\p@
   \abovedisplayshortskip \z@ \@plus\p@
   \belowdisplayshortskip 4\p@ \@plus2\p@ \@minus2\p@
   \def\@listi{\leftmargin\leftmargini
               \topsep 4\p@ \@plus2\p@ \@minus2\p@
               \parsep 2\p@ \@plus\p@ \@minus\p@
               \itemsep \parsep
   }%
   \belowdisplayskip \abovedisplayskip
}%
\def\scriptsize{%
  \@setfontsize\scriptsize\@viiipt{9.5}%
}%
\def\tiny{%
  \@setfontsize\tiny\@vipt\@viipt
}%
\def\large{%
  \@setfontsize\large\@xiipt{14}%
}%
\def\Large{%
  \@setfontsize\Large\@xivpt{18}%
```

```
}%
\def\LARGE{%
  \@setfontsize\LARGE\@xviipt{22}%
}%
\def\huge{%
    \@setfontsize\huge\@xxpt{25pt}%
}%
\def\Huge{%
  \@setfontsize\Huge\@xxvpt{30pt}%
}%
%</11pt>
```

26 The 12pt class option: the 12pt module

The file `12pt.rtx` is read in by the `revtex4` document class if `\@pointsize` has the value 12.

```
%<*12pt>
\def\normalsize{%
  \@setfontsize\normalsize\@xiipt{14pt}%
  \abovedisplayskip 12\p@ \@plus3\p@ \@minus7\p@
  \belowdisplayskip \abovedisplayskip
  \abovedisplayshortskip  \z@ plus3\p@
  \belowdisplayshortskip  6.5\p@ \@plus3.5\p@ \@minus3\p@
  \let\@listi\@listI
}%
\def\small{%
  \@setfontsize\small\@xipt{14.5pt}%
  \abovedisplayskip 8\p@ \@plus3\p@ \@minus6\p@
  \belowdisplayskip \abovedisplayskip
  \abovedisplayshortskip \z@ \@plus3\p@
  \belowdisplayshortskip 6.5\p@ \@plus3.5\p@ \@minus3\p@
  \def\@listi{%
    \leftmargin\leftmargini
    \topsep 9\p@ \@plus3\p@ \@minus5\p@
    \parsep 4.5\p@ \@plus2\p@ \@minus\p@
    \itemsep \parsep
  }%
}%
```

Same baselineskip as `\small` ?

```
\def\footnotesize{%
  \@setfontsize\footnotesize\@xpt{14.5pt}%
  \abovedisplayskip 10\p@ \@plus2\p@ \@minus5\p@
  \belowdisplayskip \abovedisplayskip
  \abovedisplayshortskip \z@ \@plus3\p@
  \belowdisplayshortskip 6\p@ \@plus3\p@ \@minus3\p@
  \def\@listi{%
```

```
    \leftmargin\leftmargini
    \topsep 6\p@ \@plus2\p@ \@minus2\p@
    \parsep 3\p@ \@plus2\p@ \@minus\p@
    \itemsep \parsep
  }%
}%
\def\scriptsize{%
  \@setfontsize\scriptsize\@viiipt{9.5pt}%
}%
\def\tiny{%
  \@setfontsize\tiny\@vipt{7pt}%
}%
\def\large{%
  \@setfontsize\large\@xivpt{18pt}%
}%
\def\Large{%
  \@setfontsize\Large\@xviipt{22pt}%
}%
\def\LARGE{%
  \@setfontsize\LARGE\@xxpt{25pt}%
}%
\def\huge{%
    \@setfontsize\huge\@xxvpt{30pt}%
}%
\let\Huge=\huge
%</12pt>
```

27 Page parameters

This code is common to both `11pt` and `12pt`.

```
%<*11pt|12pt>
%\appdef\class@documenthook{%
 \@ifxundefined\twoside@sw{\@booleanfalse\twoside@sw}{}%
 \twoside@sw{%
  \oddsidemargin   0pt
  \evensidemargin  0pt
  \marginparwidth 60pt
 }{%
  \oddsidemargin 0pt
  \evensidemargin 0pt
  \marginparwidth 44pt
 }%
%}%
\marginparsep 10pt
\topmargin -37pt
```

```
\headheight 12pt
\headsep 25pt
\topskip 10pt
\footskip 30pt
%\appdef\set@pica@hook{%
 \@ifxundefined\tightenlines@sw{\@booleanfalse\tightenlines@sw}{}%
 \textheight=665.5\p@
 \tightenlines@sw{%
  \def\baselinestretch{1}%
 }{%
  \def\baselinestretch{1.5}%
 }%
%}%
\textwidth 468pt
\columnsep 10pt
\columnseprule 0pt
\footnotesep 1pt
\skip\footins 25.25pt plus 4pt minus 2pt
\def\footnoterule{\kern-5.25pt\hrule width.5in\kern4.85pt\relax}
\floatsep        14pt plus 2pt minus 4pt
\textfloatsep    20pt plus 2pt minus 4pt
\intextsep       14pt plus 4pt minus 4pt
\dblfloatsep     14pt plus 2pt minus 4pt
\dbltextfloatsep 20pt plus 2pt minus 4pt
\@fptop 0pt plus 1fil
\@fpsep 10pt plus 2fil
\@fpbot 0pt plus 1fil
\@dblfptop 0pt plus 1fil
\@dblfpsep 10pt plus 2fil%
\@dblfpbot 0pt plus 1fil
\marginparpush 7pt
\parskip 0pt plus 1pt
\parindent 15pt
\emergencystretch8\p@
\partopsep 3pt plus 2pt minus 2pt
\leftmargini   30pt
\leftmarginii  26pt
\leftmarginiii 22pt
\leftmarginiv  20pt
\leftmarginv   12pt
\leftmarginvi  12pt
\def\@listI{\leftmargin\leftmargini \parsep 5\p@ plus2.5\p@ minus\p@
  \topsep 10\p@ plus4\p@ minus6\p@
  \itemsep 5\p@ plus2.5\p@ minus\p@
}%
```

```
\labelsep 6pt

\def\@listii{\leftmargin\leftmarginii
  \labelwidth\leftmarginii\advance\labelwidth-\labelsep
  \topsep 5\p@ plus2.5\p@ minus\p@
  \parsep 2.5\p@ plus\p@ minus\p@
  \itemsep \parsep
}%

\def\@listiii{\leftmargin\leftmarginiii
  \labelwidth\leftmarginiii\advance\labelwidth-\labelsep
  \topsep 2.5\p@ plus\p@ minus\p@
  \parsep \z@ \partopsep \p@ plus\z@ minus\p@
  \itemsep \topsep
}%

\def\@listiv{\leftmargin\leftmarginiv
  \labelwidth\leftmarginiv\advance\labelwidth-\labelsep
}%

\def\@listv{\leftmargin\leftmarginv
  \labelwidth\leftmarginv\advance\labelwidth-\labelsep
}%

\def\@listvi{\leftmargin\leftmarginvi
  \labelwidth\leftmarginvi\advance\labelwidth-\labelsep
}%

%</11pt|12pt>
```

28 The aps class option: the aps module

The file `aps.rtx` is read in by the `revtex4` document class if `\@society` has the value `aps`.

Here, code specific to APS journals is separated out from the REVTEX document class. (Other societies can customize REVTEX by supplying their own `.rtx` file.)

We first give some text entities (amounting to journal abbreviations), then some APS-specific initialisations, then code for particular APS journals. In the latter case, the choice is keyed off the macro `\@journal`.

Note on `\AtEndOfClass`: this file, like all journal substyles, is read in at `\AtEndOfClass` time, so you should not use this command in this file. Mote for LATEX developers: It would be an improvement in LATEX to `\let\AtEndOfClass` to something like `\@firstofone`. This change would be effected in `\@onefilewithoptions`.

```
%<*aps>
```

28.1 Defend Against Forseeable Errors

Protect this file from being read in by anything but REVTEX.

```
\ifx\undefined\substyle@ext
```

```
\def\@tempa{%
 \endinput
 \GenericWarning{I must be read in by REVTeX! (Bailing out)}%
}%
\expandafter\else
 \def\@tempa{}%
\expandafter\fi\@tempa
```

28.2 Abbreviations

The following macros constitute typing shortcuts for certain journal names.

```
\def\ao{Appl.\  Opt.\ }
\def\ap{Appl.\  Phys.\ }
\def\apl{Appl.\ Phys.\ Lett.\ }
\def\apj{Astrophys.\ J.\ }
\def\bell{Bell Syst.\ Tech.\ J.\ }
\def\jqe{IEEE J.\ Quantum Electron.\ }
\def\assp{IEEE Trans.\ Acoust.\ Speech Signal Process.\ }
\def\aprop{IEEE Trans.\ Antennas Propag.\ }
\def\mtt{IEEE Trans.\ Microwave Theory Tech.\ }
\def\iovs{Invest.\ Ophthalmol.\ Vis.\ Sci.\ }
\def\jcp{J.\ Chem.\ Phys.\ }
\def\jmo{J.\ Mod.\ Opt.\ }
\def\josa{J.\ Opt.\ Soc.\ Am.\ }
\def\josaa{J.\ Opt.\ Soc.\ Am.\ A }
\def\josab{J.\ Opt.\ Soc.\ Am.\ B }
\def\jpp{J.\ Phys.\ (Paris) }
\def\nat{Nature (London) }
\def\oc{Opt.\ Commun.\ }
\def\ol{Opt.\ Lett.\ }
\def\pl{Phys.\ Lett.\ }
\def\pra{Phys.\ Rev.\ A }
\def\prb{Phys.\ Rev.\ B }
\def\prc{Phys.\ Rev.\ C }
\def\prd{Phys.\ Rev.\ D }
\def\pre{Phys.\ Rev.\ E }
\def\prl{Phys.\ Rev.\ Lett.\ }
\def\rmp{Rev.\ Mod.\ Phys.\ }
\def\pspie{Proc.\ Soc.\ Photo-Opt.\ Instrum.\ Eng.\ }
\def\sjqe{Sov.\ J.\ Quantum Electron.\ }
\def\vr{Vision Res.\ }
```

28.3 APS Setup

Here we define the default procedures for APS journals. Individual APS journals may override these definitions.

28.3.1 Title block

The specifics of the title block. Apply to all APS journals; individual journals may override these settings.

`\@fnsymbol` The LaTeX kernel definition of `\@fnsymbol` is overridden.

```
\def\@fnsymbol#1{%
 \ensuremath{%
  \ifcase#1\or
   *\or                        \dagger                   \or
   \ddagger                \or \mathsection              \or
   \mathparagraph\or
   **\or                       \dagger\dagger            \or
   \ddagger\ddagger        \or \mathsection \mathsection \or
   \mathparagraph\mathparagraph\or
   *{*}*\ignorespaces      \or \dagger\dagger\dagger     \or
   \ddagger\ddagger\ddagger\or \mathsection \mathsection \mathsection \or
   \mathparagraph\mathparagraph\mathparagraph\or
  \else
   \@ctrerr
  \fi
 }%
}%
```

`\titlepage`

```
\renewenvironment{titlepage}{%
  \let\wastwocol@sw\twocolumn@sw
  \onecolumngrid
  \newpage
  \thispagestyle{titlepage}%
  \c@page\z@% article sets this to one not zero???
}{%
  \wastwocol@sw{\twocolumngrid}{\newpage}%
}%
```

`\frontmatter@abstractheading` APS Journals all set the abstract head the same way, with no head. However, if the user has specified the `preprint` class option, then the abstract will have a head.

```
\def\frontmatter@abstractheading{%
 \preprintsty@sw{%
  \begingroup
   \centering\large
   \abstractname
   \par
  \endgroup
 }{}%
}%
```

`\frontmatter@abstractwidth` All APS journals set the abstract to the same width.

```
\def\frontmatter@abstractwidth{400\p@}%
```

\frontmatter@abstractfont All APS journals set the abstract body the same way.

```
\def\frontmatter@abstractfont{%
 \small
 \parindent1em\relax
 \adjust@abstractwidth
}%
\def\adjust@abstractwidth{%
 \dimen@\textwidth\advance\dimen@-\frontmatter@abstractwidth
 \divide\dimen@\tw@
 \galley@sw{%
  \advance\rightskip\tw@\dimen@
 }{%
  \advance\leftskip\dimen@
  \advance\rightskip\dimen@
 }%
}%
```

All APS journal preprints use separate titlepage and full-width abstract.

Note that we defer polling the \preprintsty@sw Boolean until \AtBeginDocument time, because the document preamble may legitimately change the state of that variable.

```
\appdef\class@documenthook{%
  \preprintsty@sw{%
   \@ifxundefined\titlepage@sw{\@booleantrue\titlepage@sw}{}%
%  \tightenlines@sw{%
%   \def\baselinestretch{1}%
%  }{%
%   \def\baselinestretch{1.5}%
%  }%
   \def\frontmatter@abstractwidth{\textwidth}%
   \def\frontmatter@affiliationfont{\it}%
   \let\section\section@preprintsty
%  \let\@hangfrom@section\@hangfrom@section@preprintsty
   \let\subsection\subsection@preprintsty
   \let\subsubsection\subsubsection@preprintsty
  }{}%
 }%
```

\frontmatter@authorformat All APS journals set the author list the same. The leading is 11.5 points, and there is 11.5 points of extra space above the first author line (which amounts to the same thing as 11.5 points extra below the title) for a total of 23 points base-to-base.

```
\def\frontmatter@authorformat{%
 \skip@\@flushglue
 \@flushglue\z@ plus.3\hsize\relax
 \centering
 \advance\baselineskip\p@
 \parskip11.5\p@\relax
 \@flushglue\skip@
%\preprintsty@sw{}{%
```

```
% \addvspace{0\p@}%
%}%
}%
```

`\frontmatter@above@affilgroup` The default amount of space above affiliation. APS Journals have 24 points b-b above an affiliation group.

```
\def\frontmatter@above@affilgroup{%
%\preprintsty@sw{}{%
% \addvspace{11\p@}%
%}%
}%
```

`\frontmatter@above@affiliation` `matter@above@affiliation@script` The default amount of space above affiliation. APS Journals have no extra space between author group down to common affiliation.

```
\def\frontmatter@above@affiliation@script{%
 \skip@\@flushglue
 \@flushglue\z@ plus.3\hsize\relax
 \centering
 \@flushglue\skip@
 \addvspace{3.5\p@}%
}%
\def\frontmatter@above@affiliation{%
 \preprintsty@sw{}{%
% \addvspace{12\p@}%
 }%
}%
```

`\frontmatter@affiliationfont` All APS journals set the affiliation the same.

```
\def\frontmatter@affiliationfont{%
 \small\it
}%
```

`frontmatter@collaboration@above` PRL: 1.5 points extra: 13 points base-to-base above.

```
\def\frontmatter@collaboration@above{%
 \preprintsty@sw{%
 }{%
  \parskip1.5\p@\relax
 }%
}%
```

Set up the default APS style for title block authors and affiliations.

```
\@ifxundefined\groupauthors@sw{%
 \clo@groupedaddress
}{}%
```

`\frontmatter@setup` All APS journals set the title page using the same font and size. However, justification varies for the title block elements, so we assert none here.

```
\def\frontmatter@setup{%
 \normalfont
}%
```

\frontmatter@title@above
\frontmatter@title@format
\frontmatter@title@below

All APS journals set the article title the same.

Note: Spacing from title to author is 23 points base-to-base.

```
\def\frontmatter@title@above{\addvspace{6\p@}}%
\def\frontmatter@title@format{\large\bfseries\centering\parskip\z@skip}%
\def\frontmatter@title@below{}%
```

\frontmatter@makefnmark

All APS journals share this procedure for setting the titlepage footnote text.

```
\def\@author@parskip{3\p@}%
\def\frontmatter@makefnmark{%
 \@textsuperscript{%
  \normalfont\@thefnmark
 }%
}%
\def\frontmatter@authorbelow{%
\addvspace{3\p@}%
}%
```

\frontmatter@RRAP@format

All APS journals use the same format for the "Received, Revised, etc." block on the title page.

Change note: 11.5 points b-b from author/affiliation down to date.

```
\def\frontmatter@RRAP@format{%
  \small
  \centering
% \preprintsty@sw{}{\parskip.5ex\relax}%
  \everypar{\hbox\bgroup(\@gobble@leavemode@uppercase}%
  \def\par{\@ifvmode{}{\unskip)\egroup\@@par}}%
}%
\def\punct@RRAP{;\egroup\ \hbox\bgroup}%
\def\@gobble@leavemode@uppercase#1#2{\expandafter\MakeTextUppercase}%
```

\frontmatter@PACS@format

```
\def\frontmatter@PACS@format{%
   \addvspace{11\p@}%
   \footnotesize
   \adjust@abstractwidth
   \parindent\z@
   \parskip\z@skip
   \samepage
}%
```

\frontmatter@keys@format

```
\def\frontmatter@keys@format{%
   \footnotesize
   \adjust@abstractwidth
   \parindent\z@
   \samepage
}%
```

\ps@titlepage Title page style. Currently empty except for preprint header, which consists of all the \preprint arguments, stacked flush right at the right margin.

```
\def\ps@titlepage{%
  \def\@oddhead{%
   \hfill
   \produce@preprints\@preprint
  }%
  \let\@evenhead\@oddhead
  \def\@oddfoot{%
   \hb@xt@\z@{\byrevtex\hss}%
   \hfil
   \preprintsty@sw{\thepage}{}%
   \quad\checkindate
   \hfil
  }%
  \let\@evenfoot\@oddfoot
}%
\def\byrevtex{\byrevtex@sw{Typeset by REV\TeX}{}}%
```

\produce@preprints

```
\def\produce@preprints#1{%
 \preprint@sw{%
  \vtop to \z@{%
   \def\baselinestretch{1}%
   \small
   \let\preprint\preprint@count
   \count@\z@#1\@ifnum{\count@>\tw@}{%
    \hbox{%
     \let\preprint\preprint@hlist
     #1\setbox\z@\lastbox
    }%
   }{%
    \let\preprint\preprint@cr
    \halign{\hfil##\cr#1\crcr}%
    \par
    \vss
   }%
  }%
 }{}%
}%
\def\preprint@cr#1{#1\cr}%
\def\preprint@count#1{\advance\count@\@ne}%
\def\preprint@hlist#1{#1\hbox{, }}%
```

28.3.2 Stacked Heads

All APS journals put a period (.), followed by quad space, after the section number. Also, no hanging section number.

```
\def\@seccntformat#1{\csname the#1\endcsname.\quad}%
```

```
\def\@hang@from#1#2#3{#1#2#3}%
```

Note that in the following, we wish to set the section head uppercase, so we use David Carlisle's `\MakeTextUppercase`. However, because this procedure effectively parses its argument (looking for things to *not* translate), it has to be invoked in such a way that the argument of the `\section` command is passed to it as its own argument.

To accomplish this, we use the `\@hangfrom@` hook, which was developed for this purpose.

```
\def\section{%
  \@startsection
    {section}%
    {1}%
    {\z@}%
    {0.8cm \@plus1ex \@minus .2ex}%
    {0.5cm}%
    {%
      \normalfont\small\bfseries
      \centering
    }%
}%
\def\@hangfrom@section#1#2#3{\@hangfrom{#1#2}\MakeTextUppercase{#3}}%
\def\subsection{%
  \@startsection
    {subsection}%
    {2}%
    {\z@}%
    {.8cm \@plus1ex \@minus .2ex}%
    {.5cm}%
    {%
     \normalfont\small\bfseries
     \centering
    }%
}%
\def\subsubsection{%
  \@startsection
    {subsubsection}%
    {3}%
    {\z@}%
    {.8cm \@plus1ex \@minus .2ex}%
    {.5cm}%
    {%
     \normalfont\small\itshape
     \centering
    }%
}%
```

28.3.3 Runin Heads

```
\def\paragraph{%
  \@startsection
    {paragraph}%
    {4}%
    {\parindent}%
    {\z@}%
    {-1em}%
    {\normalfont\normalsize\itshape}%
}%
\def\subparagraph{%
  \@startsection
    {subparagraph}%
    {5}%
    {\parindent}%
    {3.25ex \@plus1ex \@minus .2ex}%
    {-1em}%
    {\normalfont\normalsize\bfseries}%
}%
```

\section@preprintsty
\subsection@preprintsty
\subsubsection@preprintsty

Here are the formatting procedures specific to the preprint style; the only difference is that the heads are flush left instead of centered.

```
\def\section@preprintsty{%
  \@startsection
    {section}%
    {1}%
    {\z@}%
    {0.8cm \@plus1ex \@minus .2ex}%
    {0.5cm}%
    {%
      \normalfont\small\bfseries
%     \centering
    }%
}%
%\def\@hangfrom@section@preprintsty#1#2#3{\@hangfrom{#1#2}\MakeTextUppercase{#3}}%
\def\subsection@preprintsty{%
  \@startsection
    {subsection}%
    {2}%
    {\z@}%
    {.8cm \@plus1ex \@minus .2ex}%
    {.5cm}%
    {%
     \normalfont\small\bfseries
%    \centering
    }%
}%
\def\subsubsection@preprintsty{%
  \@startsection
    {subsubsection}%
```

```
    {3}%
    {\z@}%
    {.8cm \@plus1ex \@minus .2ex}%
    {.5cm}%
    {%
     \normalfont\small\itshape
%    \centering
    }%
}%
```

Effectively invoke the `bibnote` class option, if `\frontmatter@footnote@produce` has not yet been set.

```
\@ifxundefined\frontmatter@footnote@produce{%
 \let\frontmatter@footnote@produce\frontmatter@footnote@produce@endnote
}{}%
```

28.3.4 Table of Contents

The toc will itself make an entry in the toc, but we temporarily turn off toc formatting for the duration.

```
\def\@pnumwidth{1.55em}
\def\@tocrmarg {2.55em}
\def\@dotsep{4.5pt}
\setcounter{tocdepth}{3}

\def\tableofcontents{%
 \addtocontents{toc}{\string\tocdepth@munge}%
 \print@toc{toc}%
 \addtocontents{toc}{\string\tocdepth@restore}%
}%

\def\tocdepth@munge{%
  \let\l@section@saved\l@section
  \let\l@section\@gobble@tw@
}%
\def\@gobble@tw@#1#2{}%

\def\tocdepth@restore{%
  \let\l@section\l@section@saved
}%

\def\l@part#1#2{\addpenalty{\@secpenalty}%
 \begingroup
  \set@tocdim@pagenum{#2}%
% \@tempdima 3em %
  \parindent \z@
  \rightskip\tocleft@pagenum plus 1fil\relax
  \skip@\parfillskip\parfillskip\z@
  \addvspace{2.25em plus\p@}%
  \large \bf %
  \leavevmode\ignorespaces#1\unskip\nobreak\hskip\skip@
```

```
  \hb@xt@\rightskip{\hfil\unhbox\z@}\hskip-\rightskip\hskip\z@skip
  \par
  \nobreak %
 \endgroup
}%
```

`\l@section` Determine which TOC elements are automatically indented.

```
\def\tocleft@{\z@}%
\def\tocdim@min{5\p@}%
\def\l@section{%
 \l@@sections{}{section}% Implicit #3#4
}%
\def\l@f@section{%
 \addpenalty{\@secpenalty}%
 \addvspace{1.0em plus\p@}%
 \bf
}%
\def\l@subsection{%
 \l@@sections{section}{subsection}% Implicit #3#4
}%
\def\l@subsubsection{%
 \l@@sections{subsection}{subsubsection}% Implicit #3#4
}%
\def\l@paragraph#1#2{}%
\def\l@subparagraph#1#2{}%
```

Activate the auto TOC processing.

```
\let\toc@pre\toc@pre@auto
\let\toc@post\toc@post@auto
\def\listoffigures{\print@toc{lof}}%
\def\l@figure{\@dottedtocline{1}{1.5em}{2.3em}}
\def\listoftables{\print@toc{lot}}%
\let\l@table\l@figure
```

28.3.5 Default column bottom

All APS journal styles have flush bottoms.

Note that we defer polling the `\raggedcolumn@sw` Boolean until `\AtBeginDocument` time, because the document preamble may legitimately change the state of that variable.

```
\appdef\class@documenthook{%
 \@ifxundefined\raggedcolumn@sw{\@booleantrue\raggedcolumn@sw}{}%
 \raggedcolumn@sw{\raggedbottom}{\flushbottom}%
}%
```

28.3.6 Table alignment style

`\tableft@skip@float` `\tabmid@skip@float` `\tabright@skip@float` `\array@row@pre@float` `\array@row@pst@float`

All APS publications have the same table specification: Scotch rules above and below, centered in column.

```
\def\tableft@skip@float{\z@ plus\hsize}%
\def\tabmid@skip@float{\@flushglue}%
\def\tabright@skip@float{\z@ plus\hsize}%
\def\array@row@pre@float{\hline\hline\noalign{\vskip\doublerulesep}}%
\def\array@row@pst@float{\noalign{\vskip\doublerulesep}\hline\hline}%
```

28.3.7 Footnote formatting

We customize the formatting of footnotes for all APS journals.

\@makefntext

```
\def\@makefntext#1{%
 \def\baselinestretch{1}%
 \reset@font
 \footnotesize
 \leftskip1em
 \parindent1em
 \noindent\nobreak\hskip-\leftskip
 \hb@xt@\leftskip{%
  \Hy@raisedlink{\hyper@anchorstart{footnote@\the\c@footnote}\hyper@anchorend}%
  \hss\@makefnmark\
 }%
 #1%
 \par
}%
```

28.3.8 Appendix

\appendix
\@hangfrom@appendix
\@hangfroms@appendix
\@appendixcntformat

```
\prepdef\appendix{%
 \par
 \let\@hangfrom@section\@hangfrom@appendix
 \let\@hangfroms@section\@hangfroms@appendix
 \let\@sectioncntformat\@appendixcntformat
}%
\def\@hangfrom@appendix#1#2#3{%
 #1%
 \@if@empty{#2}{%
  #3%
 }{%
  #2\@if@empty{#3}{}{:\ #3}%
 }%
}%
\def\@hangfroms@appendix#1#2{%
 #1#2%
}%
\def\@appendixcntformat#1{\appendixname\ \csname the#1\endcsname}%
```

28.3.9 Bibliography

Customize REVTEX for the journal substyle; this task requires three components: the BIBTEX `apsrev.bst` style file, customizing code for `natbib`, and customizations of the `thebibliography` environment.

`\@bibstyle` Define the argument of the `\bibliographystyle` command (if the document does not do so). The user must have installed a `.bst` file of the corresponding name. This file will then be used by BIBTEX when compiling the document's `.bbl` file.

The default bibliography style for the APS journal substyles, `apsrev.bst`, was originally called `revtex.bst`, but that name is much too generic and should be given to a `.bst` file that that applies generally to any REVTEX jounal substyle, an impossibility.

To generate `apsrev.bst`, use `custom-bib` version 4.0b2 or later. Run the `.bst` generator, `makebst.tex`, and accept all defaults, with the following exceptions:

1. ORDERING OF REFERENCES: c: seq-no (references in order of Citation);
2. AUTHOR NAMES: i: nm-init,ed-au (Initials + surname);
3. NUMBER OF AUTHORS: l: max 12, min 12 (there will be three prompts total);
4. TYPEFACE FOR AUTHORS IN LIST OF REFERENCES: u: nmft,nmft-def (User defined author font);
5. FONT FOR FIRST NAMES: u: fnm-def (First names in user defined font);
6. EDITOR NAMES IN INCOLLECTION ETC: a: nmfted (Editors incollection like authors);
7. DATE FORMAT: p: yr-par (Date in parentheses);
8. SUPPRESS MONTH: x: xmth (Date is year only);
9. TITLE OF ARTICLE: i: tit-it (Title italic)
10. ARTICLE TITLE PRESENT: x: jtit-x (No article title);
11. JOURNAL NAME FONT: r: jttl-rm (Journal title normal);
12. TECHNICAL REPORT TITLE: b: trtit-b (Tech. report title like books);
13. JOURNAL VOLUME: b: vol-bf (Volume bold);
14. VOLUME PUNCTUATION: c: volp-com (Volume with comma);
15. PAGE NUMBERS: f: jpg-1 (Only start page number);
16. JOURNAL NAME PUNCTUATION: x: jnm-x (Space after journal);
17. PUBLISHER IN PARENTHESES: d: pub-date (Publisher and date in parentheses);

18. PUBLISHER POSITION: p: pre-pub (Publisher before chapter, pages);
19. ISBN NUMBER: isbn: (Include ISBN for books, booklets)
20. ISSN NUMBER: issn: (Include ISSN for periodicals)
21. EDITOR IN COLLECTIONS: b: edby (Booktitle, edited by ...);
22. PUNCTUATION BETWEEN SECTIONS (BLOCKS): c: blk-com (Comma between blocks);
23. ABBREVIATE WORD 'PAGES': a: pp ('Page' abbreviated);
24. ABBREVIATE WORD 'EDITORS': a: ed ('Editor' abbreviated);
25. OTHER ABBREVIATIONS: a: abr (Abbreviations);
26. ABBREVIATION FOR 'EDITION': a: ednx ('Edition' abbreviated as 'ed');
27. EDITION NUMBERS: n: ord (Numerical editions);
28. STORED JOURNAL NAMES: a: jabr (Abbreviated journal names);
29. FONT OF 'ET AL': i: etal-it (Italic et al);
30. ADDITIONAL REVTeX DATA FIELDS: r: revdata, eprint, url, url-blk (Include REVTeX data fields collaboration, eid, eprint, numpages, url)
31. NEW FONT SELECTION SCHEME: n: nfss (NFSS);
32. ADDITIONAL REVTeX DATA FIELDS: y: revdata (additional data fields);
33. REFERENCE COMPONENT TAGS: y: reference component tags;
34. URL ADDRESS: n: URL as note;

A file `apsrev.dbj` file equivalent to the following should result:

```
% \input docstrip
% \preamble
% ----------------------------------------
% *** REVTeX-compatible apsrev.bst ***
% \endpreamble
% \postamble
% End of customized bst file
% \endpostamble
% \keepsilent
% \askforoverwritefalse
% \def\MBopts{\from{merlin.mbs}{%
%    seq-no%: Citation order (unsorted, like unsrt.bst)
%   ,nm-init,ed-au%: Initials + surname (J. F. Smith)
%   ,nmlm%: Limited authors (et al replaces missing names)
```

```
%     ,x10%: Maximum of 10 authors
%     ,m10%: Minimum of 10 authors
%   ,nmft,nmft-def%: User defined author font (\bibnamefont)
%   ,fnm-def%: First names in user defined font (\bibfnamefont)
%   ,nmfted%: Editors incollection like authors font
%   ,yr-par%: Date in parentheses as (May 1993)
%   ,xmth%: Date is year only
%   ,tit-it%: Title italic (\em)
%   ,jtit-x%: No article title
%   ,jttl-rm%: Journal name normal font
%   ,trtit-b%: Tech. report title like books
%   ,vol-bf%: Volume bold as {\bf vol}(num)
%   ,volp-com%: Volume with comma as vol(num), ppp
%   ,jpg-1%: Only start page number
%   ,jnm-x%: Space after journal name
%   ,pub-date%: Publisher and date in parentheses (Oxford, 1994)
%   ,pre-pub%: Publisher before chapter, pages
%   ,isbn%: Include ISBN for books, booklets, etc.
%   ,issn%: Include ISSN for periodicals
%   ,edby%: In booktitle, edited by ..  (where .. is names)
%   ,blk-com%: Comma between blocks
%   ,pp%: 'Page' abbreviated as p. or pp.
%   ,ed%: 'Editor' abbreviated as ed. or eds.
%   ,abr%: Abbreviations of such words
%   ,ednx%: 'Edition' abbreviated as 'ed'
%   ,ord%: Numerical editions as 1st, 2nd, 3rd, etc
%   ,jabr%: Abbreviated journal names
%   ,etal-it%: Italic et al
%   ,revdata,eprint,url,url-blk%: Include REVTeX data fields collaboration, eid, eprint, numpages
%   ,url,url-nt%: URL as note
%   ,bibinfo%: Reference component tags like \bibinfo in the content of \bibitem
%   ,nfss%: NFSS use \textbf, \emph, not \bf, \em
%   }}
% \generate{\file{\jobname.bst}{\MBopts}}
% \endbatchfile
%
```

To get a banner in the `.blg` file, patch by hand into `apsrev.bst`:

```
% FUNCTION {banner.log} {"REVTeX-compatible apsrev.bst 9812" warning$}
%
```

and invoke `banner.log` from within `begin.bib`

We ensure that the journal substyle has the first word in the matter by installing the (default) APS code later on (see Section 28.5).

`\bibpunct` The following commands effectively establish the style in which `\cite` commands are formatted. You can think of them as the second needed component for the bibliography.

Set up for numerical citations.

The journal substyles can override these choices.

Note that setting `\NAT@sort` at this late hour is not totally effective. You have to give `natbib` the relevant options at load time.

```
\bibpunct{[}{]}{,}{n}{}{,}%
```

`\bibsection` We define the sectioning command to use when starting the bibliography.

```
\def\bibsection{%
 \par
% \twocolumn@sw{\onecolumngrid}{}%
 \onecolumngrid@push
 \begingroup
  \baselineskip26\p@
%  \def\@currentHref{section.bib}%
  \bib@device{\textwidth}{245.5\p@}%
\endgroup
 \nobreak\@nobreaktrue
 \addvspace{19\p@}%
 \par
% \twocolumn@sw{\twocolumngrid}{}%
 \onecolumngrid@pop
}%
```

`\bib@device` We define the sectioning command to use when starting the bibliography.

`\bibpreamble`

`\bibsep`

`\newblock`

```
\def\bib@device#1#2{%
\hb@xt@#1{%
 \hfil
 \phantomsection
  \addcontentsline {toc}{section}{\protect\numberline{}\refname}%
% \hyper@anchorstart {\@currentHref }%
 \hb@xt@#2{%
  \skip@\z@\@plus-1fil\relax
                          \leaders\hrule height.25 \p@ depth.25 \p@ \hskip\z@\@plus1fil
\hskip\skip@
\hskip\z@\@plus0.125fil\leaders\hrule height.375\p@ depth.375\p@ \hskip\z@\@plus0.75fil \hskip\
\hskip\skip@
\hskip\z@\@plus0.25 fil\leaders\hrule height.5  \p@ depth.5  \p@ \hskip\z@\@plus0.5 fil \hskip\
\hskip\skip@
\hskip\z@\@plus0.375fil\leaders\hrule height.625\p@ depth.625\p@ \hskip\z@\@plus0.25fil \hskip\
% \hskip\skip@
% \hfil
 }%
%\hyper@anchorend
\hfil
}%
}%
\let\bibpreamble\@empty
\bibsep\z@\relax
\def\newblock{\ }%
```

\bibfont We define the font switch that applies to the body of the bibliography.

```
\def\bibfont{%
 \small
}%
```

28.3.10 Index

FIXME: the following call to \twocolumn appears wrong if we were in two-column grid.

```
\newenvironment{theindex}{%
%\let\wastwocol@sw\twocolumn@sw
 \columnseprule \z@
 \columnsep 35\p@
 \c@secnumdepth-\maxdimen
 \onecolumngrid@push
 \section{\indexname}%
 \thispagestyle{plain}%
 \parindent\z@
 \parskip\z@ plus.3\p@\relax
 \let\item\@idxitem
%\wastwocol@sw{\twocolumngrid}{}%
 \onecolumngrid@pop
}{%
%\onecolumngrid@pop
}%
%
\def\@idxitem{\par\hangindent 40\p@}
%
\def\subitem{\par\hangindent 40\p@ \hspace*{20\p@}}
%
\def\subsubitem{\par\hangindent 40\p@ \hspace*{30\p@}}
%
\def\indexspace{\par \vskip 10\p@ plus5\p@ minus3\p@\relax}
```

28.4 Journal-Specific Code

For APS journals, we supply code specific to PRA, PRB, PRC, PRD, PRE, PRL, and RMP. At present, they are identical, with the exception of PRB's code for superscript citations.

```
\def\journal@pra{pra}%
\def\journal@prb{prb}%
\def\journal@prc{prc}%
\def\journal@prd{prd}%
\def\journal@pre{pre}%
\def\journal@prl{prl}%
\def\journal@prstab{prstab}%
\def\journal@rmp{rmp}%
```

Note: the convention in this document class is that the substyle must not override any explicit class options declared by the document. This means that the various Booleans of Section 7 may be assigned here only if they are still undefined at this point.

For most all of the APS journals, the journal-dependent code is relatively meager and is therefore embedded in this file. However, the RMP code is sufficiently extensive that splitting it out into a separate file is more convenient.

28.4.1 pra

```
\@ifx{\@journal\journal@pra}{%
```

End of `pra` code.

```
}{}%
```

28.4.2 prb

```
\@ifx{\@journal\journal@prb}{%
```

PRB requires superscript citations. We use Patrick Daly's `natbib` package, and hyperref and other packages are already set up to cope with this.

`\bibpunct` By default, PRB selects the `bibnotes` option. We also set up for numerical citations.

```
 \@ifxundefined\bibnotes@sw{\@booleantrue\bibnotes@sw}{}%
 \bibpunct{}{}{,}{s}{}{\textsuperscript{,}}%
 \def\@onlinecite#1{\begingroup\let\@cite\NAT@citenum\citealp{#1}\endgroup}%
```

Invoke `superbib` option if the document has made no selection of its own.

```
 \appdef\class@documenthook{%
  \@ifxundefined\place@bibnumber{%
   \let\place@bibnumber\place@bibnumber@sup
  }{}%
 }%
```

End of `prb` code.

```
}{}%
```

28.4.3 prc

```
\@ifx{\@journal\journal@prc}{%
```

End of `prc` code.

```
}{}%
```

28.4.4 prd

```
\@ifx{\@journal\journal@prd}{%
```

End of `prd` code.

```
}{}%
```

28.4.5 pre

```
\@ifx{\@journal\journal@pre}{%
```

End of `prd` code.

```
}{}%
```

28.4.6 `prl`

```
\@ifx{\@journal\journal@prl}{%
```

In PRL, the default is the `bibnotes` option, and the Acknowledgments section has no head.

However, the References head is a device that may be described as a lozenge centered on the baseline, 71 points wide by 2 points thick, with the ends tapering to a half point in thickness. Space above 26 points base to base, below 31 base to base.

```
 \@ifxundefined\bibnotes@sw{\@booleantrue\bibnotes@sw}{}%
 \@booleanfalse\acknowledgments@sw
%\let\acknowledgmentsname\@empty
 \def\bibsection{%
  \par
  \begingroup
   \baselineskip26\p@
   \bib@device{\hsize}{72\p@}%
 \endgroup
  \nobreak\@nobreaktrue
  \addvspace{19\p@}%
 }%
```

Implement length checking. Use the `times` and `mathtime` packages, plus whatever other processing is required to make the formatted output be true to the metrics of the journal.

Wait. Do not use mathtime after all. APS has their own way of doing math pi, involving Adobe Mathematical Pi and other fonts.

```
%\appdef\class@documenthook{%
 \@ifxundefined\lengthcheck@sw{\@booleanfalse\lengthcheck@sw}{}%
  \lengthcheck@sw{%
   \RequirePackage{times}%
%  \RequirePackage{mathtime}%
  }{%
  }%
%}%
```

A PRL does not have numbered sections.

```
 \c@secnumdepth=-\maxdimen
```

Note: we defer this code until after type size file is read in.

```
 \appdef\set@typesize@hook{%
  \@ifnum{\@pointsize=10\relax}{%
   \lengthcheck@sw{%
    \def\large{%
     \@setfontsize\large{12.5}{14\p@}%
    }%
    \def\normalsize{%
     \@setfontsize\normalsize{10.5}\@xiipt
```

```
    \abovedisplayskip 6\p@ \@plus6\p@ \@minus5\p@
    \belowdisplayskip \abovedisplayskip
    \abovedisplayshortskip  \abovedisplayskip
    \belowdisplayshortskip \abovedisplayskip
    \let\@listi\@listI
   }%
   \def\small{%
    \@setfontsize\small{9.5}\@xipt
    \abovedisplayskip 5\p@ \@plus5\p@ \@minus4\p@
    \belowdisplayskip \abovedisplayskip
    \abovedisplayshortskip  \abovedisplayskip
    \belowdisplayshortskip \abovedisplayskip
    \let\@listi\@listI
   }%
   \DeclareMathSizes{12.5}{12.5}{9}{6}%
   \DeclareMathSizes{10.5}{10.5}{7.5}{5}%
   \DeclareMathSizes{9.5}{9.5}{7.0}{5}%
  }{%
   \def\normalsize{%
    \@setfontsize\normalsize\@xpt\@xiipt
    \abovedisplayskip 10\p@ \@plus2\p@ \@minus5\p@
    \belowdisplayskip \abovedisplayskip
    \abovedisplayshortskip  \abovedisplayskip
    \belowdisplayshortskip \abovedisplayskip
    \let\@listi\@listI
   }%
  }%
 }{}%
}%
%\appdef\set@pica@hook{%
  \@ifnum{\@pointsize=10\relax}{%
   \textheight = 694.0\p@
%  \textheight = 55\baselineskip
%  \advance\textheight by \topskip
  }{}%
%}%
```

End of `prl` code.

```
}{}%
```

28.4.7 `prstab`

```
\@ifx{\@journal\journal@prstab}{%
```

End of `prstab` code.

```
}{}%
```

28.4.8 `rmp`

Read in the code from a file.

```
\@ifx{\@journal\journal@rmp}{\input{\@journal\substyle@ext}}{}%
```

28.5 Establish APS Defaults

\footinbib@sw All APS journals except RMP invoke the `footinbib` option. We rely on the RMP-specific code to give a value to this switch.

```
\@ifxundefined\footinbib@sw{\@booleantrue\footinbib@sw}{}%
```

\place@bibnumber \@bibstyle We install code that will select the presentation for `\bibitem`s and govern the BibTeX processing. If the user has not selected any related option and if the journal substyle also has not, then the `aps` option choses these defaults.

```
\appdef\class@documenthook{%
 \@ifxundefined\place@bibnumber{%
  \let\place@bibnumber\place@bibnumber@inl
 }{}%
 \@ifxundefined\@bibstyle{%
  \def\@bibstyle{apsrev}%
 }{}%
}%

%</aps>
```

29 The rmp journal substyle: the rmp module

The file `rmp.rtx` is read in by the `revtex4` document class if `\@society` has the value `aps` and `\@journal` has the value `rmp`.

It is read at the end of the revtex4.dtxaps.rtx, so all definitions and assignments in that file are operative unless overridden here.

```
%<*rmp>
```

Protect this file from being read in by anything but REVTeX.

```
\ifx\undefined\substyle@ext
 \def\@tempa{%
  \endinput
  \GenericWarning{I must be read in by REVTeX! (Bailing out)}%
 }%
 \expandafter\else
  \def\@tempa{}%
 \expandafter\fi\@tempa
```

Protect this file from being read in as a society instead of a journal. In such a case, `\@journal` will be `\@empty`.

```
\@ifx@empty\@journal{%
 \class@warn{I have to read in the aps substyle first!}%
 \endinput
 \def\@society{aps}\def\@journal{rmp}%
 \expandafter\input\expandafter{\@society\substyle@ext}%
}{}%
```

29.1 Frontmatter

\frontmatter@setup

```
\def\frontmatter@setup{%
 \normalfont\sffamily\raggedright
}%
```

\PACS@warn Per Mark Doyle, RMP never displays the PACS, so they don't want the 'use showpacs' warning spit out.

```
\def\PACS@warn{RMP documents do not display PACS. Your \string\pacs\space will be ignored}%
```

\frontmatter@title@above
\frontmatter@title@format
\frontmatter@title@below

```
\def\frontmatter@title@above{}%
\def\frontmatter@title@format{\Large\bfseries\raggedright}% HelveticaNeue-Medium(Italic) 14pt.
\def\frontmatter@title@below{\addvspace{12\p@}}% 24pt b-b down to first author
```

\frontmatter@authorformat Set the rag to a milder value, because we want to do true ragged right typesetting, as opposed to the LaTeX default, which gives very poor results.

Note: author font is 9.8bp. 19.2bp/14.3bp above/below.

```
\def\frontmatter@authorformat{%
 \preprintsty@sw{\vskip0.5pc\relax}{}%
 \@tempskipa\@flushglue
 \@flushglue\z@ plus50\p@\relax
 \raggedright\advance\leftskip.5in\relax
 \@flushglue\@tempskipa
 \parskip\z@skip
}%
```

\frontmatter@affiliationfont The hook \frontmatter@affiliationfont controls the formatting of affiliations and affiliation groups. The hook \frontmatter@above@affilgroup is invoked just before proceeding with author/affiliation processing. The \frontmatter@above@affiliation is the amount of space above affiliations in the groupedaddress style, and \frontmatter@above@affiliation@script is that for superscriptaddress.

Note: affiliation font is 9.03/10.4bp, 14.3bp/19.2bp b-b above/below

```
\def\frontmatter@affiliationfont{% Helvetica 9/10.2
 \small\slshape\selectfont\baselineskip10.5\p@\relax
 \@tempskipa\@flushglue
 \@flushglue\z@ plus50\p@\relax
 \raggedright\advance\leftskip.5in\relax
 \@flushglue\@tempskipa
}%
\def\frontmatter@above@affilgroup{\addvspace{7.2\p@}}% additional leading above an author
\def\frontmatter@above@affiliation{\addvspace{5.3\p@}}%
\def\frontmatter@above@affiliation@script{}%
```

Set up the default RMP style for title block authors and affiliations. This command should effectively override the effect of the corresponding command in the parent substyle.

```
\@ifxundefined\groupauthors@sw{%
 \clo@groupedaddress
}{}%
```

`\frontmatter@RRAP@format` Note: in RMP, if we are not in preprint mode, the date will not be produced.
Note: Helvetica C/lc, 8.98bp, space above: 16.3bp b-b.

```
\def\frontmatter@RRAP@format{%
  \addvspace{7.3\p@}%
  \small
  \raggedright\advance\leftskip.5in\relax
% \parskip.5ex\relax
% \everypar{\hbox\bgroup(\@gobble@leavemode@uppercase}%
% \def\par{\@ifvmode{}{\unskip)\egroup\@@par}}%
% \preprintsty@sw{}{\let\@date\@empty}%
}%
%\def\@gobble@leavemode@uppercase#1#2{\expandafter\MakeTextUppercase}%
\def\produce@RRAP#1{%
  \@if@empty{#1}{}{%
   \@ifvmode{\leavevmode}{}%
   \unskip(\ignorespaces#1\unskip)\quad
  }%
}%
```

`\frontmatter@abstractheading` Space above 21.8bp b-b.

```
\def\frontmatter@abstractheading{%
 \preprintsty@sw{%
  \begingroup
   \centering\large
   \abstractname
   \par
  \endgroup
  \vspace{.5pc}%
 }{}%
}%
```

`\frontmatter@abstractfont` TimesTen 8.93bp/9.6bp X 360bp, indented 36bp, with 21.9/37.6bp b-b above/below

```
\def\frontmatter@abstractfont{%
 \footnotesize
 \hsize360\p@
 \leftskip=0.5in
 \advance\hsize\leftskip
%\rightskip=\leftskip
 \parindent\z@
%\hsize5.5in
}%
```

`\frontmatter@preabstractspace` `\frontmatter@postabstractspace` Space above and space below abstract in title block. Should be 22/36 points base-to-base.

```
\def\frontmatter@preabstractspace{7.7\p@}%
\def\frontmatter@postabstractspace{24.6\p@}%
```

FIXME: Not done: PACS. FIXME: TOC: Head is same as

30 :

HelveticaNeue 8.98. 32/22bp b-b above/below, Body: TimesTen 8/10.5.

30.1 General Text

If not in preprint mode, set the type size to 10/12 point. Note: s/b 11.6bp leading

```
\appdef\set@typesize@hook{%
 \@ifxundefined\preprintsty@sw{}{%
  \preprintsty@sw{}{%
   \def\normalsize{%
    \@setsize\normalsize{12pt}\xpt\@xpt
    \abovedisplayskip 10\p@ plus2\p@ minus5\p@
    \belowdisplayskip \abovedisplayskip
    \abovedisplayshortskip  \abovedisplayskip
    \belowdisplayshortskip \abovedisplayskip
    \let\@listi\@listI
   }%
  }%
 }%
}%
```

Footnote mods:

```
\footnotesep 9.25pt
\skip\footins 36pt plus 4pt minus 2pt
\def\footnoterule{\kern-13pt\hrule width.5in\kern15.6pt\relax}%
```

30.2 Sectioning

```
\def\thepart{\Roman{part}} %
\def\thesection{\Roman{section}}
\def\p@section{}
\def\thesubsection{\Alph{subsection}}
\def\p@subsection{\thesection.}
\def\thesubsubsection{\arabic{subsubsection}}
\def\p@subsubsection{\thesection.\thesubsection.}
\def\theparagraph{\alph{paragraph}}
\def\p@paragraph{\thesection.\thesubsection.\thesubsubsection.}
\def\thesubparagraph{\arabic{subparagraph}}
\def\p@subparagraph{\thesection.\thesubsection.\thesubsubsection.\theparagraph.}
```

In RMP, put a period (.), followed by nut space, after the section number. Also, hang the section number (the LaTeX default).

```
\def\@seccntformat#1{\csname the#1\endcsname.\hskip0.5em\relax}%
```

Note that we wish to set the section head uppercase, so we use David Carlisle's `\MakeTextUppercase`. However, because this procedure effectively parses its ar-

gument (looking for things to *not* translate), it has to be invoked in such a way that the argument of the `\section` command is passed to it as its own argument.

To accomplish this, we use the `\@hangfrom@` hook, which was developed for this purpose.

```
\def\section{%
 \@startsection{section}{1}{\z@}{0.8cm plus1ex minus.2ex}{0.4cm}%
 {%
  \small\sffamily\bfseries\selectfont
  \raggedright
  \parindent\z@
 }%
}%
\def\@hangfrom@section#1#2#3{\@hangfrom{#1#2}\MakeTextUppercase{#3}}%
\def\subsection{%
 \@startsection{subsection}{2}{\z@}{0.8cm plus1ex minus.2ex}{0.4cm}%
 {%
  \small\sffamily\bfseries
  \raggedright
  \parindent\z@
 }%
}%
\def\subsubsection{%
 \@startsection{subsubsection}{3}{\z@}{.8cm plus1ex minus.2ex}{0.4cm}%
 {%
  \small\sffamily\selectfont
  \raggedright
  \parindent\z@
 }%
}%
\def\paragraph{%
 \@startsection{paragraph}{4}{\z@}{.8cm plus1ex minus.2ex}{-1em}%
 {%
  \small\slshape\selectfont
  \raggedright
  \parindent\z@
 }%
}%
\def\subparagraph{%
 \@startsection{subparagraph}{4}{\parindent}{3.25ex plus1ex minus.2ex}{-1em}%
 {\normalsize\bfseries\selectfont}%
}%
%
 \setcounter{tocdepth}{4}% FIXME: has no effect
```

`\appendix`
`\@hangfrom@appendix`
`\@hangfroms@appendix`
`\@appendixcntformat`

```
\appdef\appendix{%
 \let\@hangfrom@section\@hangfrom@appendix
 \let\@hangfroms@section\@hangfroms@appendix
 \let\@sectioncntformat\@appendixcntformat
}%
```

```
\def\@hangfrom@appendix#1#2#3{%
 #1%
 \@if@empty{#2}{%
  #3%
 }{%
  #2\@if@empty{#3}{}{:\ #3}%
 }%
}%
\def\@hangfroms@appendix#1#2{%
 #1\appendixname\@if@empty{#2}{}{:\ #2}%
}%
\def\@appendixcntformat#1{\appendixname\ \csname the#1\endcsname}%
```

30.3 Figure and Table Caption Formatting

`\@makecaption`

```
\setlength\belowcaptionskip{2\p@}
\long\def\@makecaption#1#2{%
  \vskip\abovecaptionskip
  \vbox{%
   \flushing
   \small\rmfamily
%  \hsize\@capwidth
   \noindent
   #1\nobreak\hskip.5em plus.2em\ignorespaces#2\par
  }%
  \vskip\belowcaptionskip
}%
```

30.4 Citations and Bibliography

Customize REVTEX for the journal substyle; this task requires three components: a BIBTEX `.bst` style file, customizing code for `natbib`, and customizations of the `thebibliography` environment.

`\@bibstyle` Define the argument of the `\bibliographystyle` command (if the document does not do so).

The user must have installed a `.bst` file of the corresponding name. This file will then be used by BIBTEX when compiling the document's `.bbl` file.

To generate `apsrmp.bst`, use `custom-bib` version 3.89d1 or later. Run the `.bst` generator, `makebst.tex`, with the following options:

1. STYLE OF CITATIONS: a: ay(Author-year with some non-standard interface)
2. AUTHOR-YEAR SUPPORT SYSTEM: nat: (Natbib for use with natbib v5.3 or later)
3. ORDERING OF REFERENCES: *: (Alphabetical);

4. ORDER ON VON PART: x: vonx (Sort without von part—de la Maire after Mahone)
5. AUTHOR NAMES: a: nm-rev1 (Only first name reversed, initials—AGU style: Smith, J. F., H. K. Jones)
6. POSITION OF JUNIOR: jnrlst (Junior comes last as Smith, John, Jr.)
7. NUMBER OF AUTHORS: l: max 12, min 12 (three prompts in all);
8. TYPEFACE FOR AUTHORS IN LIST OF REFERENCES: u: nmft, nmft-def (User defined author font—`\bibnamefont`)
9. FONT FOR FIRST NAMES: u: fnm-def (First names in user defined font—`\bibfnamefont`)
10. EDITOR NAMES IN INCOLLECTION ETC: a: nmfted (Editors incollection like authors font)
11. FONT FOR 'AND' IN LIST: r: nmand-rm ('And' in normal font—JONES and JAMES)
12. FONT OF CITATION LABELS IN TEXT: u: lab, lab-def (User-defined citation font—`\citenamefont`)
13. LABEL WHEN AUTHORS MISSING: keyxyr (Year blank when KEY replaces missing author—for natbib 7.0)
14. DATE POSITION: b: dt-beg (Date after authors)
15. SUPPRESS MONTH: xmth (Date is year only)
16. DATE FORMAT: yr-com (Date preceded by comma as ', 1993')
17. TRUNCATE YEAR: note-yr (Year text full as 1990–1993 or 'in press')
18. ARTICLE TITLE PRESENT: x: jtit-x (No article title);
19. JOURNAL TITLE FONT: r: jttl-rm (Journal title normal);
20. TECHNICAL REPORT TITLE: b: trtit-b (Tech. report title like books);
21. JOURNAL VOLUME: b: vol-bf (Volume bold);
22. VOLUME PUNCTUATION: c: volp-com (Volume with comma);
23. PAGE NUMBERS: f: jpg-1 (Only start page number);
24. POSITION OF PAGES: e: pp-last (Pages at end, but before any notes)
25. NUMBER AND SERIES FOR COLLECTIONS: num-xser (Allows number without series and suppresses word "number")

26. JOURNAL NAME PUNCTUATION: x: jnm-x (Space after journal);
27. PUBLISHER IN PARENTHESES: p: pub-par (Publisher in parentheses)
28. PUBLISHER POSITION: p: pre-pub (Publisher before chapter, pages);
29. ISBN NUMBER: *: isbn (Include ISBN);
30. ISSN NUMBER: *: isbn (Include ISSN for periodicals);
31. 'EDITOR' AFTER NAMES: a: edpar ('Name (editor),' in parentheses, after name, comma after)
32. EDITOR IN COLLECTIONS: b: edby (In booktitle, edited by. . . —where . . . is names)
33. PUNCTUATION BETWEEN SECTIONS (BLOCKS): c: blk-com (Comma between blocks);
34. 'IN' WITH JOURNAL NAMES: i: injnl (Add 'in' before journal name in style for incollection)
35. ABBREVIATE WORD 'PAGES': a: pp ('Page' abbreviated);
36. ABBREVIATE WORD 'EDITORS': a: ed ('Editor' abbreviated);
37. ABBREVIATION FOR 'EDITION': a: ednx ('Edition' abbreviated as 'ed');
38. EDITION NUMBERS: xedn (Editions as in database saving much processing memory)
39. STORED JOURNAL NAMES: a: jabr (Abbreviated journal names);
40. FONT OF 'ET AL': i: etal-it (Italic et al);
41. ADDITIONAL REVTeX DATA FIELDS: r: revdata, eprint, url, url-blk (Include REVTeX data fields)
42. NEW FONT SELECTION SCHEME: n: nfss (NFSS);
43. ADDITIONAL REVTeX DATA FIELDS: y: revdata (additional data fields);
44. REFERENCE COMPONENT TAGS: r: bibinfo (reference component tags);

A file `apsrmp.dbj` file equivalent to the following should result:

```
% \input docstrip
% \preamble
% ----------------------------------------
% *** REVTeX-compatible apsrmp.bst ***
% \endpreamble
% \postamble
```

```
% End of customized bst file
% \endpostamble
% \keepsilent
% \askforoverwritefalse
% \def\MBopts{\from{merlin.mbs}{%
%    ay%: Author-year with some non-standard interface
%   ,nat%: Natbib for use with natbib v5.3 or later
%   ,vonx%: Sort without von part (de la Maire after Mahone)
%   ,nm-rev1%: Only first name reversed, initials (AGU style: Smith, J. F., H. K. Jones)
%   ,jnrlst%: Junior comes last as Smith, John, Jr.
%   ,nmlm%: Limited authors (et al replaces missing names)
%     ,x10%: Maximum of 10 authors
%     ,m10%: Minimum of 10 authors
%   ,nmft,nmft-def%: User defined author font (\bibnamefont)
%   ,fnm-def%: First names in user defined font (\bibfnamefont)
%   ,nmfted%: Editors incollection like authors font
%   ,nmand-rm%: 'And' in normal font (JONES and JAMES)
%   ,lab,lab-def%: User defined citation font (\citenamefont)
%   ,keyxyr%: Year blank when KEY replaces missing author (for natbib 7.0)
%   ,dt-beg%: Date after authors
%   ,xmth%: Date is year only
%   ,yr-com%: Date preceded by comma as ', 1993'
%   ,note-yr%: Year text full as 1990--1993 or 'in press'
%   ,jtit-x%: No article title
%   ,jttl-rm%: Journal name normal font
%   ,trtit-b%: Tech. report title like books
%   ,vol-bf%: Volume bold as {\bf vol}(num)
%   ,volp-com%: Volume with comma as vol(num), ppp
%   ,jpg-1%: Only start page number
%   ,pp-last%: Pages at end but before any notes
%   ,num-xser%: Allows number without series and suppresses word "number"
%   ,jnm-x%: Space after journal name
%   ,pub-par%: Publisher in parentheses
%   ,pre-pub%: Publisher before chapter, pages
%   ,isbn%: Include ISBN for books, booklets, etc.
%   ,issn%: Include ISSN for periodicals
%   ,edpar%: 'Name (editor),' in parentheses, after name, comma after
%   ,edby%: In booktitle, edited by ..  (where .. is names)
%   ,blk-com%: Comma between blocks
%   ,injnl%: Add 'in' before journal name in style for incollection
%   ,pp%: 'Page' abbreviated as p. or pp.
%   ,ed%: 'Editor' abbreviated as ed. or eds.
%   ,xedn%: Editions as in database saving much processing memory
%   ,jabr%: Abbreviated journal names
%   ,etal-it%: Italic et al
%   ,revdata,eprint,url,url-blk%: Include REVTeX data fields collaboration, eid, eprint, numpages
%   ,bibinfo%: Reference component tags like \bibinfo in the content of \bibitem
%   ,nfss%: NFSS use \textbf, \emph, not \bf, \em
%   }}
% \generate{\file{\jobname.bst}{\MBopts}}
```

```
% \endbatchfile
%
\appdef\class@documenthook{%
 \@ifxundefined\@bibstyle{%
  \def\@bibstyle{apsrmp}%
 }{}%
}%
```

\bibpunct The following commands effectively etablish the style in which \cite commands are formatted. You can think of them as the second needed component for the bibliography.

Set up for author-year citations: when \NAT@set@cites executes (at \begin{document} time), the \@biblabel will be set to \NAT@biblabel.

Per Karie Friedman (friedman@phys.washington.edu), multiple citations are separated by semicolons, e.g., (Jones, 1999; Abbott and Smith, 2000; Wortley, 2001a), and multiple citations by the same author by commas, e.g., Abela et al. (1995, 1997a, 1997b). The third argument of \bibpunct handles the former.

The fifth argument puts a comma after the author when the year is not in parens: (Lee et al., 1996).

```
\bibpunct{(}{)}{;}{a}{,}{,}%
```

\cite Per Mark Doyle, \cite is mapped to \citep in RMP.

```
\let\cite\citep
```

\bibsection We define the sectioning command to use when starting the bibliography.

\bibpreamble \newblock \bibhang \bibsep

```
\def\bibsection{%
 \let\@hangfroms@section\@hang@froms
 \section*{\refname}%
 \@nobreaktrue
}%
\let\bibpreamble\@empty
\def\newblock{\ }%
\bibhang10\p@
\bibsep\z@
```

\footinbib@sw Footnotes in bibliography are inconsistent with author-year references, and are particularly nast under natbib: the packcage will automatically change to numbered references if any \bibitem commands lack the optional argument. Therefore, we must uninvoke it now, even if invoked by the document.

```
\@ifxundefined\footinbib@sw{}{%
 \footinbib@sw{%
  \class@warn{%
   Footnotes in bibliography are incompatible with RMP.^^J%
   Undoing the footinbib option.
  }%
 }{}%
}%
\@booleanfalse\footinbib@sw
```

`\eprint` RMP requires the `\eprint` field in the bib entry to be set off with the word "eprint".

```
\def\eprint#1{eprint #1}%
```

30.5 Table of Contents

We set up for auto-sizing of certain TOC elements.

To do this, we override certain definitions for the default TOC font (`\toc@@font`) and spacing (dotsep), and define formatting for the needed elements (`\l@...`). Finally, we activate the autosizing by assigning `\toc@pre` and `\toc@post`.

`\toc@@font` Set the formatting characteristics of the auto-indenting part of the TOC.

```
\def\toc@@font{%
 \footnotesize\rmfamily
 \def\\{\space\ignorespaces}%
}%
\def\@dotsep{5.5pt}%
```

`\l@section` Determine which TOC elements are automatically indented.

```
\def\tocleft@{\z@}%
\def\tocdim@min{5\p@}%
\def\l@section{%
 \l@@sections{}{section}% Implicit #3#4
}%
\def\l@subsection{%
 \l@@sections{section}{subsection}% Implicit #3#4
}%
\def\l@subsubsection{%
 \l@@sections{subsection}{subsubsection}% Implicit #3#4
}%
%\def\l@subsubsection#1#2{}%
\def\l@paragraph#1#2{}%
\def\l@subparagraph#1#2{}%
```

Activate the TOC processing.

```
\let\toc@pre\toc@pre@auto
\let\toc@post\toc@post@auto
```

```
%</rmp>
```

Here ends the programmer's documentation.

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